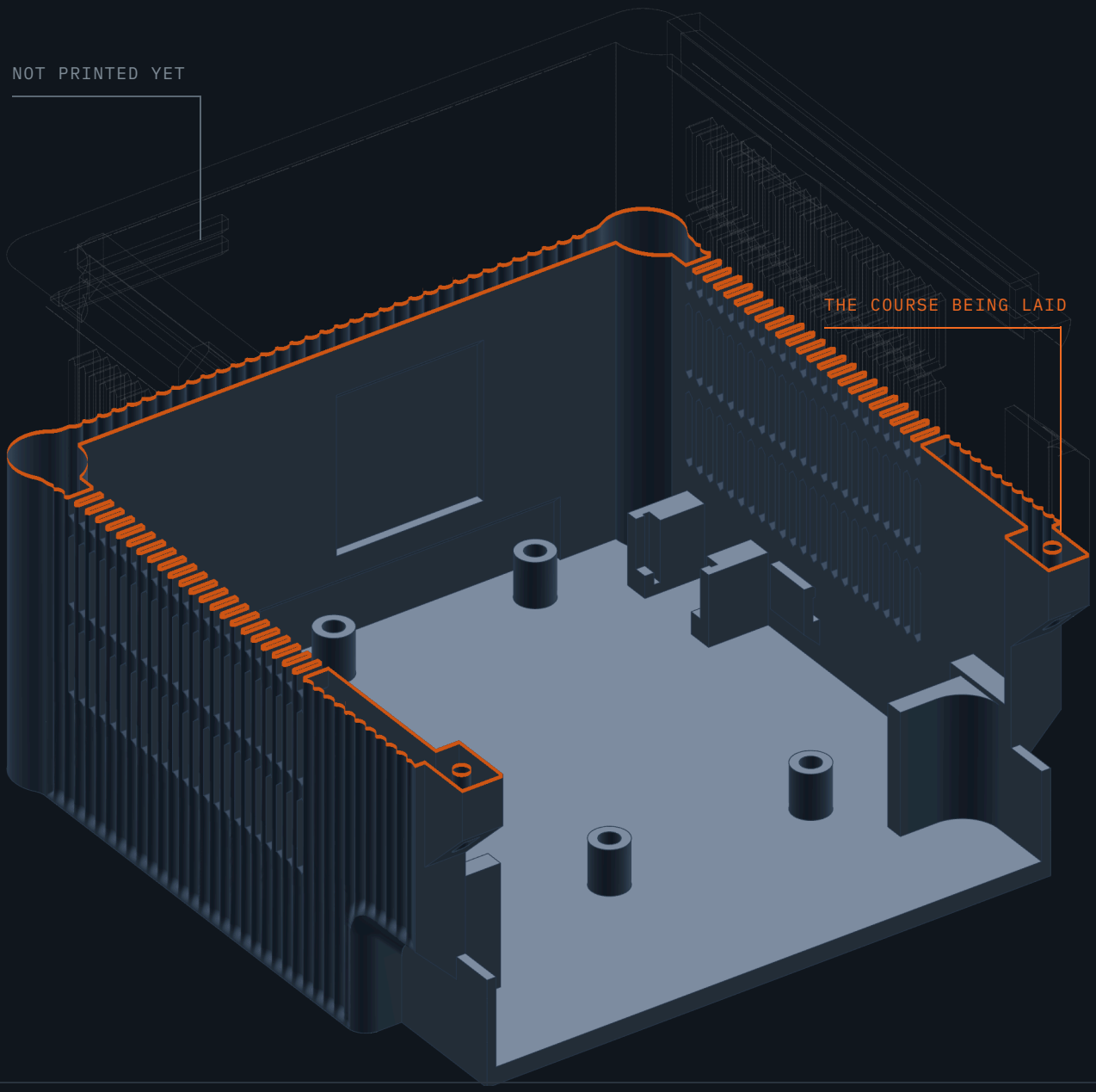
HOME SODA MACHINE

Printing the Machine

What nine months of printing a kitchen appliance taught us about FDM, one plastic at a time.



WHAT THIS BOOK IS

Every page is one of our prints.

Nearly everything you can see or touch on the Home Soda Machine comes off two desktop printers. Getting there took nine months of prints that failed for reasons. This book is the reasons, each shown on the print that taught it.

On 5 April 2026, a month into the project, a section of general advice about 3D printing was deleted from the repository. An AI had written it, and it was not about anything we had printed.

Everything here is the opposite. Each page starts from a print that exists: its date, its printer, its plastic and a photograph. The rule at the top is what that print taught, and the page says how far the lesson has been tested.

It is sorted by plastic

Most of what a print teaches belongs to the plastic it was made of. PET-GF wants its fan off; soft TPU curls without it. A first layer that is right for one is wrong for the next. So after a short part on what is plain geometry, the book has a part for each plastic, and every page says which one it is about.

It says what it does not know

STILL OPEN marks what one print suggested and nothing has confirmed, or a result that changed two things at once.

Its numbers do not go stale

A setting in the current recipe is read from its project file each time the book is put together. A number from a trial stays as it was measured, with its date.

It is written for agents too

The machine is designed and printed with AI agents, who start each day knowing none of this. Every page is a plain file in the repository, so what a person reads here and what an agent reads there are the same words.



1. **The rule.** One line you could say across a workbench.
2. **Where it holds.** Which plastic, printer, plate and nozzle the evidence comes from.
3. **The print that taught it.** Photographed, with its date and printer.
4. **Derek's words.** Exactly as he typed them, beside an orange bar.
5. **In the record.** Links to the files that hold the evidence.

HOW TO READ A DRAWING

Plastic already printed.

The course being laid, still soft.

Printed to be thrown away: a fin or a support.

Something that was cut away.

CONTENTS

One plastic at a time.

Before any filament

Every course lands on something 05
A bridge needs two landings 06
Height moves one layer at a time 07
A coupon has to print like the part 08
Change one thing 09
The same question, a different answer 10

PET-GF

It hides its layers and breaks along them 11
One file holds the recipe 12
A first-layer trim does not travel 13
Season the edge of the plate, or stay off it 14
Fan only where the plastic is over air 15
A small part needs time between layers 16
A matte wall hides the layers and nothing else 17
Layer height is not a knob 18
Start a round with a chamfer 19
Blocks do not come out of a cavity 20
One layer of air held on, two let go 21
The support that fell off was the failure 22
Support D, a solid roof and a small gap 23
Plan a support from its way out 24
On a tilted underside, check the roof is there 25
Paint supports by hand and every underside is yours 26
One pair of supported faces 27
Orientation comes first 28
The roof is the first layer 29
The skin is a printing decision 30
A clearance is surface, not air 31
PET-GF does not stretch 32
A snap bends within its layers 33
Never load a peg across its layers 34

PETG

Plain PETG holds the flavoring 35
It weeps where the floor meets the wall 36
A floor made of loops leaks along its seams 37
Slow and hot were not what sealed it 38
Five hundredths of a millimetre 39

TPU

Choose the hardness by the fit 40
The nozzle goes around, the string goes straight 41
Less fan made it curl more 42
The setting said 10, the lip ran at 16 43
One island at a time 44
Every underside at 45 degrees or steeper 45
A thin soft wall starts on a cap 46
Two brands, one hardness 47

ASA Aero

It prints lighter than it feeds 48
Pause, seat the magnet, print on 49
Every jump carries a bit along 50

PET-CF

Twenty-one tries at one faucet 51
A plastic's lessons stay with it 52

The H2C

Each plastic has its side, plate and feed 53
Measure where the second nozzle lands 54
Start the second printer three minutes after the first 55
A send is proven only by a new job number 56
One overload message, three causes 57

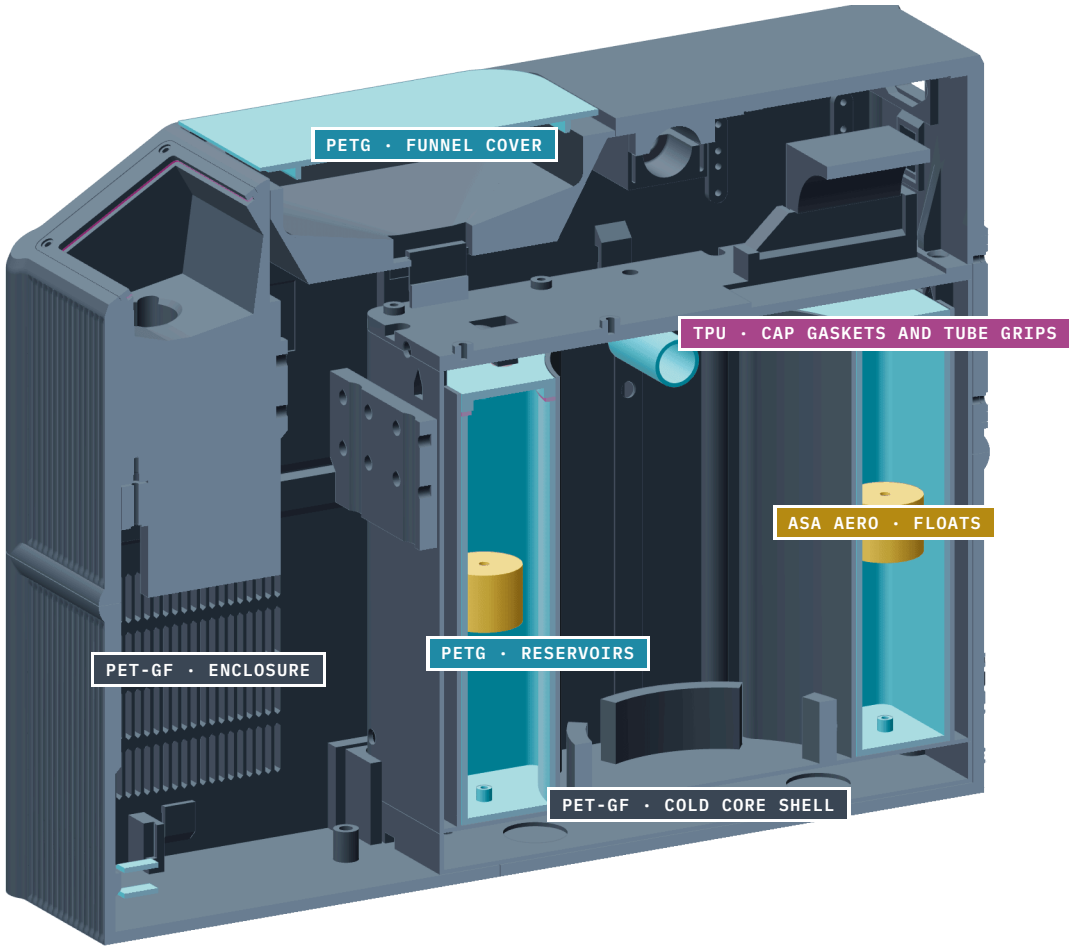
At the back

Seven things that had to be said again 58
Nine months 59
How this book is kept 60

THE MACHINE

Four plastics, each doing one job.

The machine cut in half, with every printed part coloured by the plastic it is made of. Which plastic a part gets was decided by what the part has to do.



BOUGHT PARTS, TUBES AND WIRING ARE HIDDEN
THE FAUCET, ALSO PET-GF, STANDS ON THE COUNTER ABOVE

PET-GF	**Everything stiff and everything you see.**	The four enclosure quadrants, the cold core's shell and caps, the faucet, the funnel frame, the snap caps, rings and labels.
PETG	**Whatever holds liquid, and a few small parts.**	Both flavor reservoirs and their caps, the funnel cover, the status plate, a shroud, a bridge and a clamp. The mold for the silicone funnel.
TPU	**Whatever has to stretch or seal.**	Reservoir cap gaskets, tube grips, the display gasket, the drain adapter, and the pump elbows in development.
ASA AERO	**The one part that has to float.**	Three level floats, each printed around a magnet.
PET-CF	**Nothing now.**	The faucet's first attempts, in the spring.

IN THE RECORD

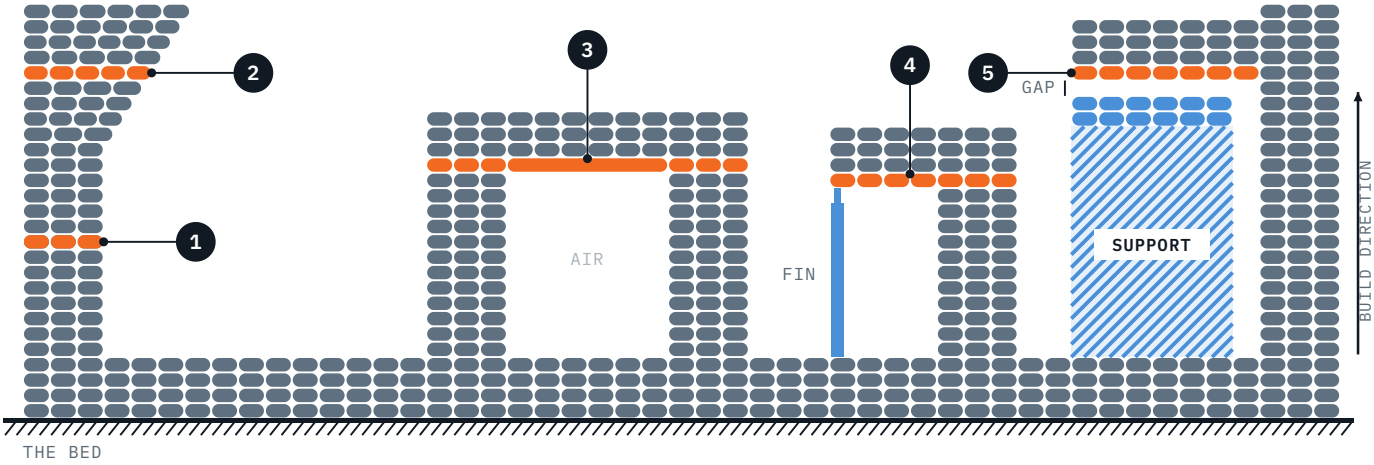
Every printed part and its plastic · The machine in 3D

THE ONE IDEA

Every course lands on something.

A printer builds a part one course at a time, from the bed up. Each road of plastic comes out soft and stays put only if something is already underneath. There are five things it can land on.

HOLDS FOR any filament: it is geometry · each plastic's own limits are in its part



1 Straight up

The course below, directly underneath. A plain wall. Everything else here is a way of getting back to this.

IN PET-GF
Every wall of the enclosure.

2 Lean out

Step each course out a little, so most of every road still sits on the one below.

IN PET-GF
The lifting handholds, the chamfer under a bed-facing round, the pointed roof of a teardrop hole.

3 Bridge

Pull the road tight between two landings. Both have to be there before the road arrives.

IN PET-GF
The connector pockets, the receiver catches, the roofs of the insert pilots.

4 Fin

A thin wall we draw into the part under an edge, to be snapped off at the bench.

IN PET-GF
The visible lower edge of the faucet base.

5 Support

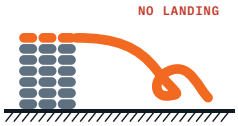
Scaffold from the slicer, with a gap so it lets go. Only where a tool can get in to lever it out.

IN PET-GF
The two rail catches of front-bottom. Nothing else on that part.

The first three leave nothing to remove.

A wall, a lean and a bridge are all part. A fin has to be snapped off and a support levered out, so each needs a way in. In PET-GF a bridge leaves the roughest underside of the five, so it goes where nobody looks or no tool can reach. Before any of them: can the part be turned so the roof becomes a floor?

Air is not on the list. A road laid over nothing does not hang there waiting for the next one. It droops, and everything printed onto it afterwards goes with it.





12 NOVEMBER 2026 · PET-GF
A pocket roof laid across a cutout that had taken its landing away.

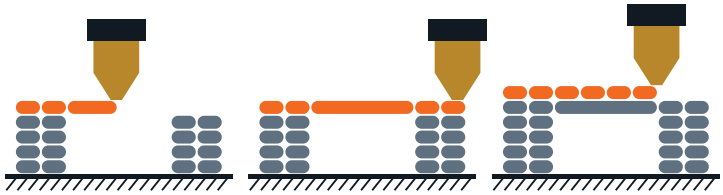
LANDING 3 · BRIDGE

A bridge needs two landings.

HOLDS FOR any filament: it is geometry · shown here in PET-GF

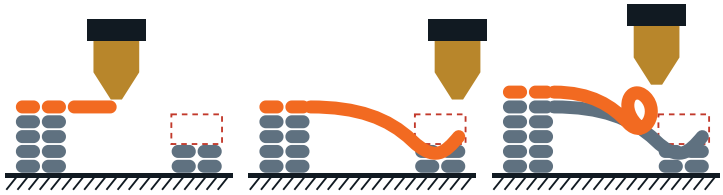
A bridge is the one time the printer lays plastic over air on purpose. It works because the road is fixed at one end, pulled across, and pressed down at the other while it is still soft. It hangs like a line pulled tight. Take away either end and it is just a road laid over nothing.

TWO LANDINGS

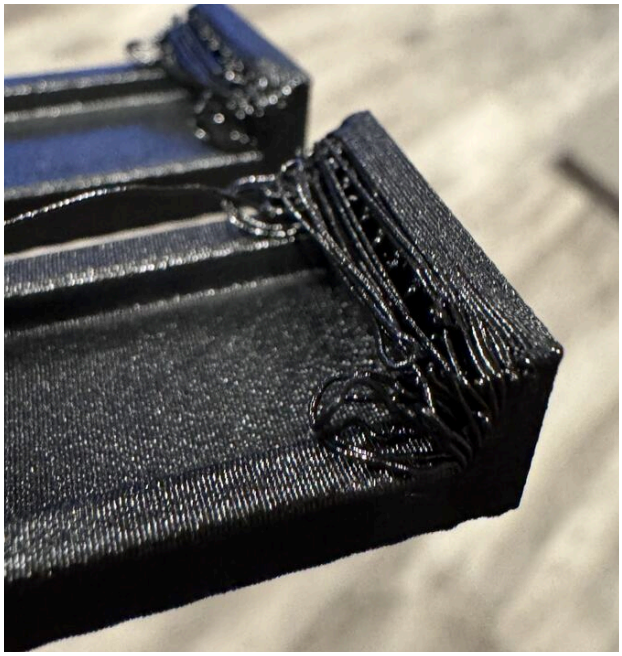


Leaves the first landing. Pulled across, pressed onto the second. The next course has a floor.

ONE LANDING CUT AWAY



Leaves the first landing. Nothing to press onto. It sags. The next course is laid on the sag.



12 NOVEMBER 2026 · MARK2 · PET-GF
A fit coupon for the connector pockets in front-top. A cutout added to clear the connector's levers went through the wall the pocket roofs rested on. The slice still showed those roofs as bridges, and that was taken as proof they would print.

> *"The only reason a bridge works is because it *is* a bridge that actually has a column supporting both ends."*
>
> DEREK · ON SEEING THAT COUPON

How it gets in

Nobody draws a bridge to nowhere on purpose. It arrives as a side effect, when a relief, a chamfer or a cutout made for some other reason passes through the wall a bridge was resting on. On the rear receiver it was a service opening. Five days later, on these pockets, it was clearance for a lever.

Before you print

In the slice, find the first road across every opening and follow each of its ends straight down. If either end is over air it is not a bridge, whatever the slicer calls it. After any cut near an opening, look at the roofs beside it again.

IN THE RECORD

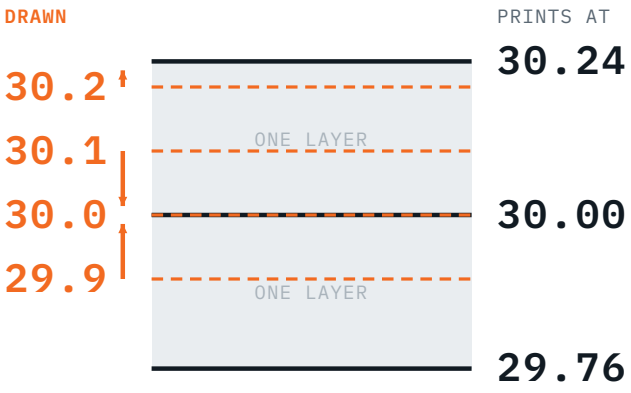
The coupon and its photographs · The receiver, five days earlier

HEIGHT

Height moves one layer at a time.

The printer cannot lay half a layer. A height in the drawing is rounded to a whole number of layers, so a change smaller than a layer either vanishes or arrives as a whole layer.

HOLDS FOR any filament: it is arithmetic · the example is a PET-GF pocket, 12 November 2026



ONE POCKET ROOF, FOUR DRAWINGS
Three of the four drawn heights print as the same pocket. The fourth prints a whole layer taller.

What was asked

Two test pockets for a connector, one a tenth of a millimetre shorter than the other, to feel which grips better.

What the slice did

A tenth is less than half of a 0.24 mm layer. Both wide pockets came out the same height. The narrow pair came out a whole layer apart.

POCKET	DRAWN	PRINTED
Wide pocket, three dots	30.0 mm	**30.00 mm**
Wide pocket, four dots	29.9 mm	**30.00 mm**
Narrow pocket, three dots	30.2 mm	**30.24 mm**
Narrow pocket, four dots	30.1 mm	**30.00 mm**

What went wrong next

To make the tenth show, the plate was re-sliced with thinner layers around the pockets, and sent. Derek stopped it.

> *“Do not use finer layers anywhere - that is not an option - that is a completely different variable”*
>
> DEREK · 12 NOVEMBER 2026

> *“if the slices come out the same with what I requested then just tell me that.”*
>
> DEREK · THE SAME MINUTE

A different variable

Layer height is part of the process, like temperature. Change it and an overhang prints differently, a support sits a different distance under its roof, and each course has a different time to cool. A fit trial that changes it is testing two things.

What we do

Keep the plastic’s own layers. Read the opening from the sliced file, from the top of the last floor course to the underside of the first roof course. When two drawings print the same, say so. To compare heights, draw them a whole layer apart.

IN THE RECORD

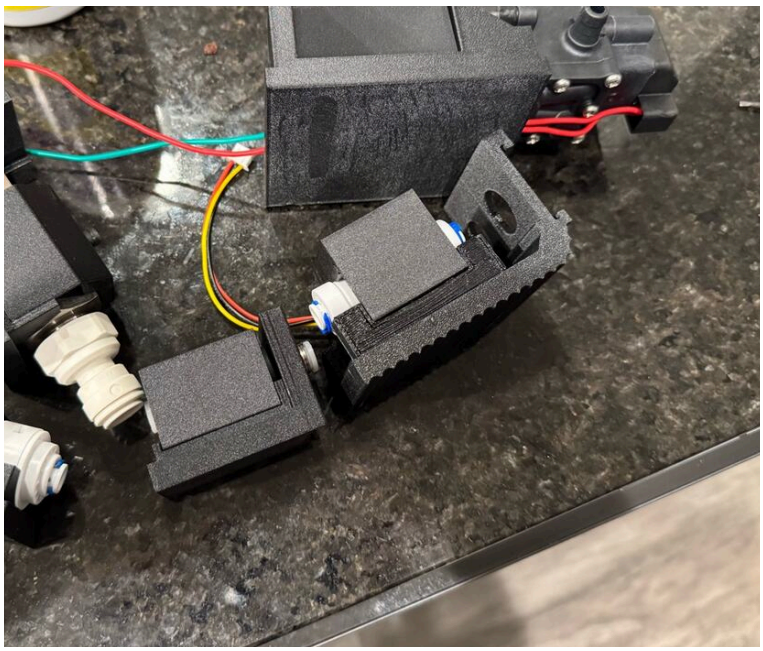
Drawn and sliced openings · The cancelled plate with thinner layers · How the fit came out

TEST PIECES

A coupon has to print like the part.

A coupon is a small piece cut out of a big part, so a fit can be tried in an hour instead of a day. It tells the truth only if it keeps everything that decides how the feature prints.

HOLDS FOR any filament: it is method · every example on this page is PET-GF



2 NOVEMBER 2026 · PET-GF
Five coupons, each a crop of the real part with its own wall, holding the hardware it was cut to hold. Two were kept as they were. Three went back to be changed.

What a coupon keeps

1. **The pose.** The same way up as the full part, so its layers run the same way and its overhangs face the same way.
2. **The layers.** The plastic's own first layer and layer height, the same nozzle and that printer's own trim.
3. **The wall it sits in.** A pocket cut free of its wall is a different pocket. Keep the wall and its full thickness.
4. **Both landings.** Every bridge in the coupon needs both of the landings it has in the part.
5. **Every stop.** Whatever the mating part touches in the machine, it should touch in the coupon.

What a coupon cannot tell you

WHETHER A TOOL CAN REACH

Tiny supports came out of the coupons with a screwdriver. In the full rear quadrant the walls were in the way.

> *"I can't reach in that far to remove those tiny little supports."*
>
> DEREK · 5 NOVEMBER 2026

HOW IT FEELS AT FULL LENGTH

A screw hole that passed a screw freely in a short sample dragged in a longer one. Drag grows with the length of plastic the screw passes through.

> *"On the 12 mm sample it pushed through fine, on the 40 mm sample I got some resistance in some cases and had to screw to get past it."*
>
> DEREK · 26 OCTOBER 2026

HOW LONG A LAYER TAKES

A small coupon printed alone finishes a layer in a few seconds, and the next goes down on plastic that is still soft. On 26 October 2026 a cable-clip coupon printed that way deformed and was cancelled. The full part gives every layer minutes.

IN THE RECORD

The five coupons · The full quadrant, where the tool did not reach · The screw-hole ladder · The coupon that printed too fast

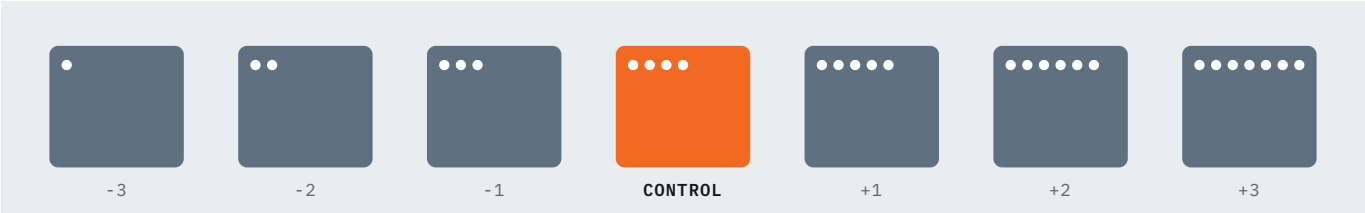
TRIALS

Change one thing.

A print has dozens of settings and every one of them moves the result. A trial that changes two cannot say which one did it.

HOLDS FOR any filament: it is method · the quotes are from PET-GF, PETG and TPU trials

ONE PLATE · ONE SETTING STEPPED · DOTS PRINTED INTO EACH PIECE



> *"...have one control, and have variations in both directions, with whatever you can fit in about a 30 minute to 120 minute print..."*
>
> DEREK · 11 OCTOBER 2026 · PET-GF

> *"Please don't keep doubling down on changes already tried, I would prefer you get back to our normal settings, and pick one specific thing to change..."*
>
> DEREK · 9 OCTOBER 2026 · PET-GF

> *"And I don't want you to keep wasting my time printing the same control over and over again."*
>
> DEREK · 15 OCTOBER 2026 · PET-GF

One control, once

The control is the current recipe, printed on the same plate as its variations. Once it has been printed and judged it is not printed again.

About an hour

A coupon plate that takes an hour gets tried the same evening. A trial that takes a day gets tried once.

> *"This amount of TPU costs less than your tokens, so please just start a print once you are ready."*
>
> DEREK · 13 OCTOBER 2026 · TPU

Both printers

When two printers are loaded and clear, the plate is split across them. One sitting loaded and idle is a wasted evening.

> *"Both printers are available and its my bed time. Print as you see fit."*
>
> DEREK · 15 OCTOBER 2026

Then write down what happened

Every print keeps its photographs, the file that was sent, and what Derek said when he held it, word for word. When he says a part is good enough, that goes on record against the part, with the flaw he named.

Where this book says Still open, it is usually because a trial changed two things at once and worked. The result is real. Which change did it is not known.

IN THE RECORD

What Derek has tried in hand and said · A plate with one control

> *"That worked. There's still a gap, so make a note of it, but I don't want to iterate any further on this right now. It's fine. It's good enough."*
>
> DEREK · 25 OCTOBER 2026

> *"Fixtures we may print again. Trials will never be reprinted, we'll just make new trials instead."*
>
> DEREK · 2 NOVEMBER 2026

BEFORE THE PLASTICS

The same question, a different answer.

Four recipes side by side, each read from the project file that owns it. Hardly a row agrees. Settings that are right for one plastic have failed prints in another, which is why the rest of this book is sorted by plastic.

READ FROM one part's project file in each plastic · filled in each time the book is bound

	PET-GF the enclosure	CLEAR PETG a reservoir	TPU 85A a sleeve	ASA AERO a float
Plate	**Textured PEI Plate**	**Engineering Plate**	**Engineering Plate**	**Engineering Plate**
Bed	**80 °C**	**70 °C**	**35 °C**	**90 °C**
Nozzle width	**0.4 mm**	**0.8 mm**	**0.6 mm**	**0.4 mm**
Nozzle heat	**280 °C**	**260 °C**	**225 °C**	**270 °C**
First layer / the rest	**0.2 / 0.24 mm**	**0.3 / 0.24 mm**	**0.3 / 0.18 mm**	**0.2 / 0.2 mm**
Part fan, least / most	**0 / 0 %**	**10 / 20 %**	**100 / 100 %**	**30 / 50 %**
Fan over an overhang	**70 %**	**90 %**	**100 %**	**50 %**
Flow	**0.9555**	**1.05**	**1**	**0.52**
Most plastic per second	**18 mm³**	**21 mm³**	**2.2 mm³**	**12 mm³**
Walls / fill	**2 / 15%**	**5 / 100%**	**1 / 0%**	**300 / 0%**

FAN
PET-GF prints with its fan off except over air, and went brittle when the fan was left on. Soft TPU prints with the fan full on, and its overhanging lips curled the moment it was turned down.

FLOW
The reservoir's recipe feeds a little more plastic than the lines call for. The float's feeds about half, because that plastic foams as it leaves the nozzle.

PLATE
PET-GF goes on bare textured plate. The other three print on a smooth plate with glue, and clear PETG got there by lifting off the textured one.

PACE
Each plastic has a limit on how much can be pushed through the nozzle each second, and the limit sets the speed. Soft TPU's is a small fraction of PET-GF's.

> *"There's just so many variables, and you really have to look at the slice and how the lines actually print and what they are connected to and how all the lines of filament are actually laid down and connected and it is just a lot."*
>
> DEREK · 20 OCTOBER 2026

IN THE RECORD

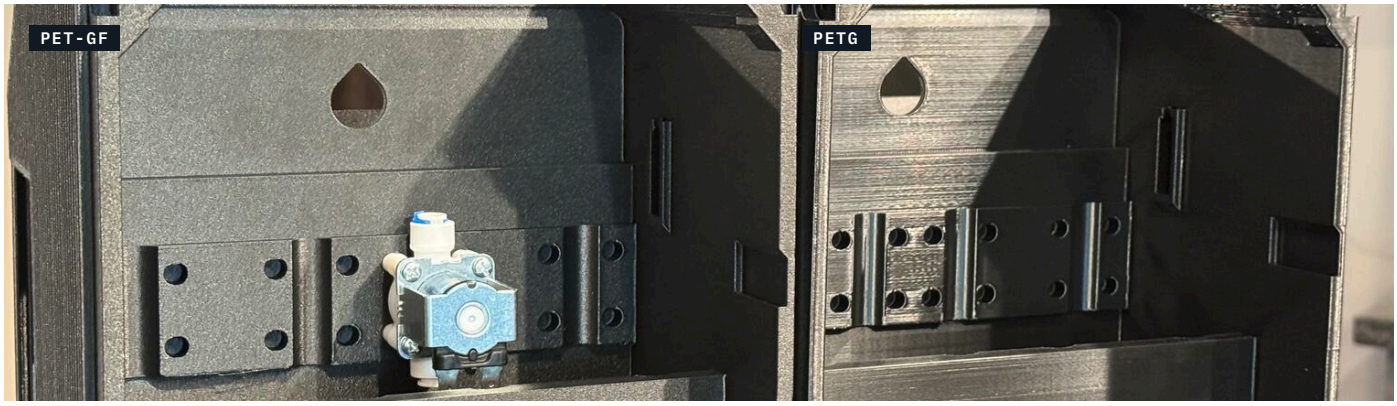
PET-GF project · Reservoir project · TPU sleeve project · Float project

THE PLASTIC

It hides its layers and breaks along them.

PET-GF is PET with short glass fibres mixed in. The glass makes a wall stiff and gives it a flat, grainy face on which the layers do not show. It also takes the stretch away: bent the wrong way, a thin wall splits between two layers.

SEEN IN Polymaker Fiberon PET-GF15 · Mark 1 and Mark2 · textured PEI · hardened 0.4 mm nozzles Not tried in: any other maker's glass-filled PET



26 AUGUST 2026 · THE SAME ENCLOSURE FRONT, PRINTED IN TWO PLASTICS
Low sun across the inside faces. The PET-GF is one even grain. The PETG shows every layer, most of all on the sloped ceiling. Edges and small features look the same in both. It is the large flat faces that give the plastic away.

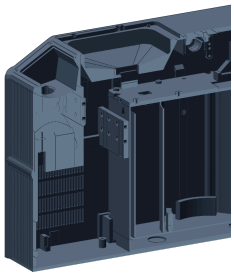
Stiff
A wall of it does not give. A clip has to be a long, broad, thin leaf before it will spring at all.

Matte
A fine grain with no direction in it. On a flat face the layers do not show.

Weak between layers
Bent across its layers, a thin wall splits. On 17 October 2026 two white covers cracked as a display was pressed in.

No stretch
A hard part pushed into a pocket a hair too small does not ease in. On 12 November 2026 the pocket's edge snapped off.

Abrasive
Its profile asks for a nozzle of hardness HRC 40 or more. Ours are hardened steel, tungsten carbide and diamond.



WHERE IT IS
The enclosure and the shell round the cold core, cut open. Everything drawn here is PET-GF.

Most of the machine
By weight it is most of the plastic we print: the four quadrants of the enclosure, the frame under the funnel, the cold core's shell, and out of this picture the faucet, the snap caps and the tube collars.

What it replaced
Until August 2026 the box was printed in PETG and the faucet in carbon-filled PET, whose printed dowels had sheared off along their layers. The first enclosure piece in PET-GF went on the printer on 15 August, and the first faucet on 17 September.

> *"very beautiful wall surface."*
>
> DEREK · 15 AUGUST 2026 · SIX HOURS INTO THAT FIRST PIECE

IN THE RECORD
The two fronts, and what each photograph holds · The first piece · The maker's figures beside carbon-filled PET · Every printed piece and its plastic

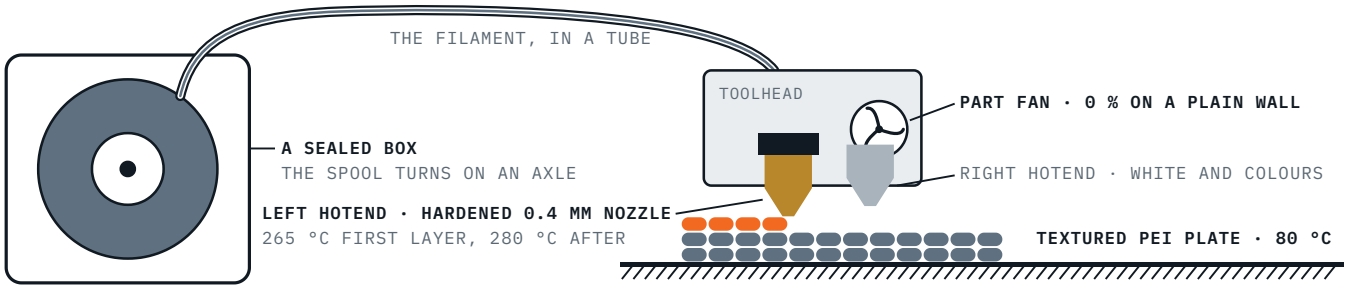
STILL OPEN Nothing about the strength of our printed PET-GF has been measured. What this part says about it is parts that held or broke in the hand, and the maker's own data sheet.

THE RECIPE

One file holds the recipe.

A printer does not know what plastic is in it. How hot, how thick, how fast and when to blow are all saved in one project file, petgf.3mf. Every PET-GF job starts from that file, and what a print teaches goes back into it.

SEEN IN Polymaker Fiberon PET-GF15 · Mark 1 and Mark2 · textured PEI · hardened 0.4 mm nozzles



WHAT PETGF.3MF HOLDS TODAY

Nozzle	**0.4 mm, HRC 40 or harder**	Infill	**15% gyroid**
First layer	**0.2 mm, laid at 265 °C**	Most plastic per second	**18 mm³**
Every layer after	**0.24 mm, laid at 280 °C**	Flow ratio	**0.9555**
Plate	**Textured PEI Plate, 80 °C**	Part fan, plain roads	**0 % to 0 %**
Chamber heater	**0 °C, which is off**	Part fan, over air	**70 % from 50% unsupported**
Walls	**2 loops**	Auxiliary fan	**0 %**

Where it started

On 15 August 2026 the first enclosure piece ran on a 0.8 mm nozzle in 0.4 mm layers: 290 °C, a plate at 100 °C, the chamber held at 50 °C, no fan. Ten days later the maker's sheet was read beside it: nozzle 280 to 310, plate 70 to 80, room air, fan off. The nozzle had been in range. The plate and the chamber had not.

Fed and dried

Black PET-GF, nearly all of it, feeds the left hotend, the only side that takes carbide and diamond. White and coloured PET-GF has fed the right, through hardened steel. Polymaker gives one drying cycle, for a spool that has taken on moisture. It is hotter than either AMS reaches, so it runs in a separate dryer. A 3 kg spool turns too stiffly in that dryer to print from, and feeds from the sealed box.

The printer calls it PET-CF

The slot was copied from Bambu's carbon-fibre preset. Its type still reads PET-CF, so the screen and the send dialog say PET-CF. Its density still reads 1.29 where this plastic is 1.43, so the grams the slicer shows for a plate are not this spool's.

> *"I don't know why bambu defaults 0.24 layer height to a 0.20 first layer, but if I don't have a reason to muck with a bambu default choice, I don't muck with it."*
>
> DEREK · 25 OCTOBER 2026

Why these numbers are read from the file. On 29 August 2026 an ordinary re-save in the slicer moved this file's nozzle from 290 to 265 and changed its first-layer preset. Nobody had asked for it and nothing showed it. So this page reads the file each time the book is bound, and a sliced job is read back before it is sent.

IN THE RECORD

petgf.3mf · The first piece's settings · The maker's sheet read against them · What dries where · The re-save

STILL OPEN The file sets no pressure advance of its own (its switch reads 0). Each job leaves that to the printer's automatic flow calibration, and nobody has compared the result with a value measured for this plastic.

FIRST LAYER · TEXTURED PEI

A first-layer trim does not travel.

The printer finds the plate by touching it with its nozzle. On a textured plate it touches the tops of the bumps, so how hard a first layer is pressed in depends on that one plate. The trim is the small correction, found by printing.

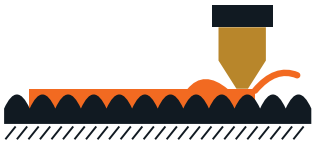
SEEN IN PET-GF · Mark 1 and Mark2 · textured PEI · left 0.4 mm nozzle Not tried in: smooth plates. Carried to PETG on one, this trim failed

HOW IT FINDS THE PLATE



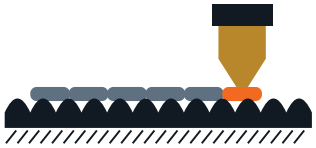
Before each print the nozzle touches the plate. On texture it finds the peaks, so the printer then lowers it a hair. Our trim lifts it again.

TOO CLOSE



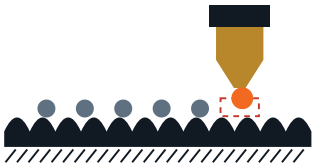
Forced down into the texture. One plate, 30 August 2026: “more melty pressed in material from first layer into plate”.

RIGHT



Flat roads, joined to each other and keyed to the texture, which stays printed in the face of the part.

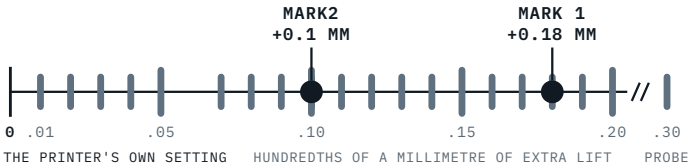
TOO FAR



Round roads resting on the peaks. They lift, and loose plastic balls up on the nozzle (20 September 2026, next page).

Three days, twenty rungs, two answers

Bambu’s start code says why a textured plate gets a correction: “the nozzle was touching topmost of the texture when homing”. Ours is found by printing. Between 28 and 30 August 2026 a ladder of empty project files was built, z-trim-0.01.3mf to z-trim-0.20.3mf, a hundredth of a millimetre of lift apart, and one at 0.30 to prove the setting did anything. On the first night Derek thought the answer was 0.02, then 0.01. Two days later the ladder had been climbed to its top.



Mark2 prints PET-GF at +0.1 mm and Mark 1 at +0.18 mm: the same model of printer, the same plastic, the same size of nozzle and the same kind of plate.



20 SEPTEMBER 2026 · MARK2 · BLACK AND WHITE PET-GF
A first layer that went down right. This plate was printed face down, so the letters and the code are its first layer. “The QR code scans clean from 2 feet away”. (The nameplate has since moved to PETG.)

> *“it seems one of the confounding variables is which textured PEI plate is used.”*
>
> DEREK · 30 AUGUST 2026

Not to another printer. On 8 October 2026 a lower trim, tried once on Mark 1, had been written down as Mark 1’s new rule. “The +0.04 trim was a *special* case for a problem with a *specific* print - not a change to the rule for Mark1.”

Not to another plastic or plate. On 6 November 2026 Mark 1’s PET-GF trim rode along onto a PETG job on a glued smooth plate. “lol, okay, print stopped, it lifted right off.” Then: “No. That should be documented as for pet-gf on textured plate.”

IN THE RECORD

How the trim works, and the ladder · Each printer’s trim · The nameplate · The PETG job that lifted

What we do now. Each printer’s trim is written in one place. A job’s file name carries it, and the trim line in the sliced file is read back before the job is sent.

STILL OPEN Nobody wrote down what each rung looked like: the record holds the two answers and not the prints that chose them. Mark2 also printed two whole faucets at the higher trim on 18 and 19 September 2026, and the first “Came out great”, so how wide the good band is on either printer is not known. The plates are not marked, so no trim can be tied to the plate it was found on.

FIRST LAYER · THE PLATE'S EDGE

Season the edge, or stay off it.

Small parts are printed in the middle of the plate, so the middle gets used and the last few centimetres at the edge do not. PET-GF held on the used part of a textured plate and let go where nothing had been printed. A part that fills the bed finds that out in its first layer.

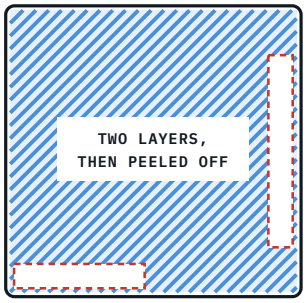
SEEN IN black PET-GF · Mark2's textured PEI · left 0.4 mm nozzle · 20 September, 4 October and 10 October 2026 Not tried in: Mark 1. On 6 October 2026 the rule also stopped clear PETG on Mark2's textured PEI (left 0.8 mm), and no failure was named

SENDS 1 TO 4
OUR TRIM



20 September 2026: a bench stand as big as the nozzle can reach, 5 mm from the edge. Two clumps on the nozzle, then a peel on a fresh hotend. Pulled in to 15 mm: "It was lifting already."

SEND 5
THE PRINTER'S OWN OFFSET



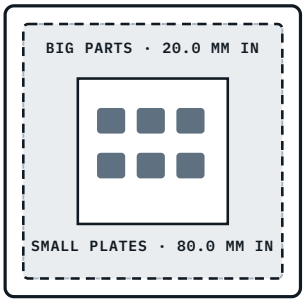
Back at the edge without our trim, so the nozzle ran lower. It pressed flat and held, rough at the right edge and the front left. Derek stopped it and peeled it off.

SEND 6
OUR TRIM AGAIN



The file of the third send, on the same hotend. The first layer went down "clean and uniform out to the plate's edge", and the print ran all 917 layers.

NOW
THE MARGINS WE KEEP



The cold-core shell keeps every road 20.0 mm in from the edge. Small test plates are centred 80.0 mm in. Both are measured on the sliced paths, brim and supports included.



14 OCTOBER 2026 · MARK2 · BLACK PET-GF
What staying off it looks like: twenty-five test pieces grouped in the middle, the nearest road 82 mm from an edge.

> *"The seasoning worked. I mean even if it fails after this there is still a very obvious effect so far!"*
>
> DEREK · 21 SEPTEMBER 2026 · THE SIXTH SEND

What seasoned means. A place on the plate where PET-GF has been printed before. The record says nothing more exact.

Two weeks later. "On mark2 you had pieces close to the edge that did not get a good first layer, so that plate is not well seasoned there yet. We could keep printing until it works, or you could avoid going that far to the edge here." (4 October 2026, black PET-GF, nearest road 15.5 mm in.)

The one time it was not PET-GF. On 6 October 2026 the rule stopped a plate of clear PETG on Mark2's textured plate, its nearest piece 17.5 mm in: "You can't print that close to the edge of the bed on a plate that hasn't been seasoned properly first." The margins we keep come from that message. The record names no piece that failed that day, and PETG has since moved to a glued smooth plate.

IN THE RECORD

The six sends · The stand, and its seasoning layer · 4 October · 6 October · The margins

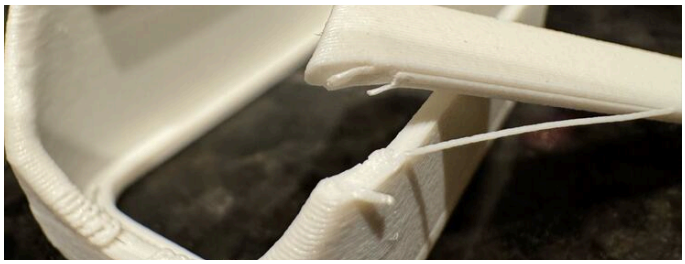
STILL OPEN Nobody has looked at what a first layer leaves on the plate, so why this works is not known. The plates are not marked and there is no map of where each has been printed on. The stand reached to 5 mm from Mark2's edge on 21 September, and on 4 October an edge on Mark2 would not hold: the record cannot say whether that was another plate or whether seasoning wears off.

COOLING

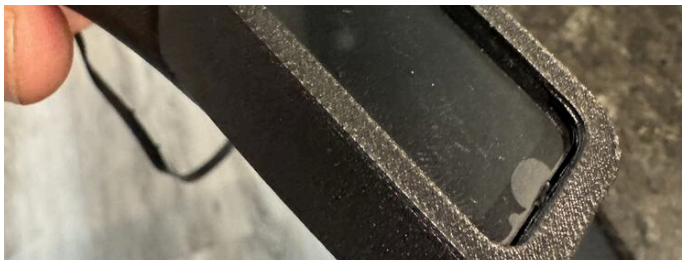
Fan only where the plastic is over air.

The part fan blows room air at the plastic as it leaves the nozzle. A road laid over air needs that, to set before it sags. A plain wall of PET-GF does not. Printed with the fan on everywhere, thin covers cracked along their layers in the hand.

SEEN IN white and black PET-GF · Mark2 · textured PEI · right and left 0.4 mm nozzles Not tried in: PETG. Soft TPU wants the opposite: its sleeve prints with this fan at 100 % (the TPU part)



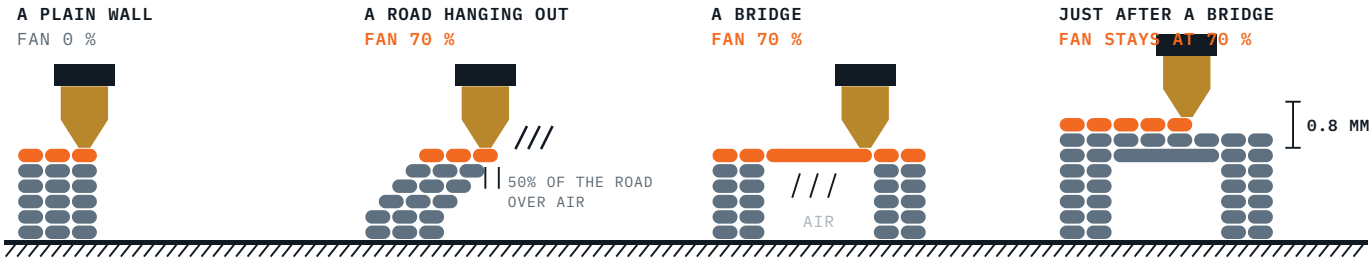
17 OCTOBER 2026 · MARK2 · WHITE PET-GF · RIGHT NOZZLE
Printed with the fan at 70 percent on every road above the third layer. It split along its layers as the display was pressed in. The cover before it had done the same.



25 OCTOBER 2026 · MARK2 · BLACK PET-GF · LEFT NOZZLE
A later cover, reshaped at its front corners, printed with the fan on its overhangs only. It took its display and went on. “That worked.”

> *“That new printed cover works great. In fact, it worked so well that the other parts of the faucet broke while I was trying to put that cover on, which I think really indicates that we have a brittleness issue with those cooling settings that we need to be aware of.”*
>
> DEREK · 17 OCTOBER 2026 · THE SAME COVER, FAN OFF

One change. That afternoon the cracked cover’s file was printed again with every fan setting at zero. Shape, temperatures, speeds and supports stayed as they were. It took the display and went on. By evening the fan was back, on bridges and overhangs only.



Walls, infill and top faces are laid with the fan at rest. On the first faucet printed this way, 17 October 2026, over 99 percent of the cover’s outer wall was laid with it off.

The slicer asks how much of a road must hang over air before the fan comes on. Ours is 50%. At zero, Bambu Studio cools every outer wall, overhang or not. The cracked covers’ file carried a zero.

A road pulled across a gap is laid slowly, with the fan on. With no fan at all, on 24 September 2026, a rounded underside in fine layers “turned out worse. Exploded basically.”

The slicer cools a bridge and then stops. On 5 November 2026 the course laid on a fresh bridge came off “stringy and loose”: fan off, at up to 200 mm/s. Now fan and bridge speed are kept for 0.8 mm above.

> *“the cooling, high cooling, is totally fine. We can use 100% cooling if we want, but only on the places that actually need it, and everywhere else we shouldn’t be using that cooling.”*
>
> DEREK · 17 OCTOBER 2026

IN THE RECORD

The cracked covers · The fan-off cover · The cooling we keep · The rail, read course by course · The round with no fan

STILL OPEN Why the covers cracked is not established: the fan was the one thing changed between the cover that broke and the one that held, and no strength was measured. The cooling kept above a bridge has been read in each file’s commands since 6 November 2026. No print has yet been judged for it.

COOLING · SMALL PARTS

A small part needs time between layers.

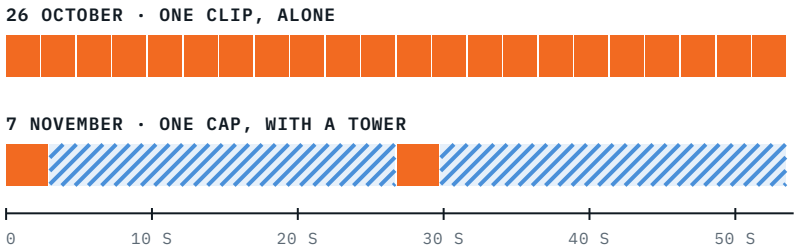
Each layer is laid on the one before while that one is still giving up its heat. On a big part the nozzle is away for a minute or more. On a part the size of a thumb it is back in a second or two, laying hot plastic on plastic that has not set.

SEEN IN black and white PET-GF · Mark2 and Mark 1 · textured PEI · left 0.4 mm nozzle Not tried in: any other plastic. PET-GF has no fan to fall back on, so time is all it has



26 OCTOBER 2026 · MARK2 · BLACK PET-GF
A clip for a ribbon cable, printed upright and alone: a wall 24 mm wide and 3 mm thick, ten minutes of printing. Derek stopped it part way.

> *"So that last one came out pretty melty there. I stopped it partway, as you'll see. So we definitely can't do anything quite like that again without some major part fan cooling and time between layers."*
>
> DEREK · 26 OCTOBER 2026



ONE SCALE OF SECONDS
Orange is the nozzle printing the part. Blue is the nozzle away, printing a tower that is thrown out.

1 to 3.5 s
a whole layer, high on the clip that melted. Read from its file.

24 s away
after each layer of the lone cap, whose shortest layer was 3 s of printing.

The file does not wait by itself. Many profiles slow down when a layer is short. This one will not slow a layer that takes more than 1 s.

THREE WAYS IT GETS THE TIME

1 A batch

Other parts on the plate keep the nozzle busy somewhere else. On 7 November two regulator caps shared a plate with their test receivers, and the file gave each cap 19 seconds or more between layers with no tower. "Like that gives us the layer time already."

2 A tower

Alone on the plate, the nozzle is sent away after every layer to print a tower, 50 mm wide at 10 mm/s. The printer's smooth timelapse move does the sending. 7 November 2026, one cap, Mark2: "The print turned out fine".

3 A wait

11 November 2026, one white display cover, Mark 1: a tower on all 86 layers, and the nozzle parked above it for ten seconds before each. "Just the cover. With a prime tower and wait between layers so its not a melty blob."

> *"But, sometimes I want to print just one."*
>
> DEREK · 7 NOVEMBER 2026

IN THE RECORD

The melted clip and its photographs · The retry that night · The lone cap, timed · The batch, timed · The white cover · Small parts in the cooling rule

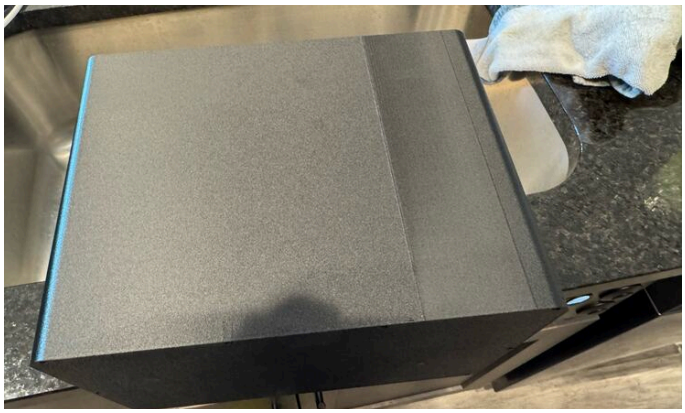
STILL OPEN The retry on the night of 26 October changed two things at once: three parts shared the plate with a 20-second floor on each layer, and the fan ran at 100 percent. The lone cap of 7 November printed fine on time alone, with the fan off except over air. No small part has been printed alone and unhelped since, so there is no control, and nobody has measured a temperature. Twenty seconds was a target borrowed from other people's practice.

SURFACE

A matte wall hides the layers and nothing else.

On a flat PET-GF wall the layers vanish into the grain. That leaves a face so even that anything which breaks it stands out: the seam where every layer starts and stops, a patch of ripples, a flat in the model. Some of it shows only when the light comes from the side.

SEEN IN black PET-GF · Mark2 · textured PEI · left 0.4 mm nozzle Not tried in: white or coloured PET-GF, which nobody has set beside the black



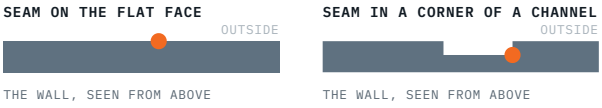
PRINTED 26 OCTOBER 2026 · MARK2 · BLACK PET-GF
The shell that wraps the cold core. The line down its face is the seam, and the band beside it is where the seam was feathered. “It’s really quite small, just very noticeable.”



PRINTED 29 OCTOBER 2026 · MARK2 · BLACK PET-GF
The next shell, with the seam moved into a channel on another flank. The face is clean. What is left is a thumbnail of ripples at the top edge. “In some lights it’s not visible at all, but in others, as you can see, it is quite visible there.”

Give the seam a corner

Each layer of a wall starts and stops somewhere, and on PET-GF that spot shows. Blending it along the face made a band instead of a line. The shell of 29 October has a shallow channel down one flank, and every layer but the first starts in the channel’s inside corner.



On the outside of the machine the standing walls are fluted. The grooves lay their own light and shadow, finer and far stronger than a layer line, and the seam between front and back sits in one.

Three things that showed

Flats. 18 September 2026, the first PET-GF faucet: “Some of those defects (low resolution polygons on what should be smooth surfaces) are showing through in the final print.”

The seam. 28 October 2026, the first shell: “if this were an exterior surface, we’d have to keep iterating for a while.”

Ripples. 1 November 2026, the second shell. Derek’s guess is the tall layers near the top and the hole cut through the wall beside them: “you can see it lines up pretty precisely with where we cut our hole there.”

> *“But this is a defect that we are okay shipping with.”*
> DEREK · 1 NOVEMBER 2026

That patch is now the first entry in a register of accepted defects. Its photographs define what ships.

IN THE RECORD

The first shell and its seam · The second shell and its ripples · Where its seam starts, layer by layer · Accepted defects · Fluted and flat, in two plastics · The faucet’s flats

STILL OPEN The cause of the ripples is Derek’s guess and has not been tested. On two failed prints the bad courses came out glossy where sound PET-GF is matte: the melted clip of 26 October and the courses above a bridge on 5 November. Nobody has tested whether gloss can be read as a sign of heat.

LANDING 1 · STRAIGHT UP

Layer height is not a knob.

A print is a stack of courses, and the layer height is how thick each one is. It sets the step on every lean, the gap over every support, and how often the nozzle passes the same edge. Change it and all three change.

SEEN IN PET-GF · Mark 1 and Mark2 · textured PEI · left 0.4 mm Not tried in: any other plastic

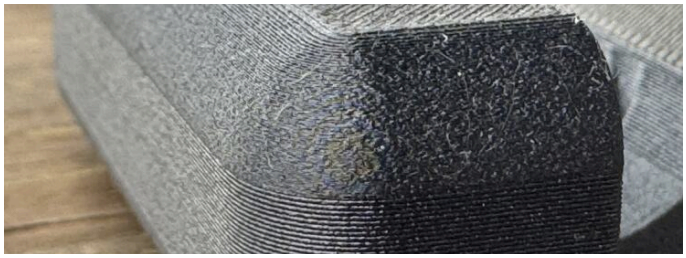
THE LANDINGS, IN PET-GF · THE FIFTH, SLICER SUPPORT, HAS THE NEXT CHAPTER

1 Straight up
One first layer and one layer height, on every part. This page.

2 Lean
Drawn at 45°. A round that starts on the bed starts as a chamfer.

3 Bridge
Slow, with extra plastic and the fan on. Out of sight.

4 Fin
A thin wall under a visible edge, one layer short of touching.



24 SEPTEMBER 2026 · MARK2 · PET-GF
The round at the top of the print, in fine layers. It turns inward as it rises. The band shows: the texture changes where the fine layers begin.



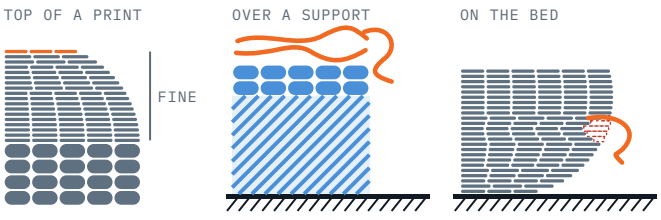
THE SAME PRINT, THE BED SIDE
The same radius in the same fine layers, growing outward off the bed. About twenty courses went down clean. Then the edge lifted and the nozzle dragged through it.

0.2 mm
the first layer, on the bed

0.24 mm
every layer after it

Both live in the shared PET-GF project and every part inherits them. The first is the slicer's default. Nobody has had a reason to move it.

WHERE A FINER LAYER WAS EVER ALLOWED



ALLOWED A show round that turns inward as it rises.

NEVER Over a support.

NEVER On an edge that grows outward.

Turning inward, every thin course lands on the one below. Over its supports, the fine courses of the front-top printed on Mark 1 on 3 October 2026 came down as spaghetti. The edge that grows outward is the bed-side round above, and the next page.

STILL OPEN One recipe still carries fine layers: front-top, on two show rounds at the top of its print. Whether the instruction of 12 November, given about a fit coupon, retires them has not been asked.

HOW THE FINE LAYERS WENT AWAY

> *"...thin layers create a visible artifact that should only be used when absolutely necessary..."*
>
> DEREK · 4 OCTOBER 2026

4 OCT 2026 The pump cartridge and both bottom pieces of the enclosure go back to ordinary layers.

11 OCT Faucet lip coupons, Mark2. The same support gap came out 0.60 mm under fine layers and 0.48 mm under ordinary ones, because a gap is a whole number of layers. On the first layer over a support: "0.24 separates every time I think, it just recovers better as more heavy layers are added."

25 OCT Both faucets go to one layer height: "...that didn't turn out so good, so I'd like to go back to the same layer height everywhere..."

9 NOV The handholds are reshaped so that nothing needs them: "We don't need any fine layers."

12 NOV A fit coupon is sliced in finer layers to make a tenth of a millimetre show, and stopped: "Do not use finer layers anywhere - that is not an option - that is a completely different variable". That print has its own page, *Height moves one layer at a time*.

IN THE RECORD

The shared PET-GF project · The tee carrier's two rounds · Front-top, 3 October · The gap under two layer heights · The stopped coupon · Front-top's fine bands

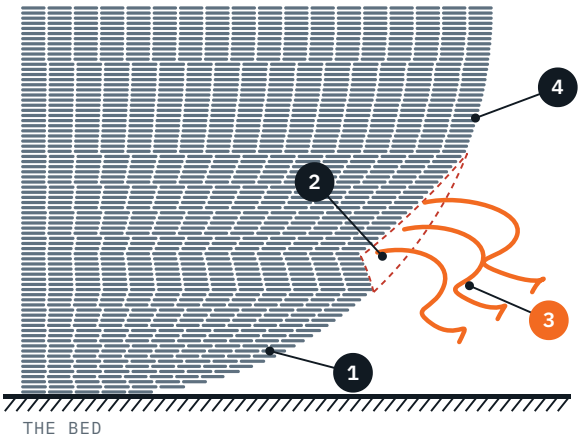
LANDING 2 · LEAN · FROM THE BED

Start a round with a chamfer.

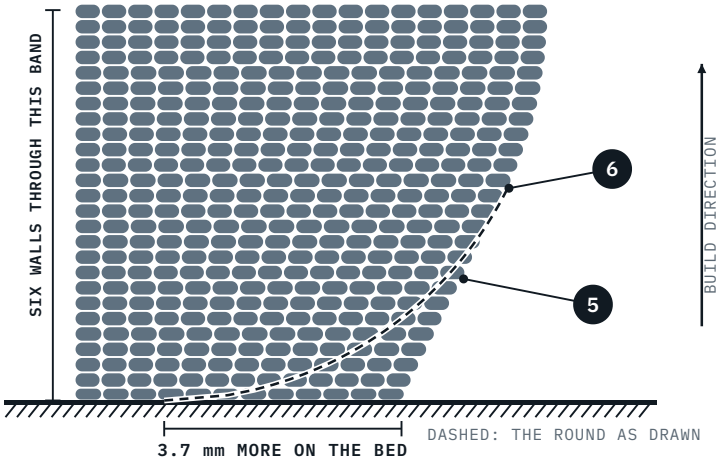
A round edge that begins on the bed grows outward as it rises: each course is wider than the last. Printed in fine layers, its edges drew back and the next course came down on air. The fan did not fix it. The shape did.

SEEN IN PET-GF · Mark2 and Mark 1, September 2026 · textured PEI · left 0.4 mm Not tried in: any other plastic

24 SEPTEMBER 2026 · MARK2 · FINE LAYERS, 0.08 mm



29 SEPTEMBER 2026 · MARK 1 · ORDINARY LAYERS, 0.24 mm



1. About twenty courses go down clean.
2. Edges already laid draw back.
3. The next road out has nothing under it.
4. Higher, where the wall is steeper, the print recovers from the inside.
5. The chamfer starts steep. Each course steps out half its own height.
6. It meets the round here, tangent. Above this the curve is as drawn.



24 SEPTEMBER 2026 · MARK2 · PET-GF · FAN OFF

The second try. The edge that grew off the bed is the ragged one.

> *"The layers already laid down end up shrinking, pulled back, pulled back just at the edges, and the next layer that goes down ends up printing in air..."*
>
> DEREK · 24 SEPTEMBER 2026

WHAT WAS TRIED, IN ORDER

FINE LAYERS	Mark2, 24 September 2026. No supports. The top round clean, the bed round torn part way up.
FAN OFF	Same file, same day. "That turned out worse. Exploded basically."
FAN STEADY	Stopped when the edge failed again. Derek saw where: between the places the infill reached the wall. Where it reached, the edge held.
SIX WALLS	Through the fine band, infill laid before walls. "We have no spaghetti explosion. It's not a great curve, it's not even a chamfer, it actually clearly angles the wrong direction... Still, probably good enough for now."
A CHAMFER	Mark 1, 29 September. Ordinary layers from the bed, six walls through the band, walls before infill, no supports. The record: the bottom curve turned out great.

What we do now

An edge that grows outward from the bed is drawn with a chamfer under it, tangent to the round and stepping out 0.5 mm for each millimetre of rise. It prints in ordinary layers with a thicker band of walls and no support. On the enclosure that is back-top, whose roof edges start on the bed.

Our reading

Nobody measured why the edge drew back. A thin course is a thin ribbon with little stiffness of its own, and as it shrinks there is not much to hold its free edge out. Infill tying into the wall did. So did six walls. That fits what Derek saw. It is not proof.

STILL OPEN The accepted print changed the shape, the layer height, the order of walls and infill, and the printer, all at once. No print separates them. The record has no photograph of it, and no result yet for the same chamfer on another part.

IN THE RECORD

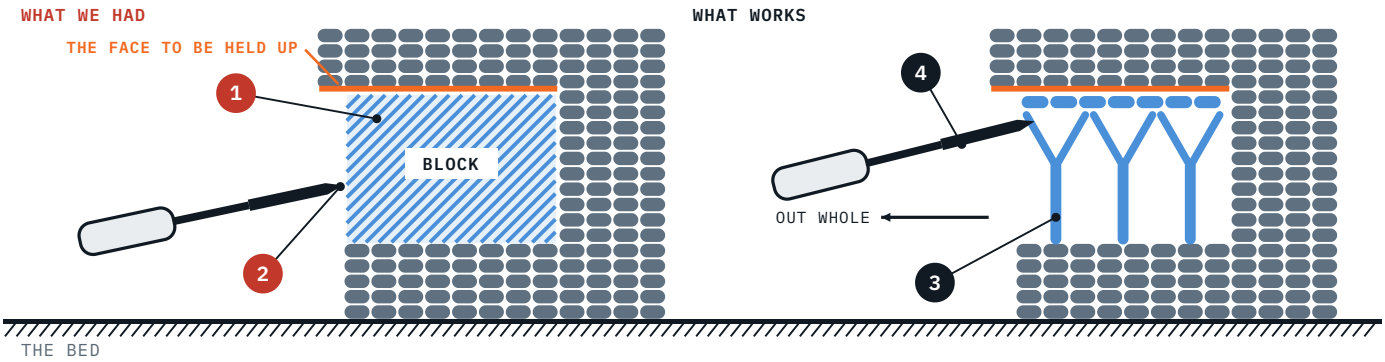
The print log, 24 September · Fan off: the result and Derek's account · Six walls · The chamfer, sliced and printed · Its acceptance · The slope today

SUPPORTS · INSIDE A CAVITY

Support blocks do not come out of a cavity.

Where a part hangs over air, the slicer prints scaffold under it. In PET-GF that scaffold is the part's own plastic, from the same nozzle, and nothing dissolves it. It must be levered out, and a block that fills a cavity leaves a tool nowhere to go.

SEEN IN PET-GF, support included · Mark 1 · textured PEI · left 0.4 mm Not tried in: any other plastic



1. It fills the cavity and touches the floor, the back wall and the roof at once.
2. A blade cannot get behind it, and it is too big to turn.
3. It stands on the floor of the same cavity, so it is short, and it meets the part only at a thin roof.
4. A small flat screwdriver goes in at the open side and levers the whole tree out.

Two full-size prints lost in a week

3 October 2026, Mark 1. Front-top, nearly 26 hours. Tall blocks, which the slicer calls normal supports, had fused with the trees beside them into masses bigger than the cavities they had to leave through. A cover holder buried in one snapped. Rejected the next day.

8 October 2026, Mark 1. Back-top, 27 hours. Every support had been switched to blocks, to keep the print inside the margin we hold at the plate's edge. Thrown away the next day.

10 October 2026, Mark 1. Back-top again, on trees. For one groove they came up from far below, long and thin, and stuck in the walls on the way.

> *"lol, lmfao, it is *every* support here … yeah, there's no way I'm going to be able to do anything with this except throw it away"*
>
> DEREK · 9 OCTOBER 2026, ON THE SECOND PRINT

> *""Auto" is fine … I've yet to see bambu's auto put a normal support in an impossible to remove place … it's when you choose to put normal somewhere that it introduces problems."*
>
> DEREK · THE SAME DAY

What we do now

The shared recipe asks for **tree(auto)**. No cavity gets a rectangular support, whether a person asked for it or the slicer chose it. On 6 November 2026 Derek made it flat: "So we can't have normal supports and cavities."

A tree has to start in the cavity it serves, and allowing that does not make the slicer do it, so we read each start in the sliced file. In the front-bottom sent to print on 11 November 2026, 2 of 2 supports start inside their cavity, 7.92 mm under the face they carry.

IN THE RECORD

Front-top, rejected 4 October · Back-top, thrown away 9 October · Back-top, the long trees · Where the rail trees start · The rule as agents read it

STILL OPEN Nobody has measured how open a cavity must be before a block will come out of it. In the back-top printed on 5 November 2026, blocks in an open rail groove did come out, once the screwdriver found its angle. The rule stayed.

SUPPORTS · THE GAP

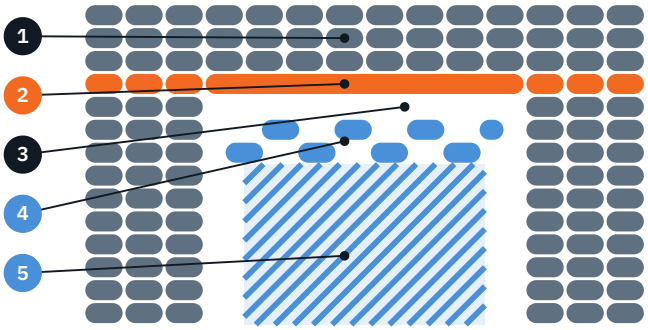
One layer of air held on. Two let go.

A support does not touch the part. The slicer leaves a little air between the support's roof and the first course of the part, so that the two can be pulled apart, and that first course is laid across the air like a bridge. How much air is the whole question.

SEEN IN PET-GF, support included · Mark2 · textured PEI · left 0.4 mm Not tried in: any other plastic

ONE LAYER OF AIR

ASKED FOR 0.30 MM. THE FILE HAD 0.24.

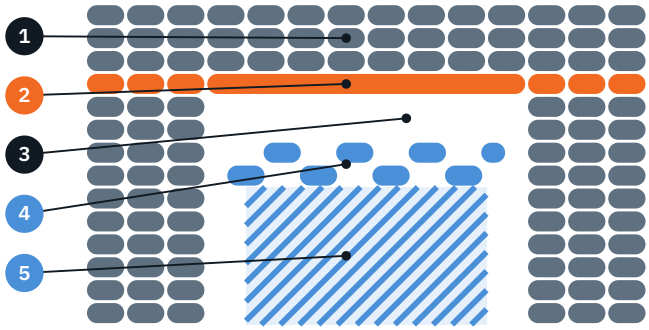


0 of 8

coupons with one layer of air made Derek's group of cleanest releases.

TWO LAYERS OF AIR

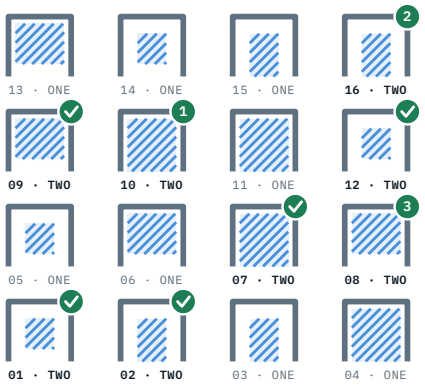
ASKED FOR 0.45 MM. THE FILE HAD 0.48.



8 of 8

with two layers did. The support came away and the roof's own strands stayed on the roof.

1 The part. 2 Its first course, pulled from wall to wall like a bridge. 3 Air. 4 The support's roof: two courses of spaced lines. 5 A tree.



FRONT OF THE BED · EACH COUPON'S OPEN SIDE FACES IT

15 SEPTEMBER 2026 · MARK2 · PET-GF
Sixteen small ceilings on one plate, seen from above. Eight asked the slicer for 0.30 mm of air and eight for 0.45 mm. The trees we print with count the gap in whole layers. Hatching is the support's roof. Green marks Derek's cleanest eight, with 1, 2 and 3 for his best.

The edges changed nothing

The first reports had the support sticking at its edges. So twelve of the sixteen had the support's roof pulled back 0.6 mm from the walls, from the open edge, or from both. The clean ones are the two-layer ones, whatever was done at their edges. Derek's best three used three different edge treatments. The best of all had none.

What stayed stuck was the part

A second plate that afternoon had four coupons: a crossed roof, straight lines, straight lines cut apart at their ends, and no roof at all, only the tips of the trees. The same strands stayed under all four. Under the one with no roof there was nothing else they could be. In the sliced file they are the part's own first course.

What the gap costs

Even where the support let go, that first course hung loose. On all eight clean coupons it looked and felt loose beside the course above it. A face printed over support is rougher and lower than it was drawn, so every fit beside one is given 0.25 mm more room. The pages on fits show where that goes.

IN THE RECORD

The sixteen coupons and their ranking · How they were made · The four coupons and the strands · The allowance

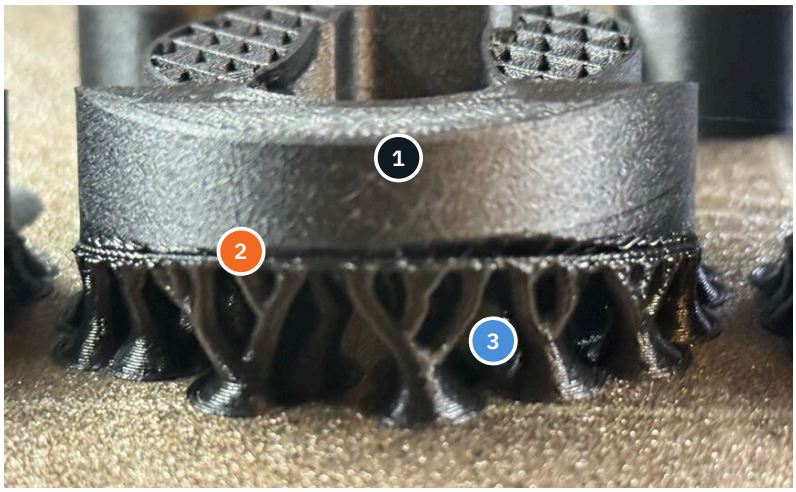
STILL OPEN Nobody measured how far the loose strands hang. The allowance is Derek's judgement from coupons in his hand, set before Support D closed the gap again. On 10 November 2026 a rail printed with D fitted "a little bit loose".

SUPPORTS · JUDGING A RESULT

The support that fell off was the failure.

A support has two jobs that pull against each other. It has to hold up the first course of the part while that course is soft, and afterwards it has to let go. One that lets go with no effort at all has usually skipped the first job.

SEEN IN PET-GF, support included · Mark 1 and Mark2 · textured PEI · left 0.4 mm Not tried in: any other plastic



10 OCTOBER 2026 · MARK2 · PET-GF
A section of the faucet's base, still on the plate. **1** The part. **2** A slit where the part should be joined to itself. **3** The trees and the roof they carry. All three copies on the plate had the slit.

These supports had the gap that won in September. Derek had been calling the slit lifting. Then he saw what it was.

> *"In these faucet base prints, at that "lifting" area, the first layer just falls off, it is not connected to the layers above it all. So this lifting is actually "perfect removal" for the removal I was initially seeking, before I realized that was removing the first layer."*
>
> DEREK · 11 OCTOBER 2026

The layers in that band were fine ones, which is trouble of its own and has its own page. The lesson here is about the judging. For a month, easy removal had been the thing we were looking for.



9 NOVEMBER 2026 · MARK 1 · PET-GF
The undersides of A, B and C.

Three ceilings that let go by themselves

Three small ceilings, A, B and C, each on a denser support roof than the last, all with two layers of air. Nobody removed their supports.

> *"Like as soon as the plate cooled a little bit, the prints just detached. They just slid off. There was no removal of the supports."*
>
> DEREK · 9 NOVEMBER 2026

Removal cannot get easier than that. The roofs they left are in the photograph: separate strands, some standing proud. None was acceptable. What he asked for next was the opposite of easy.

> *"Give me three samples of some things that will perhaps be more difficult to remove, but also potentially support our first layer a lot better."*
>
> DEREK · THE SAME EVENING

What we judge now

The removed support is kept beside its part, and the underside it left is looked at first. Are the first roads joined to each other and to the course above, or do they lie there as separate strands? Only then does it matter how the support came away. One that drops off by itself is marked down.

IN THE RECORD

The faucet coupon · The three ceilings · What to look at, as written for the bench

STILL OPEN All of this is judged by eye and by hand. No coupon in this chapter has a measured sag, roof thickness or pulling force.

SUPPORTS · SUPPORT D

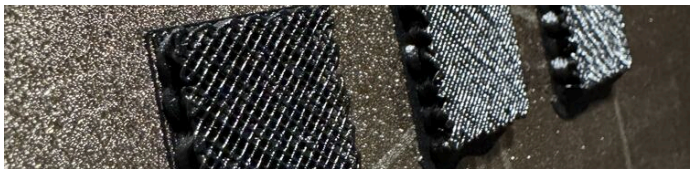
Solid roof, small gap, out in one piece.

On the evening of 9 November 2026 Mark 1 printed six small ceilings, lettered A to F. Three kept the wide gap and made the support's roof denser. Three made the roof solid and closed the gap up. Every PET-GF part now prints with the fourth.

SEEN IN PET-GF, support included · Mark 1 · textured PEI · left 0.4 mm · flat ceilings a tool can reach Not tried in: any other plastic

A

roof lines **0.5 mm apart**
2 roof layers
0.45 mm of air asked



FELL OFF All three slid off their supports as the plate cooled and left roofs of separate strands. These are the supports, still on the bed. Which is which was not recorded.

B

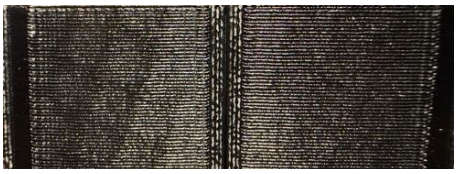
roof lines **0.15 mm apart**
2 roof layers
0.45 mm of air asked

C

roof lines **touching**
2 roof layers
0.45 mm of air asked

D

roof lines **touching**
3 roof layers
0.24 mm of air asked



LET GO BY HAND "D and E both removed without pulling the first layer with them." Which is which was not recorded.

E

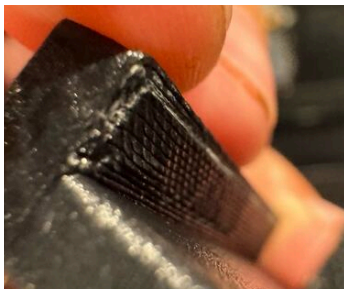
roof lines **touching**
3 roof layers
0.12 mm of air asked

F

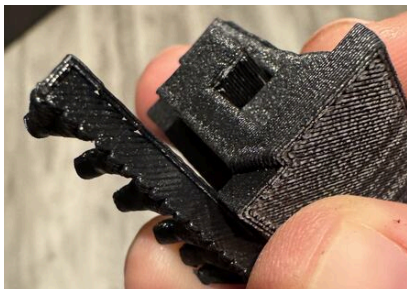
roof lines **touching**
3 roof layers
0.06 mm of air asked

NO PHOTOGRAPH

NO RESULT Printed. Never reported.



E · 9 NOVEMBER 2026 · MARK 1
Harder to get out than rail supports usually are. It broke up, a tree or two at a time, and one corner had fused pale.



D · 10 NOVEMBER 2026 · MARK 1
The roof and its trees, out as one comb. The mating rail then slid in with room to spare.

Then the real cavity

A ceiling between three walls is kind to a support. D came clean off four less sheltered shapes the same evening. Then E and D were each printed as the front-bottom's own rail cavity, on its short trees.

> *"This support, this D support on this rail cavity removed perfectly. It was great. It came off as a single piece and it pried out just as easily as the others."*
>
> DEREK · 10 NOVEMBER 2026

He had put E's breaking up down to its denser roof. D has the same roof.

Support D today

Support	**tree(auto)**
Space between roof lines	**0 mm**
Roof layers	**3**
Air asked for	**0.24 mm**
Bridges	**1.25 of normal flow, 15 mm/s**

Asked for is not printed



A bridge at D's flow is a fatter bead than its layer, and it hangs lower. In D's file the 0.24 mm asked for left 0.18. E's 0.12 left 0.06 on the ceiling. On the rail the trees rounded E a whole layer closer, and its bead sat inside the roof. Read the gap in the sliced file.

IN THE RECORD

A, B and C · D, E and F: the file read back · D on the rail · E on the rail · Where D is applied

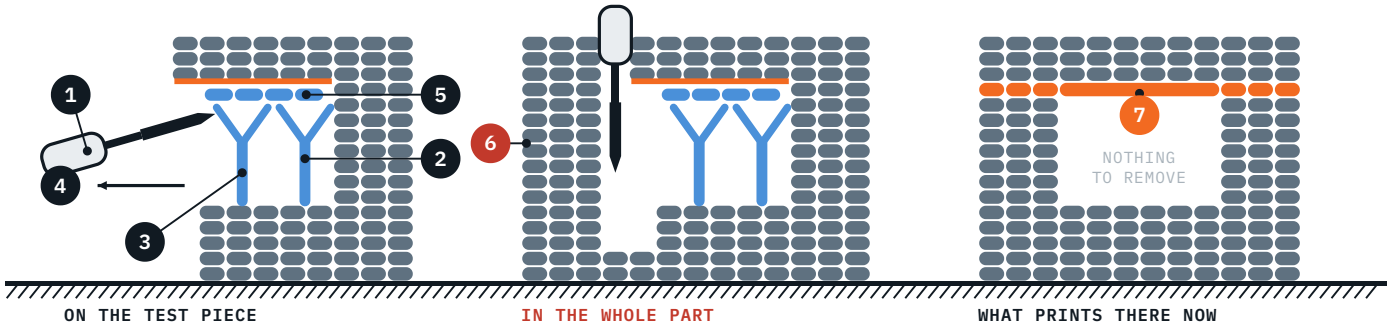
STILL OPEN D is not one change. From C to D the gap, the roof's layers, the bridge flow, the bridge speed, the tree style and the coupon itself all changed, and nothing has separated them. No full-size enclosure piece printed with D has a recorded result yet.

SUPPORTS · THE WAY OUT

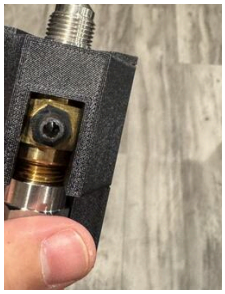
Plan a support from its way out.

Whether a support will come out is settled before the print starts, by the shape around it. A test piece is small and open on every side. The whole part puts the same pocket deep inside, behind its own walls.

SEEN IN PET-GF, support included · Mark 1 and Mark2 · textured PEI · left 0.4 mm Not tried in: any other plastic



1. **A way in for a tool.** A small flat screwdriver, and room to swing it.
2. **A tree, not a block.** Thin where it meets the part.
3. **A root inside the cavity.** A tree from far below arrives long, thin and stuck to the wall.
4. **An open lane.** One side of the cavity stays open until the support is out.
5. **One piece.** A roof that ties the trees together, so the first pry brings all of them.
6. **The whole part's own walls.** The pocket is the same. The way to it is gone.
7. **A bridge instead.** Two full shoulders, a bar pulled across, nothing underneath.



1 NOVEMBER 2026 · MARK 1
One of the ten. Open on every side.

Easy on the test piece

1 November 2026, both printers. Ten test pieces for the rear fittings, each small enough to hold in one hand. Their tiny cavity supports came out with a small flat screwdriver: "just an extra step".

5 November 2026, Mark2. The whole back-top, 27 hours, with the same receivers inside it and the same supports inside them. Its own walls stood in the way.

> *"I can't reach in that far to remove those tiny little supports."*
> DEREK · 6 NOVEMBER 2026

A test piece answers for a support only where the whole part leaves the same way in.

Where no tool reaches

> *"But when there's not a clear path for the screwdriver, like there's not on the Pisco there and on the ASSE there, then I need those to be printed without supports entirely."*
> DEREK · 6 NOVEMBER 2026

Take the overhang away. The receivers' catches became bars bridged between two full shoulders, and bores across the rear wall grew pointed roofs. The back-top part study shows both.

Fill it. Derek, 25 October 2026, of pockets beside the display: "inner supported areas that could be filled entirely solid instead of being impossible to remove supports". Saving plastic is not a goal.

Change the joint. 6 November 2026: "So I'd like four screws for this display cover instead of these impossible-to-remove supports." The snap-in cover became a screwed one.

What stays supported. A face that has to be flat to do its job keeps its support, where all five needs are met. The front-bottom part study has exactly two such faces, its rail catches. A support that comes out cleanly is not a debt, and a working face is never sloped to be rid of one.

STILL OPEN No full-size enclosure piece printed since these changes has a recorded cleanup. One was stopped, one result is pending and one has not been written down.

IN THE RECORD

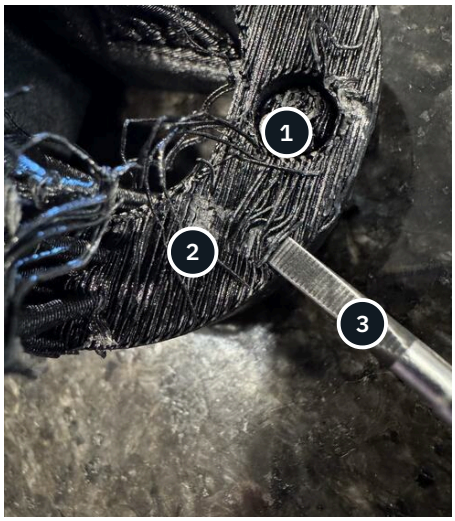
The ten test pieces · The whole back-top · The display cover · The enclosure's rule

SUPPORTS · TILTED UNDERSIDES

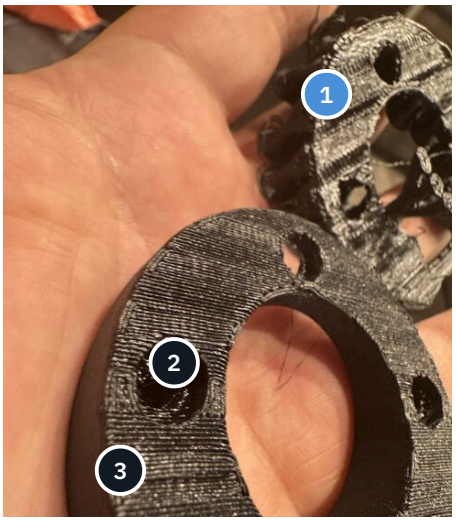
Before blaming the gap, check the roof is there.

The faucet's base prints leaning over, so its underside is a slope that curves. Under a flat ceiling a support's roof is one sheet. Under a slope it has to be built again, a step further out, on every layer.

SEEN IN PET-GF, support included · Mark 1 and Mark2 · textured PEI · left 0.4 mm · the faucet's base Not tried in: any other tilted part



10 NOVEMBER 2026 · MARK 1 · PET-GF
Slim trees, so the roof was there. **1** A stem snapped off inside a pocket meant for a threaded insert. **2** Roof that tore instead of parting. **3** The screwdriver.



11 NOVEMBER 2026 · MARK 1 · PET-GF
Four prints later. **1** The support, off as one ring. **2** A pocket, open and empty. **3** The underside it left.

A different problem than we thought

The first faucet printed with Support D looked as if its roof had welded on: the stems broke and something stayed behind. Read back, the sliced file had almost no roof on the slope. The default trees met the part there with bare tips.

> *"Still a huge problem … but a different problem than we thought, very different."*
>
> DEREK · 10 NOVEMBER 2026

0.3 m
of roof in 42.5 m of support, as that faucet was sliced. All of it lay in one low band.

9.1 m
with the tree style set to Slim and nothing else changed.

2026	WHAT CHANGED	WHAT HAPPENED
10 NOV · MARK2	Default trees, with D's roof and gap.	Stems snapped. Their tops stayed on the part.
10 NOV · MARK 1	Slim trees, so that a roof is there.	Would not come off, even with a screwdriver.
10 NOV · MARK 1	The gap doubled to 0.48 mm.	Easier: "they were all joined together". Underside no better.
10 NOV · MARK 1	Gap back at 0.24 mm. Courses beside it bent to follow the slope.	A better underside. Stems snapped inside the insert pockets.
10 NOV · MARK 1	Pockets bridged and kept empty. Fins at the rim, under the roof.	Stopped part way. The roof came off in one unit.
11 NOV · MARK 1	The same recipe on the Sculpted faucet's base.	"…remove the entire bottom support in one piece"

What we do now. Read the sliced file for the roof before judging a support. Keep supports out of blind holes, and bridge each hole's shoulder and roof instead. Derek on why the fifth print let go: "it's the removal of the heat sets that really made that possible".

IN THE RECORD

Roof paths, two tree styles · The screwdriver · Stems in the pockets · One unit · The Sculpted base

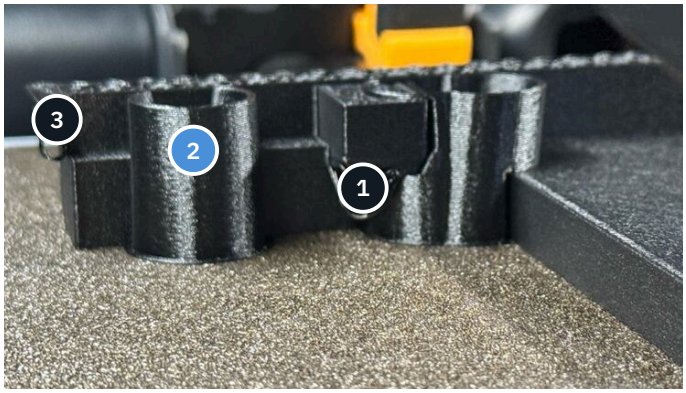
STILL OPEN The last two prints changed several things together, and the shared rules still call tilted and curved contacts unqualified. A trees-only print on 11 November has no recorded result. Since these changes every faucet tip has printed with a rough band near its end.

SUPPORTS · PAINTED BY HAND

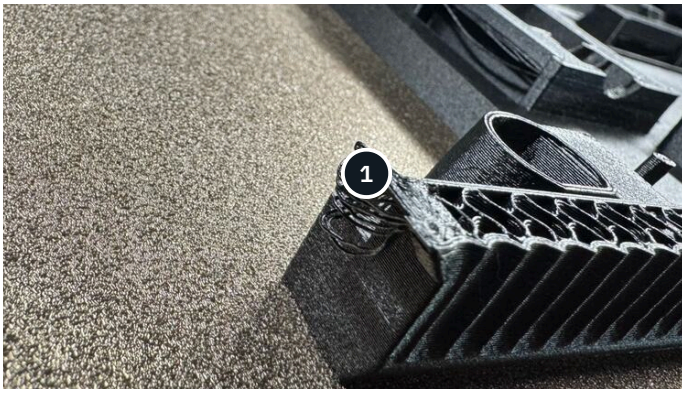
Paint supports by hand and every underside is yours.

Left alone, the slicer puts support under everything that overhangs. To keep supports out of pockets no tool reaches, back-top was switched to supports painted on by hand: none unless asked for. From then on, whatever nobody painted was printed over air.

HOLDS FOR any filament: it is method · seen here in PET-GF on Mark 1, textured PEI, left 0.4 mm



10 NOVEMBER 2026 · MARK 1 · PET-GF
1 A ledge nobody painted, laid over air. **2** Hollow trunks on their way up to something that was painted. They pass the ledge and do not touch it. **3** A second spot.



THE SAME PRINT
1 The third: the end of a fluted wall.

> *"… at least three spots here that are not supported and pretty obviously should be, and it's printing an error and loops are falling down. Do you see what I'm talking about? What the hell happened here?"*
>
> DEREK · 10 NOVEMBER 2026, TWELVE HOURS INTO A 37-HOUR PRINT

What had been checked

Four regions were painted: the rails and the rear outside. Before the print was sent, the sliced file was read for supports in forbidden pockets, for room at the edge of the bed, for the right walls and fill in the right places. All of it passed. Nobody had asked the other question: does every underside have something to land on? He stopped the print.

Every underside gets an owner

The list made that afternoon has 84 downward-facing surfaces on this one piece, in 22 groups. Each group is given one of the five landings, and each support a way out.

UNDERSIDE	LANDS ON	WAY OUT
Rail bearing faces	Support	The open rail run
Funnel-frame ledges	Support	The open flank
Raked joint ends	Support	The open front
Receiver catches	Bridge	Nothing to remove
Bores in the rear wall	Lean, to a point	Nothing to remove

Before you print with painted supports: walk the model for every face that points down, not for the ones you remember. For each, find its first course in the sliced file and follow it down to what it rests on. A review that passes has answered only the questions it asked.

IN THE RECORD

The loops · The list of undersides · Why that list is not ready · The rule as agents read it

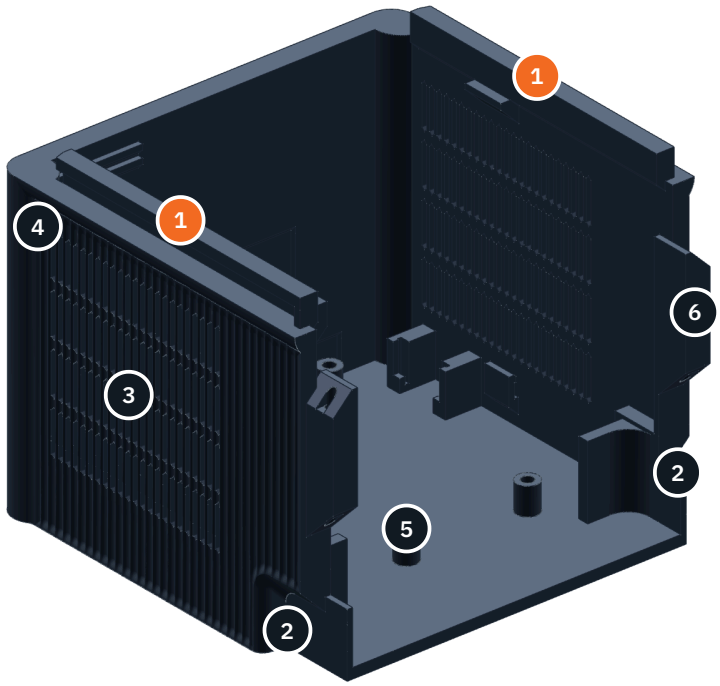
STILL OPEN No back-top has been printed from the list. The one sliced on 11 November 2026 passed its review, as the stopped one had.

PART STUDY · ENCLOSURE FRONT-BOTTOM

One pair of supported faces.

It takes about a day to print and it overhangs all over: handholds, vents, rails, seam knuckles. All but two of those places land on plastic. The two are flat because their job needs them flat.

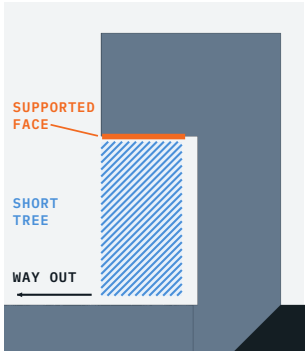
SEEN IN PET-GF · the print sent to Mark2 on 11 November 2026 · textured PEI · left 0.4 mm



AS IT SITS ON THE BED
BUILD DIRECTION ↑

754
layers in that print

2
supports, both standing on the part



SECTION THROUGH A RAIL
The lane is open along its whole length, so a screwdriver levers the tree out sideways in one piece.

1 **Rail catches** SUPPORT
The only two supported faces. They are flat on purpose: they hold the top half down, and a sloped catch would cam open. Each gets a short tree that stands on the ledge just below it.

2 **Handhold roofs** LEAN
A 45° corbel, printed in ordinary layers. It replaced a flat ceiling that had needed tree supports, a band of six walls and snap-off fins along its edge.

3 **Vents** LEAN
The vent is the flutes, pierced. Each slot runs up the floor of a groove and ends in a 45° point, so nothing has to bridge.

4 **Flutes** STRAIGHT UP
Grooves that run straight up the build cost nothing. They stop short of the bed so the first layer is a plain outline, and they hide the seam between front and back in a shadow.

5 **Bosses** STRAIGHT UP
Stand on the floor slab and are bored along the build direction, which keeps the holes round.

6 **Seam knuckles** LEAN
Every hidden underside rises at the same 45°. Where a screw hole crosses one it runs on as a slot, because a round hole through a slope leaves a skin a third of a millimetre thick.

> *"...precisely one pair of supported faces, that are quite intentional and the absolute easiest to remove (with tool) support removal situation we've found so far."*
>
> DEREK · 12 NOVEMBER 2026

A support is not a defect. In September a tool counted supports and scored fewer as better. To lose one, a catch on the rear rail became a ridge, which can cam open. The flat went back, with its support. A count of supports falls whenever the thing being supported is deleted.

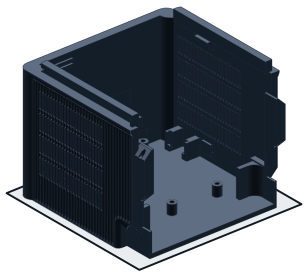
STILL OPEN The record holds this piece's reviewed slice and a rail coupon whose support came out whole on 10 November. The full-size print's own result is not written down yet.

ORIENTATION

Orientation comes first.

Before a part has a hole in it, it has a face that will lie on the bed. That choice fixes which surface comes out flat, which way grooves and bores run, and where the undersides fall.

SEEN IN PET-GF · the four enclosure quadrants as prepared in November 2026 · textured PEI · left 0.4 mm



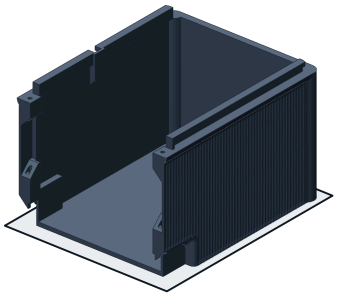
Front-bottom

FLOOR DOWN

ON THE BED, DRAWN IN ORANGE
The floor slab.

KEPT CLEAN
The walls. Flutes and vent slots run straight up the build.

UNDERSIDES
Two rail catches on short trees. The handhold roofs lean at 45°.



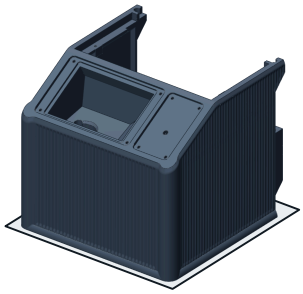
Back-bottom

FLOOR DOWN

ON THE BED, DRAWN IN ORANGE
The floor slab.

KEPT CLEAN
The rear wall and its raised warning letters, each raked 45° underneath.

UNDERSIDES
Two rail catches, the handhold mouths and the seam lips, on trees.



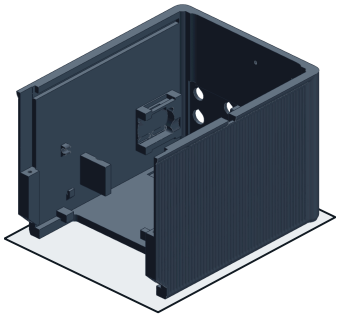
Front-top

MOUTH DOWN

ON THE BED, DRAWN IN ORANGE
The rim that meets front-bottom. A ring.

KEPT CLEAN
The roof and display surround, printed last. Their rounds close inward and carry themselves.

UNDERSIDES
The rail channels close as 45° gables. The pump mount is solid to the roof.



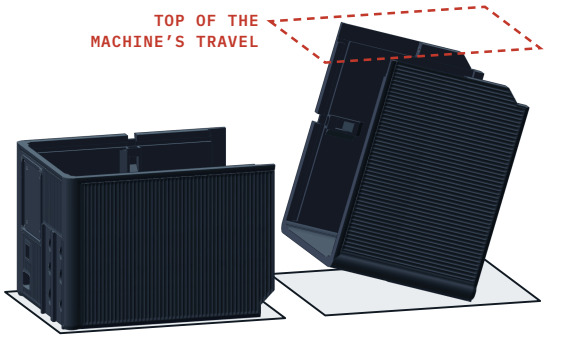
Back-top

CEILING DOWN

ON THE BED, DRAWN IN ORANGE
The whole outer roof, as the first layer.

KEPT CLEAN
The roof, flat and textured by the plate.

UNDERSIDES
Few. What hung from the ceiling stands on it. Catches bridge. Rails take support.



The day the pose was traded away

On 6 November 2026 a back-top came off Mark2 with supports no tool could reach. That afternoon the next plate was prepared with the part stood on its rear wall and tilted, so its catches and rails would grow as plain walls. That solved the supports. It also laid the flutes on their sides, hung the rear show face over trees in fine layers, and stood the part 319.08 mm tall in 325.0 mm of travel. The slicer's estimate went from 27 hours to 82. It was never sent.

> *"Please fix it. Please fix the enclosure. Restore our correct print orientation and fix the support problems the right way."*
>
> DEREK · 7 NOVEMBER 2026

The right way was in the part: a full landing under each end of each bridged catch, pointed roofs on the sideways bores, and every remaining support named. That is the next page.

One face is free. The face on the plate comes out flat with nothing under it. The last face printed can close however it likes. Every other face is a wall, or it overhangs.

Grooves and holes pick a side. Along the build they print as drawn. Across it, a groove is a row of ledges and a round hole needs a pointed roof.

Fits follow. Each supported face in a joint adds 0.25 mm to its gap. The same rail is drawn looser in the back column, where two of its faces print on support, than in the front, where one does.

IN THE RECORD

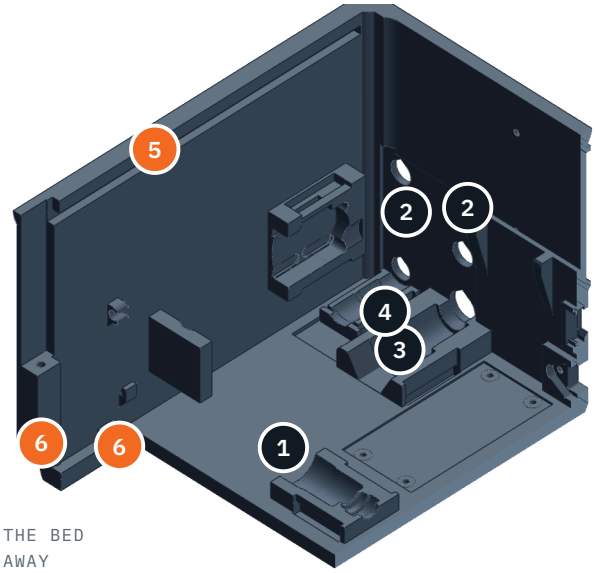
How each piece is laid down · The tilted plate, prepared and not sent · The restored pose · Back-top first printed on its ceiling

PART STUDY · ENCLOSURE BACK-TOP

The roof is the first layer.

Back-top is the lid over the plumbing and electronics. It prints upside down. Its outer roof lies on the plate, and what will hang from the ceiling grows up from it.

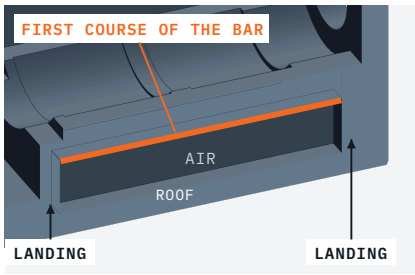
SEEN IN PET-GF · back-top as prepared for Mark2 on 11 November 2026 · textured PEI · left 0.4 mm



AS IT SITS ON THE BED
WEST SIDE CUT AWAY
BUILD DIRECTION ↑

28 places the slicer is forbidden to support

23 undersides it is told to support



SECTION THROUGH A RECEIVER
The bar a cap snaps under, bridged over its pocket. On 7 November 2026 a service opening beside it was shortened so the fore end lands on full stock too.

1 **Roof** ON THE BED
The outer roof is the first layer, so it takes the plate's texture and needs nothing under it. Its side edges leave the bed as a chamfer, 0.5 mm out per millimetre up, not a round.

2 **Fitting bores** LEAN
Six holes cross the rear wall sideways. Each keeps its round seat and gains a pointed roof. Coloured plates cover the points.

3 **Receivers** STRAIGHT UP
The cradles for the regulators stand on the slab like furniture on a floor. Under a ceiling, each one was an overhang.

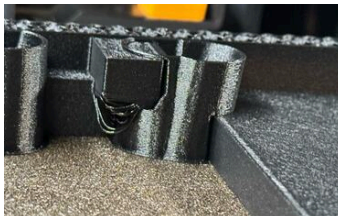
4 **Catches** BRIDGE
The bar a cap snaps under is pulled along its pocket between two full shoulders. Nothing is printed beneath it, so there is nothing to dig out.

5 **Rails** SUPPORT
The face that holds this piece down is flat because it bears. It prints on strips that stand on the rail's own floor and leave through the open lane.

6 **Ledges** SUPPORT
Small undersides at the open end, where a screwdriver does reach. Each one is named for the slicer by hand. One left off the list prints over air.



10 NOVEMBER 2026 · MARK 1
A few millimetres up. The slab is the roof.



THE SAME PRINT
A ledge nobody had named, laid over air.

No tool, no support. The back-top that finished on Mark2 on 6 November 2026 had slicer supports inside its receivers. On a coupon they had come out with a small screwdriver. In the full part the walls were in the way.

> *"I can't reach in that far to remove those tiny little supports."*
>
> DEREK · 6 NOVEMBER 2026

So those interiors print with none, and every other support is chosen by hand. The first print prepared that way was stopped on 10 November with loops hanging under three ledges.

STILL OPEN No back-top has come off the bed since the missing undersides were named. The preparation of 11 November has been sliced and read, not printed.

IN THE RECORD

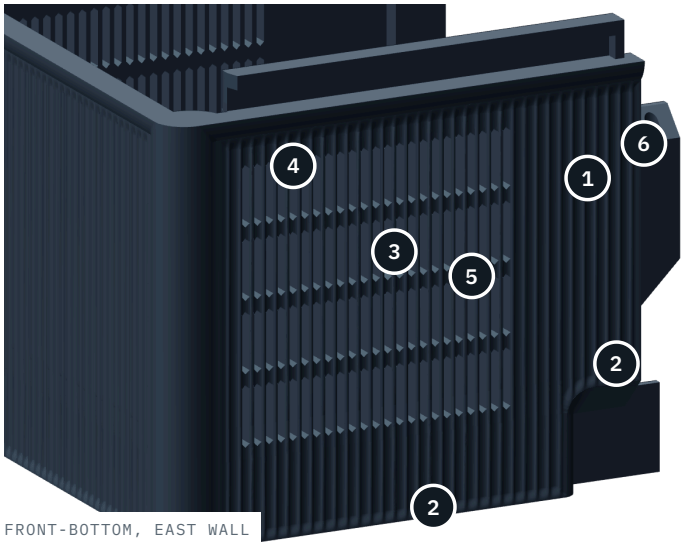
What gets support and what must not · The print of 5 November and what came back · The loops, 10 November · The bar's fore landing · A receiver on the live model

THE SKIN

The skin is a printing decision.

The outside of the box is fluted from floor to roof. A printer draws each layer's outline in one pass, so a groove that runs straight up is simply part of that outline, layer after layer. Turned sideways, the same groove is a row of ledges.

SEEN IN PET-GF · every quadrant printed since September 2026 · Mark 1 and Mark2 · left 0.4 mm Not tried in: the coupons that chose the groove were drawn for PETG



FRONT-BOTTOM, EAST WALL
BUILD DIRECTION ↑



AT A TRANSOM
Slots end in a point and begin on one. The grooves run on through the band unbroken.



CUT LEVEL THROUGH THE VENT
Each slot goes through the floor of a groove. What is left between two slots is a bar.

1 Flutes STRAIGHT UP
Half-round grooves every 5.0 mm, running the way the part grows. Each layer's outline simply has the grooves in it. Nothing overhangs and nothing steps.

2 Fades STRAIGHT UP
Every groove shallows to nothing before the bed, the seam above, an opening or a rounded corner. The first layer is a plain outline, and no groove meets an edge as a row of teeth.

3 Vent slots LEAN
A slot 3.1 mm wide is cut through the floor of each groove in front of the condenser's fan. Seen from the side, the wall still reads as flutes.

4 Pointed ends LEAN
Each slot closes at the top and opens at the bottom on a 45° point, inside the groove's own shadow. Nothing bridges.

5 Transoms STRAIGHT UP
Three bands are left unpierced. Every bar between two slots is tied to its neighbours instead of standing free for the height of the fan.

6 The seam STRAIGHT UP
The joint between the front and back pieces runs the full height of the wall. It lands in a groove, off its centre, in a shadow that is already there.



24 SEPTEMBER 2026 · PET-GF
A printed front-top. The flutes fade out below the display surround.

Four walls behind every groove

The slicer lays this wall as two loops inside and two outside, 1.74 mm in all. A groove may take only what leaves all four. Whatever disturbs those loops from behind shows on the outside. On 11 October 2026 a back-top came off Mark 1 with lengths of flute missing, each opposite a screw pad that the slicer had been told to reinforce with extra walls.

> *"...a bunch of features from the interior side have resulted in parts of the flutes cut off, like different lengths of vertical lines removed from the exterior surface..."*
>
> DEREK · 11 OCTOBER 2026

Reinforcement now changes infill only, and nothing inside the box is cut into the back of a fluted wall.

STILL OPEN The bars between slots and the wall under a groove are the thinnest sections on the box. Both were approved as drawn on 21 October 2026. No one has yet written down how the vent printed, or whether the missing flutes came back.

IN THE RECORD

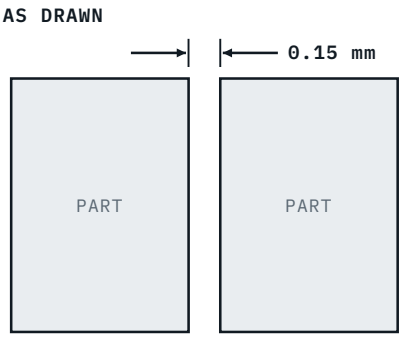
The reeded skin and the vents · The flutes cut off, 11 October · The texture coupons, August · The vents on the live model

FITS · PRINTED FACE AGAINST PRINTED FACE

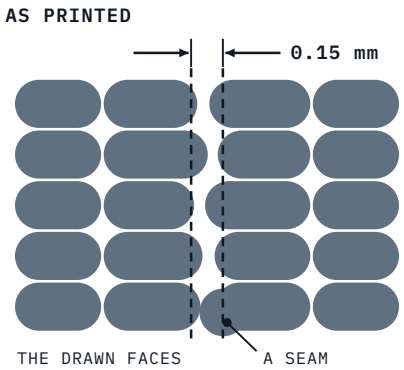
A clearance is surface, not air.

A printed face is not the flat plane that was drawn. It is the sides of roads, the ridges between courses and a seam on every loop. Press two together and those touch first. The gap we draw between parts is the room they take.

SEEN IN PET-GF against PET-GF · Mark 1 and Mark2 · textured PEI · left 0.4 mm Not tried in: other pairs



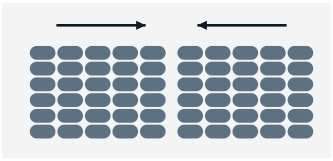
HOW IT GETS READ
Two flat faces and nothing between them. Read that way, the gap looks like room to spend somewhere else in a stack.



WHAT IT IS
Both faces stand a little past where they were drawn. Seated under a screw or its own weight, a part still rests about where it was drawn.

> *"You do realize our clearances are not filled with free space but are actually filled with rough edges of materials? ... if you put a 1mm object next to a 1mm object pressed with several lbs of fource you still don't get a 2 mm object, you get a 2.15 mm one."*
>
> DEREK · 22 OCTOBER 2026

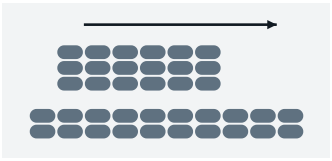
> *"Our clearance rules are not "we need more free space", they are "this is the reality of what these surfaces are""*
>
> DEREK · THE SAME DAY



A seat
0.15 mm

Two faces that only have to close. It is a friction fit and hard to slide.

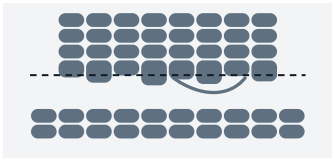
The faces that meet head-on when front and back come together. The back of a nameplate.



A slide
0.15 + 0.1 mm

One face has to travel past the other. It is still a tight slide.

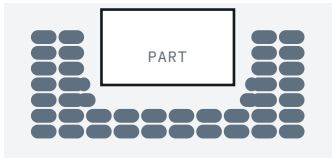
Rails, tongues and pin passages. The pump cradle going into its bay.



Under a supported face
+ 0.25 mm

An underside printed over support sags into bumps and keeps loose strands. Each such face in a gap adds this once.

The front rail catch has one. The rear catch has two.



At the floor of a pocket
+ 0.1 mm

The first courses of a wall round the corner where it leaves the floor, so a pocket is narrowest at the bottom.

The floor under the pump cradle. The floor under the cold core.



10 NOVEMBER 2026 · MARK 1 · PET-GF
A slice of the sliding rail in a slice of the wall it slides into, cut from the real parts with their own allowances.

What we do now

Every allowance stays on the face it describes. A face that seats, bears or stops a stroke gets one too. Derek, 16 October: "SEATING FACES ARE NOT EXEMPT FROM THE RULES!"

The slicer's hole and contour corrections stay at 0, so every fit is in the drawing, where it can be read.

7 October 2026. Three quarters of the enclosure, the foam shell, the pump cradle and cap, a tee carrier and a funnel frame were on the bench: "And all those prints fit together right now".

The pocket floor came first. On 23 September Derek said the bottom of a printed cavity "is almost guaranteed to narrow at the tip because of the layer transitions resulting in very slight rounding".

STILL OPEN The rail slice went together with "plenty of clearance" and "might be a little bit loose". Derek left it alone until a full-length rail is tried. That result is not in the record.

IN THE RECORD

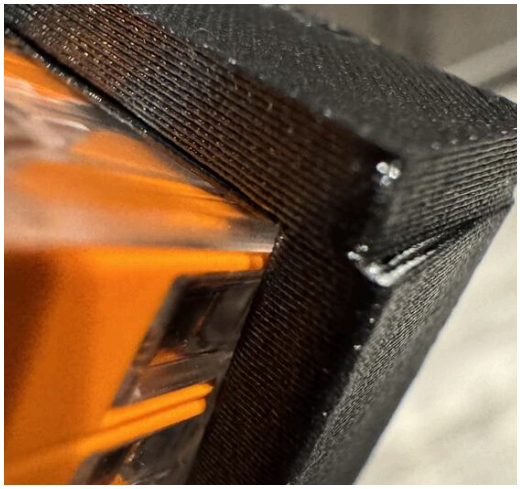
The allowances and where each is used · The file that owns them · The rail slice · The pocket floor, 23 September · What fitted on 7 October

FITS · A BOUGHT PART IN A PRINTED POCKET

PET-GF does not stretch.

Glass-filled PET is stiff. A pocket wall gives a few tenths of a millimetre and then an edge lets go. A hard part held by friction gets a pocket drawn at the part's own size, give or take a tenth of a millimetre.

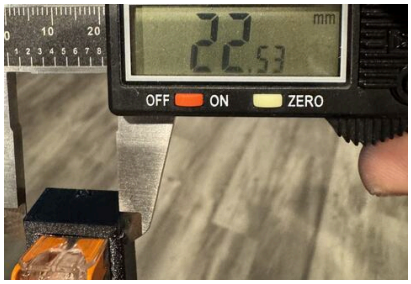
SEEN IN PET-GF, black Fiberon PET-GF15 · Mark 1 · textured PEI · left 0.4 mm Not tried in: PETG, PET-CF



12 NOVEMBER 2026 · MARK 1 · PET-GF
A fit coupon for the two lever connectors in front-top, its pockets drawn narrower than the connectors. Pressed against the edge, a connector split it.



AT REST
21.83 mm across the coupon where nothing pushes on it.



SPREAD
22.53 mm across a pocket with a connector shoved in. Walls three and a half millimetres thick had moved 0.7 mm.

> *"Prints turned out well. My guesses on adjustments were way off. It looks likethe PET-GF does flex enough at some points there to allow me to corner shove in a wago, but when I pressed the wago to the edge, you can see it snapped the edge of the receiver."*
>
> DEREK · 12 NOVEMBER 2026

THE POCKET'S WALLS AGAINST THE CONNECTOR'S CATALOG SIZE, EACH SIDE · 12 AND 13 NOVEMBER 2026 · TIGHTEST FIRST

●●
0.60 under
Derek's first guess, times one and a half.

●
0.35 under
His first guess. Neither coupon took a connector straight.

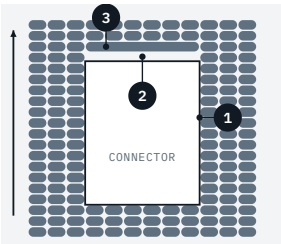
●●●●
0.10 under
"it's possible to get in there, but it's too tight." Its height differed too.

●●●●●
0.05 under
The next morning: "Size wise I think we're getting very good".

●●●
0.00
"the WAGO fits in very easily, but is still held pretty well. It's awesome."

EARLIER FRONT TOP
0.15 over
The seat allowance of the page before. "in Y they are a bit loose."

THREE-DOT FIT COUPON · SEEN THROUGH THE POCKET'S MOUTH · NOT TO SCALE



1. **Wall to wall: the catalog size.** No gap is drawn.
2. **Under the roof: a fifth of a millimetre,** as sliced. Up the build a pocket moves a whole course at a time. See "Height moves one layer at a time".
3. **The roof is a bridge** from wall to wall.

Twice before

On 5 October 2026, on Mark 1, Derek tried ladders of pockets for a ring magnet and for the four posts of a valve. He chose the pocket at the part's nominal size both times.

What we do now

A ladder for a hard part starts at its catalog size and steps a twentieth or a tenth of a millimetre either way. Each coupon is cut from the real part at the real depth, with dots in the plastic.

Where it stands

Derek accepted the tighter pocket dimensions for the production front top. Both roofs bridge between the sidewalls. The single six-dot coupon came out distorted when printed without a prime tower.

STILL OPEN Nobody put a caliper on a connector: the catalog size is the maker's figure. The record does not say which coupon cracked or which pocket was too tight.

IN THE RECORD

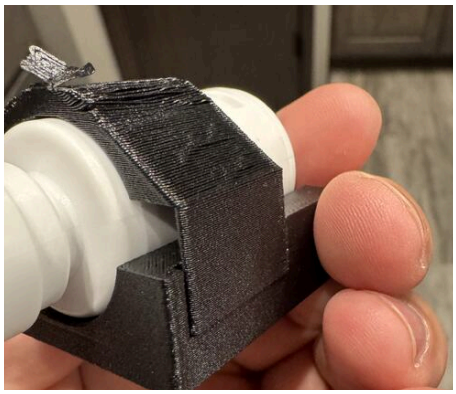
The cracked coupon and the caliper · The coupon that fitted · Five dots · Six dots and fit acceptance · The connectors' catalog sizes · Magnet and valve ladders

FITS · SNAPS

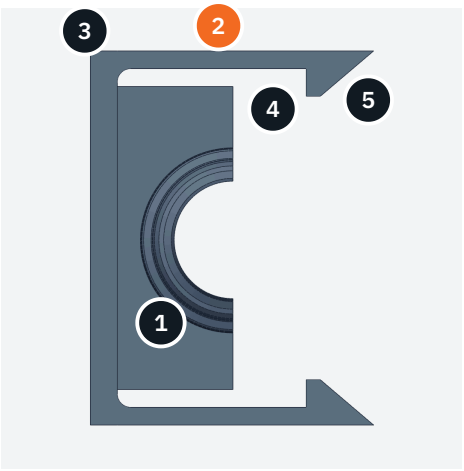
A snap bends within its layers.

A snap is a leaf that bends out of the way and springs back. Printed so that every road runs the length of the leaf, the plastic itself takes the bend. Printed in courses across the leaf, the bend pulls one course off the next.

SEEN IN PET-GF · Mark 1 and Mark2 · textured PEI · left 0.4 mm · caps printed solid Not tried in: PETG, TPU

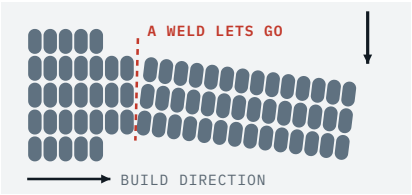


1 NOVEMBER 2026 · MARK2 · PET-GF
A first-round cap, printed as it sits. Its layers run across both leaves. Fitting it over the flow meter tore it.



THE CAP NOW · LOOKING DOWN AT THE BED
It stands on its end, so every layer is this shape.

1. **Body**
   A thick block shaped to the part it holds. It does not flex.
2. **Leaf**
   The spring. Two millimetres thick and about twenty-two long since 2 November. Thinner tore. Thicker bent only at its root.
3. **Root**
   A rounded inside corner, so the bend spreads along the leaf.
4. **Catch**
   A flat face square to the pull. It will not cam open. To let go, the leaf is spread by hand.
5. **Lead-in**
   One solid slope to a point, 6.0 mm long, that wedges the leaf aside on the way in.



ACROSS
The bend is carried by the welds between courses.



ALONG
The bend is carried by the roads themselves.

> *"Those snaps need to be, the wings need to be at least twice as thick."*
>
> DEREK · 1 NOVEMBER 2026

> *"Can they all be designed to be printed on their end like the pump cap?"*
>
> DEREK · THE SAME DAY

ONE WEEK OF CAPS · 1 TO 8 NOVEMBER 2026 · MARK 1 AND MARK2 · PET-GF

1 NOV
One thin arch
The pump's first cap was all spring. "...this has basically 0 hold." It became a thick box with two separate leaves.

1 NOV
Thin leaves, across
Flat catches on leaves 1.3 mm thick, the caps printed as they sit. Two of five tore while fitting.

2 NOV
Thick leaves, on end
2.6 mm, nearly twice as long, every cap on its end. They held, and bent only at the root.

2 NOV
In the middle
2.0 mm with a rounded root. That night three fits were "perfect, or close enough to perfect".

3 NOV
No tunnels
A screwdriver tunnel under each catch bent the leaf only where the blade touched. "Please just remove those."

8 NOV
A long point
The lead-in doubled. "The longer pointed tips on the snaps there worked beautifully."

What we do now

Stand a snap part on the bed so its leaves lie in the layers. Give it a stiff body and separate leaves. Leave the leaves where fingers can spread them. Keep the catch flat and put the ease in a long lead-in.

STILL OPEN Every verdict here is Derek's hand. No snap has been measured for force, cycled or shaken on a running pump. The long point was approved on the water regulator's cap. The other caps now carry it and have not been tried.

IN THE RECORD

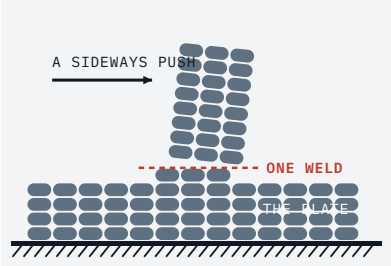
"basically 0 hold" · The round that tore · Thick leaves · In the middle · No tunnels · The long point · The caps today

FITS · PEGS AND BOSSES

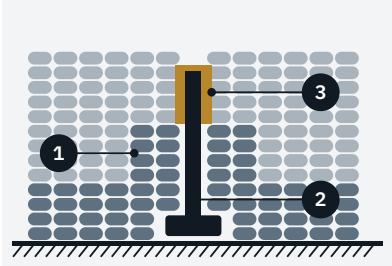
Never load a peg across its layers.

Each layer is welded to the one below, and that weld is the weakest thing in a print. A peg standing up off a plate joins it along one such weld. Push the peg sideways and that weld takes the whole load.

SEEN IN PET-CF first: a faucet plate, May 2026, 0.6 mm nozzle · then PET-GF: a pump base, 2 November 2026, Mark 1 Not tried in: a PET-GF peg, broken on purpose



MAY 2026 · A PEG
Two pegs four millimetres across stood on the faucet's plate to press into its shell. One snapped off on the first try.



JUNE 2026 AND SINCE · A PEDESTAL AND A SCREW
The pedestal is short and wide, and nothing pulls on it.

> *"they still don't fit, and they snap so easy because of the orientation of the layer lines. This is just not workable."*
>
> DEREK · MAY 2026 · AFTER THE SECOND TRY

The second try made the pegs a tenth of a millimetre slimmer, as loose locating pins with nothing to hold. They snapped again. The fault was the direction of the layers, not the fit.

1 a broad pedestal, which only locates 2 a steel screw, which holds 3 a brass insert melted into the other part



2 NOVEMBER 2026 · MARK 1 · PET-GF
The pump's test base, printed flat. Prying its cap off, the base let go before the snaps did. It opened along a layer.

Which plastic

The pegs were not PET-GF. The faucet was PET-CF that spring and its print log files them with its PET-CF prints. The plate's own notes that week say "PETG or PET-CF", and nothing names the spool.

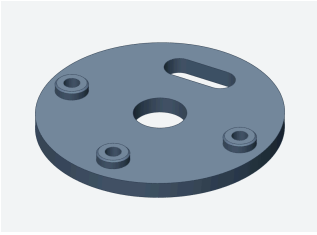
In PET-GF

The photograph is the nearest we have. Derek asked for no change on account of it. The first round of snap caps, printed with their leaves across the layers, is the other case, on the page before.

27.9 of 65.9 MPa. PET-CF across its layers, against along them.

48.2 of 59.9 MPa. PET-GF, the same two figures.

The maker's own printed test bars, as noted in the faucet's print log on 24 August 2026, the day the faucet moved to PET-GF.



THE PLATE TODAY · PET-GF
Three pedestals, three screws.

What we do now

A feature that will be pushed sideways or pulled is laid so that its roads run through the load, as a snap's leaves are. Anything that has to stand up off a plate is kept short and broad and is only asked to locate. A screw in a heat-set insert carries the joint.

STILL OPEN Nobody has broken a PET-GF peg on purpose. By the maker's figures PET-GF keeps four fifths of its strength across its layers and PET-CF less than half, so the rule may bind less hard here. Nothing in the record tests that.

IN THE RECORD

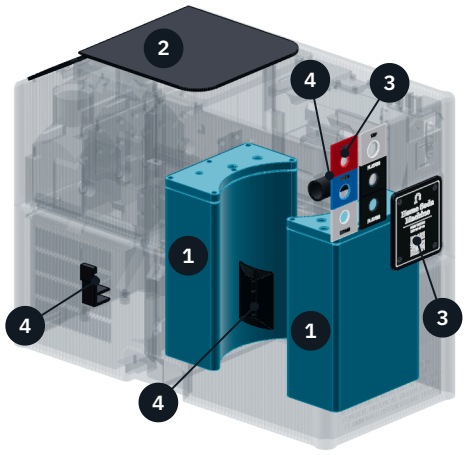
The pegs, in the faucet's print log · The commit that removed them · The base that split · The plate today

PETG · THE PLASTIC

Plain PETG holds the flavoring.

The enclosure is stiffened with glass fibre. The two reservoirs are not. They and their lids are plain clear PETG, laid a little fat so that every road is pressed into its neighbours.

SEEN IN Bambu PETG Translucent Clear · Mark 1 · glued Engineering Plate · left 0.8 mm Not tried in: Mark2



ONE MACHINE FROM THE REAR, ENCLOSURE FADED
Teal is clear PETG. The rest is PETG Basic, drawn in the colour it prints in.

1 Reservoirs and their lids
CLEAR

Printed, and holding water. They hold the flavoring, and the level shows through the wall. The glass-filled PET-GF stays in the shells around them.

2 Funnel cover
BASIC, BLACK

Sliced, never sent. A solid plate that closes the fill funnel between fills.

3 Chips and nameplate
BASIC, FOUR COLOURS

Printed 8 November 2026. Black, white, blue and red through 0.2 mm nozzles. The status plate on the front shared their bed, with a clear window for its light.

4 Small parts inside
BASIC, BLACK

No print on record. The relief-valve shroud, the reed bridge and the fuse clamp.

WHICH PETG, WHICH PRINT

SunTop clear, an off-brand spool: the first two reservoirs, 22 and 25 May 2026. The first leaked. Not used since. **Bambu PETG Translucent Clear:** every reservoir and lid from 30 May 2026, both halves of the funnel mold, and the status light's window. **Bambu PETG Basic:** the chips, the nameplate and the status plate.

One family, two recipes, and neither is PET-GF's.

	CLEAR PETG · RESERVOIR BODY	CLEAR PETG · MOLD HALF	PET-GF · ENCLOSURE
Nozzle	**Mark 1, left 0.8 mm**	**Mark 1, right 0.4 mm**	**left 0.4 mm**
Plate	**Engineering Plate, glued, 70 °C**	**Textured PEI**	**Textured PEI Plate, 80 °C**
Lift added to the first layer	**none:** `G29.1 Z0`	**0.04 mm**	**its own for each printer and plate**
Brim	**0 mm**	**8 mm**	**auto_brim**
Layers, first / rest	**0.3 / 0.24 mm**	**the preset's**	**0.2 / 0.24 mm**
Nozzle heat, first / rest	**255 / 260 °C**	**the preset's**	**265 / 280 °C**
Flow	**1.05**	**0.88**	**0.9555**
Wall loops	**5, arachne, 0.8 mm each**	**6**	**2, classic**
Inside the loops	**100% zig-zag**	**15% gyroid**	**15% gyroid**
Fill pushed into the loops	**50% left body · 30% right body**	**the preset's**	**15%**
Part fan on plain walls	**10 to 20%, off for 3 layers**	**the preset's**	**0%**

> *"I have found in other testing that over extrusion (flow ratio) helps with air tightness / water tightness."*
>
> DEREK · 22 MAY 2026, BEFORE THE FIRST RESERVOIR

IN THE RECORD

The project that holds this recipe · Which print each plate is bound to · Mold recipe · Why no glass touches the drink

STILL OPEN The project file holds the reservoir column. Two notes beside it still describe the file as it stood during October's trials: 30% and 265/270 °C through the right nozzle. The right body is still bound to 30%, the last value with a water test written against that side, though its 50% print finished on 26 October 2026 and Derek's word of 12 November covers it. No printed PETG surface has been through the food-contact screen the project wrote for it.

WATERTIGHT · THE FLOOR

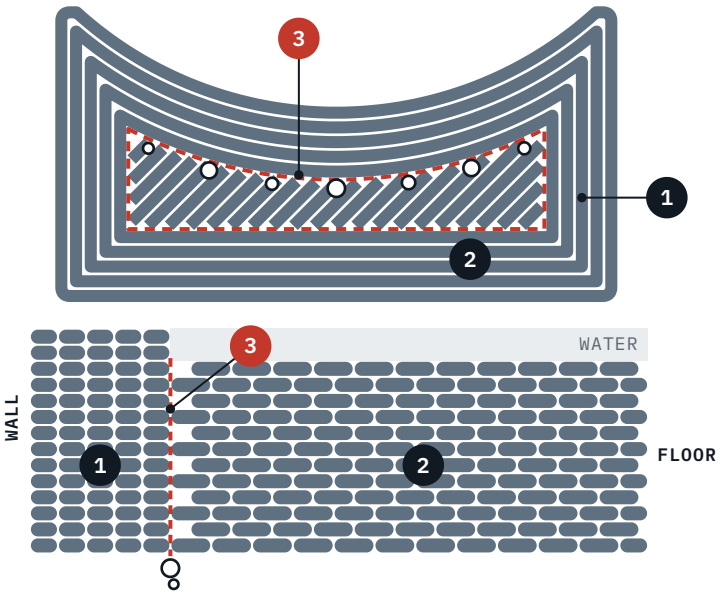
It weeps where the floor meets the wall.

A floor is printed in two parts: loops that follow its outline, then straight fill inside them. Unless the fill is pushed into the loops, the line where the two meet lets water through.

SEEN IN Bambu PETG Translucent Clear · Mark 1 · 0.8 mm nozzle · flat 3 mm floor Not tried in: PETG Basic



21 OCTOBER 2026 · MARK 1 · BAMBU CLEAR PETG
The underside of a reservoir, filled and held up to the light. Loops follow the scooped side and fill runs at an angle. The beads of water sit in a row on the line between them.



THAT FLOOR FROM ABOVE, AND CUT THROUGH AT THE WALL
1 five loops, laid first. **2** fill, laid inside them afterwards. **3** the line where they meet, which runs through every course of the floor.

One setting moved, in three steps: how far the fill is pushed into the loops.



LEAKED **15%, 21 October 2026.** The slicer's own number. Beads in a row: "Leaks between infill and wall."



ONE WEPT **30%, 22 to 25 October.** Changed with flow, loop count and heat. Bodies held for days. Then the newest wept specks in two hours.



HOLDS **50%, 26 October.** The only setting changed. That body: "It's holding water so far."

> *"It does appear to be precisely in the line between wall and infill. I know we increased that overlap some already, can we increase it more?"*
>
> DEREK · 26 OCTOBER 2026, OF THE SPECKS

An agent prepared 40%. "Please make that 50% instead, thanks."

What we do now

5 loops, solid fill between them, and the fill pushed 50% of a road into the loops on the left body. The PET-GF enclosure stays at 15%. Nothing asks it to hold water.

IN THE RECORD

The beads in a row · The specks at two hours · 30% to 50%, nothing else · Derek, 12 November

STILL OPEN The photographs show where water came out, not the path it took. Nobody has cut a floor open. The first step, 15% to 30%, changed with three other settings. Every result is a fill-and-watch test with no fill height written down. On 12 November 2026 Derek wrote that every reservoir printed since the change was still holding.

WATERTIGHT · A FLOOR OF LOOPS

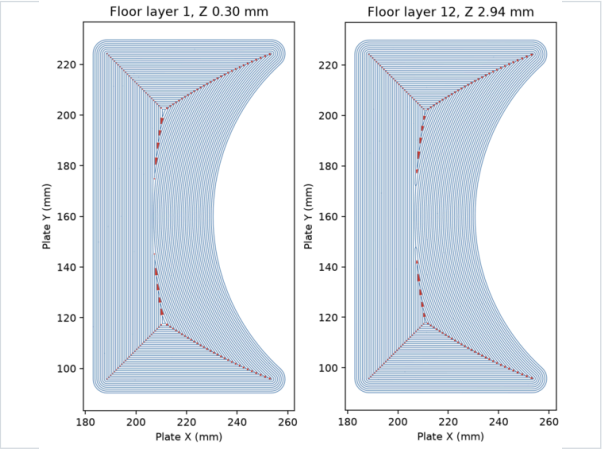
A floor made of loops leaks along its seams.

Loops laid side by side run out of room where they turn a corner and where they meet in the middle. Each of those places is a small gap, and it comes back on every layer.

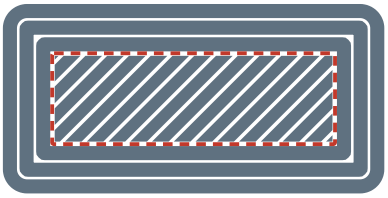
SEEN IN Bambu PETG Translucent Clear · Mark 1 · right 0.8 mm · textured PEI Not tried in: a flow above 1.02



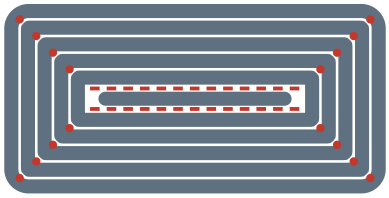
22 OCTOBER 2026 · MARK 1 · BAMBU CLEAR PETG
A reservoir whose floor is loops and nothing else, full of water, seen from below. The beads sit along a branching line across the floor and at its corners.



FROM THE RECORD · FIRST AND TWELFTH FLOOR LAYER
Blue is a loop. Red is floor no loop covers, counted from the file sent to the printer. Nobody has laid the photograph over this plot.



Loops round fill. One seam, a closed line, and one setting pushes the fill across it. The floor on the page before.



Loops all the way in. No fill to push. A gap wherever a loop cannot follow its neighbour: at each turn, and along the middle.

45
loops in each floor layer of that print.

10 mm²
of floor left uncovered in each of its twelve layers. About a thousandth of the floor.

72% less
uncovered in the slice once a setting added centre roads where loops meet. Printed on 22 October, it leaked in the same place.

> *"I think I'd rather continue the walls forever, as we did in the sphere pressure vessel test. Another agent suggested at some point infill works better, but I haven't seen that."*
>
> DEREK · 21 OCTOBER 2026, ASKING FOR THIS PRINT

> *"Leaks along the seam"*
>
> DEREK · 22 OCTOBER 2026

Of the second try: "Not sure that made any difference at all."

What we do now

The floor is solid fill inside five loops. The loops-only floor has not been printed again.

IN THE RECORD

Result and photograph · Gaps, layer by layer · The second try

STILL OPEN Both loops-only floors were printed at flow 1.02, before the flow went up to the value that held. A floor of loops has not been tried at that flow, so this page shows that these two floors leaked, not that no floor of loops can hold. The sphere Derek remembered was printed in April 2026, and its settings were never found.

WATERTIGHT · WHAT WE KNOW

Slow and hot were not what sealed it.

Between the reservoir that leaked everywhere and the one that holds, the print got three times quicker and, for one body, ten degrees hotter. The heat came back out and it still held. Flow, loop count and the first lap step moved together.

SEEN IN Bambu PETG Translucent Clear · one reservoir · printer, nozzle and plate named in each row

PRINT	PRINTER · NOZZLE · PLATE	FLOOR	TIME	FLOW	LOOPS	LAP	°C	WATER
30 May 2026	Mark 1 · 0.6 mm · textured	thin, sloped, on supports	none kept	1.02	6	15%	255 / 260	**HELD** for hours
5 and 8 September	**Mark2 · 0.8 mm · textured**	**solid, sloped inside**	**20 h and 26 h**	**0.97**	6	15%	**255**	**HELD** both
19 October 2026 · the floor becomes one flat slab 3 mm thick, with nothing through it								
21 October	**Mark 1 · right 0.8 mm · textured**	**flat, 3 mm**	19 h	0.97	6	15%	255	**LEAKED** “many places”
21 October	Mark 1 · right 0.8 mm · textured	flat	**5 h 42 min**	**1.02**	6	15%	**255 / 260**	**LEAKED** along the fill line
21 and 22 October	Mark 1 · right 0.8 mm · textured	**flat, loops only**	5 h 46 min	1.02	**all**	**no fill**	255 / 260	**LEAKED** at loop seams, twice
22 October	Mark 1 · right 0.8 mm · textured	**flat, fill again**	5 h 42 min	**1.05**	**5**	**30%**	**265 / 270**	**HELD** and two days on
24 October	Mark 1 · right 0.8 mm · textured	flat	5 h 42 min	1.05	5	30%	**255 / 260**	**HELD**
25 October	**Mark 1 · left 0.8 mm · glued plate**	flat	5 h 42 min	1.05	5	30%	255 / 260	**HELD** at first report
26 October · the newest 30% body lets a few specks through in two hours. The record does not say which print it was.								
26 October	Mark 1 · left 0.8 mm · glued plate	flat	5 h 42 min	1.05	5	**50%**	255 / 260	**HOLDS** and so does the next

Orange is what changed from the print above it. Lap is how far the fill is pushed into the loops, °C is the nozzle on the first layer and after, and time is the slicer’s estimate for a body and its lid.

What did not matter

Going slow. September’s recipe asked for 30 mm/s everywhere and took 20 hours. It held on September’s solid floor and leaked in many places on the flat one. The recipe that holds asks for the slicer’s ordinary speeds and is done in under six.

Ten more degrees. Heat went up with three other things on 22 October. It came back down on the 24th, alone, and the body held. Derek had asked for that: the hotter print held, and he did not like the look of it.

What was never isolated

Flow. 0.97, then 1.02, then 1.05, each time with something else. No body has been printed at two flows with the rest left alone.

Loop count. Six became five by reading slices at two, three, four and five loops. Five was never water-tested against six.

Plate and nozzle. The glued plate and the left nozzle came in to stop lifting and fouling. 50% has only ever run on them.

> *“What I’d like is:*
> *- 1.05 flow rate*
> *- enough walls for our actual walls to be entirely walls, no more*
> *- double the XY overlap (does bambu slicer call this wall/infill overlap? Its at 15% now?)*
> *- higher temp”*
>
> DEREK · 22 OCTOBER 2026 · FOUR AT ONCE, AND IT HELD

STILL OPEN The test is water, a look, and Derek’s report. No fill height, temperature, pressure or dye is written down, and the longest duration on record is “holding water for an hour” followed by “still holding” two days later. One body at 30% wept and nobody knows why that one.

IN THE RECORD

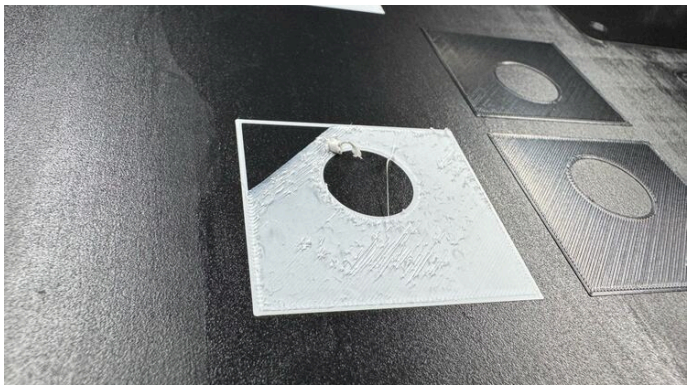
Every attempt, in order · Each hold and leak · Heat alone: twelve commands differ · Five loops, by reading slices · September’s slow recipe

FIRST LAYER

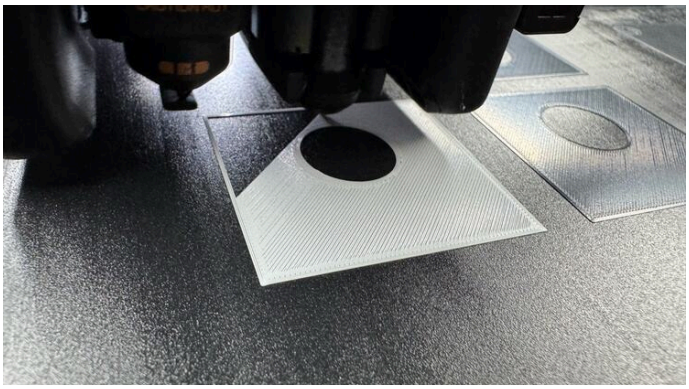
Five hundredths of a millimetre.

The first layer is pressed into the plate through a gap the printer sets for itself. A little too small and the plastic has nowhere to go. It tears and smears, and the printer stops to say its extruder is overloaded, which sounds like a different problem.

SEEN IN Bambu PETG Basic, white · Mark 1 · glued Engineering Plate · right 0.2 mm nozzle · 8 November 2026 Not tried in: any other plastic, plate or nozzle



8 NOVEMBER 2026 · MARK 1 · WHITE PETG
The fourth try. By then the spools had been dried twice, the hotend swapped for a spare and the filament rethreaded. It stopped at the same place.



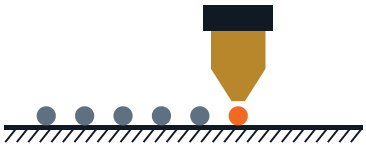
THE SAME DAY · THE SAME FILE · THE SAME SPOT
The right-side nozzles lifted by 0.05 mm, and nothing else changed. “It’s looking a lot better at the same spot, so far at least.”



TOO CLOSE
Nowhere to go. It tears and backs up.



RIGHT
Pressed flat, joined to its neighbour.



TOO FAR
Round lines that sit on top.

> *“Let’s see if we can change just that one variable of the Z offset and try again.”*
>
> DEREK · 8 NOVEMBER 2026

That print ran to the end. The next day: “So the multicolor print turned out beautifully that time.”

It belongs to one setup

The lift is set only while a right-side filament is printing, and taken off again before each tool change. The black from the left nozzle prints with none. It is the number for this plastic on this plate through these nozzles, and for nothing else.

STILL OPEN One print. The record calls the cause unconfirmed: the lift was the only change and the print finished, which does not rule out something else helping.

PET-GF’s number is not PETG’s

Two days earlier this same job had been sent with the lift Mark 1 uses for PET-GF on textured plate, 0.18 mm. On the glued plate, Derek: “lol, okay, print stopped, it lifted right off.” Asked whether that lift was the printer’s own: “No. That should be documented as for pet-gf on textured plate.”

IN THE RECORD

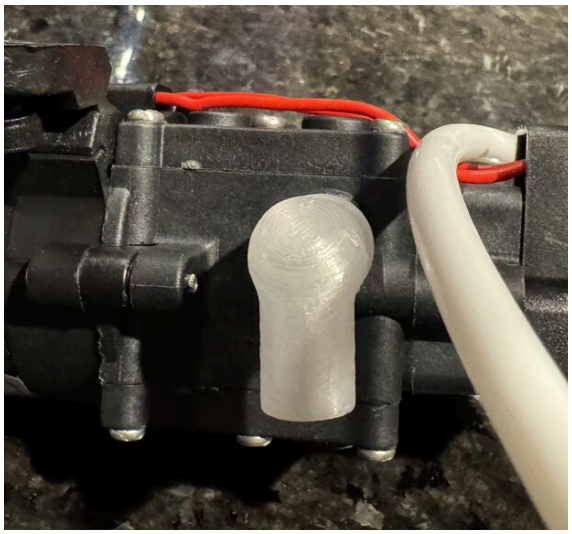
Sent with PET-GF’s lift, 6 November · The fourth stop · How the lift is applied · The same spot, lifted · The finished print · Trims, and what each belongs to

THE PLASTIC

Choose hardness by the fit.

TPU is the rubbery one: the pump's elbows, the faucet's sealing sleeve, the gaskets. The first two must stretch over something rigid, and only the softest grade we own would.

SEEN IN Bambu TPU 85A, 90A, 95A and TPU for AMS · Siraya TPU 85A · Mark 1 and Mark2 · right 0.6 mm nozzle



23 OCTOBER 2026 · SIRAYA TPU 85A
A printed elbow on a port of the water pump. The port is a barbed stub wider than the elbow's bore, so the elbow has to stretch to go on and then grips it.

STIFFER, AND QUICKER TO PRINT ↑ · SOFTER ↓

68D — **TOO STIFF** Bambu TPU for AMS · the pump elbow · 13 October 2026
"That was too stiff." Its maker lets 12 mm³ a second through the nozzle, and the slicer gave the elbow 43.5 minutes. In 85A the same elbow was given 62.

95A — **LOCKED UP** Bambu TPU 95A · the faucet sleeve · 24 May 2026
It went over the tube alone and into the faucet's port alone, "barely any force required". "But attempting both at the same time? Nah, not happening."

90A — **NEVER STRETCHED** Bambu TPU 90A · flat gaskets · October 2026
Printed only as flat gaskets, which are squeezed and never stretched. How those fitted was not written down.

85A — **FITS** Bambu and Siraya TPU 85A · sleeve, elbow · May, October 2026
The sleeve seated in the port with the tube in it "for the first time". The elbow went onto the pump by hand: "nice firm friction stretchy fit".

> *"85A is the only one that flexes enough here."*
> DEREK · 19 OCTOBER 2026

The recipe

	PUMP ELBOW SIRAYA 85A	FAUCET SLEEVE BAMBU 85A
Nozzle, first layer / rest	**225 / 215 °C**	**225 / 225 °C**
Layers, first / rest	**0.20 / 0.12 mm**	**0.3 / 0.18 mm**
Bead width	**0.62 mm**	**0.62 mm**
Walls	**6, and filled solid**	**1, as a spiral**
Flow ratio	**0.96**	**1**
Flow limit	**2.2 mm³/s**	**2.2 mm³/s**
Part fan, after layer one	**100%**	**100%**
Auxiliary fan	**0%**	**100%**

Flat gaskets take the sleeve's plastic settings with 2 walls and 100% fill. No supports in any of them.

STILL OPEN The 85A sleeve that seated also had a thicker wall than the 95A one that locked, so that pair changed two things. 90A has never been tried on a stretch fit. Nothing compares the two plates: gaskets printed on textured PEI on 8 October "turned out fine".

The flow limit is the speed

Each maker caps how much plastic may pass through the nozzle in a second: 2.2 mm³ for 85A. On a solid part that cap decides the speed. The gasket profile asks for 200 mm/s walls. The gasket printed on 23 October 2026 ran them at 21.

How it is fed

Through the right hotend, from a spool outside the printer. No AMS feeds TPU. The spool turns on an axle in a sealed box. To dry it, the Bambu 85A spool went into the AMS HT on 13 October 2026 at 75 °C for 18 hours, the cycle published for 90A.

Its bed is not PET-GF's

A glued Engineering Plate at 35 °C, with a Z trim of 0. PET-GF prints on a Textured PEI Plate at 80 °C, with a trim of its own on each printer. On 12 October a TPU liner printed at that trim fell over. The next evening Derek changed the plate: "So not our usual 0.18 offset, but with zero offset."

IN THE RECORD

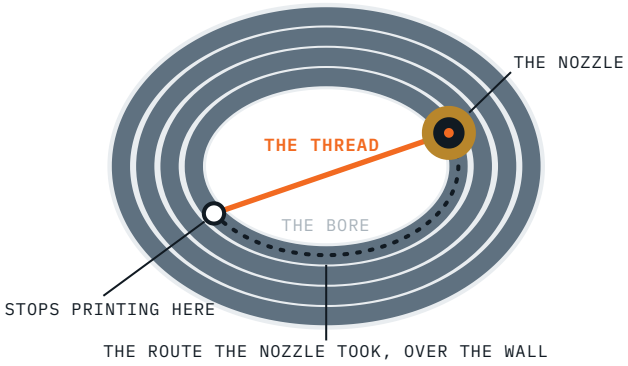
The sleeve's print log · Too stiff · The elbow's fit · The elbow recipe · The 23 October gasket · What dries where · The fallen liner · The wrong plate

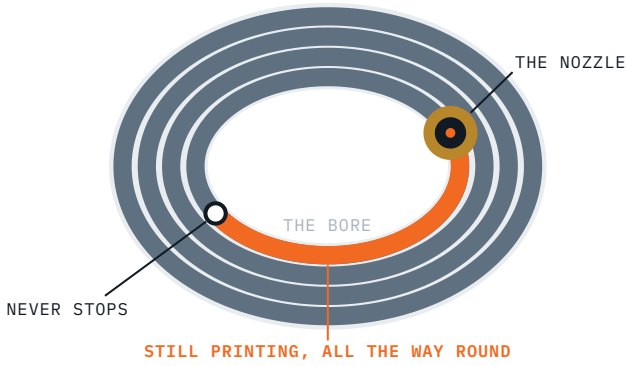
STRINGS

The nozzle goes around. The string goes straight.

Soft TPU never quite stops coming out. When the nozzle stops printing and moves, a thread stays attached where it left off and follows. Sending the nozzle the long way round, over the walls, does nothing. The thread is not on that route. It is pulled tight between its two ends.

SEEN IN TPU 85A, both brands · Mark 1 and Mark2 · right 0.6 mm nozzle Not tried in: PET-GF or PETG, which do not string like this here





WHAT WE HAD
Every move between walls was routed to stay over plastic. In the file the nozzle never crosses the bore. The thread does.

WHAT WORKS
Print through every turn. Any hop that is left is short, straight and over plastic, so a thread has nowhere to cross.



16 OCTOBER 2026 · MARK 1 · BAMBU TPU 85A
A pair of pump elbows joined by a web. Seam routing, wall-following travel, wiping laps and fins had each been tried by then. The bores still came out like this.

> *"...if you stop printing and you travel south and then you travel east, you end up drawing a diagonal thread in the southeast direction."*
>
> DEREK · LOOKING AT THAT PRINT

78 of 118
rerouted moves in that file could still pull a thread across a bore, though the nozzle stayed over plastic on every one.

0 of 6,118
in the reworked file. No retraction and no Z-hop anywhere in the part.

22 October 2026, Mark2, Siraya TPU 85A: "That worked very well. I see no strings, and seal seems as good as before."

STILL OPEN The clean print also changed the brand of TPU. The reworked path has not yet been run on the first brand.

IN THE RECORD

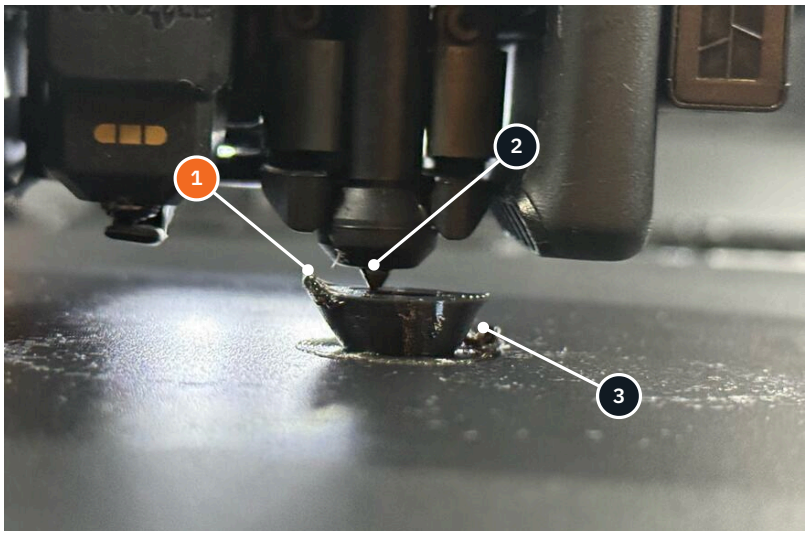
The print with the threads · Which moves could leave a thread · The reworked file · The clean print

COOLING

Less fan made it curl more.

Where a wall leans out, the newest ring of plastic has air under its outer edge. In soft TPU that edge curled upward while the fan was blowing at full. Derek halved the fan to see what would happen. Both lips curled at once.

SEEN IN Bambu TPU 85A, black · Mark 1 · glued Engineering Plate · right 0.6 mm nozzle Not tried in: Siraya 85A



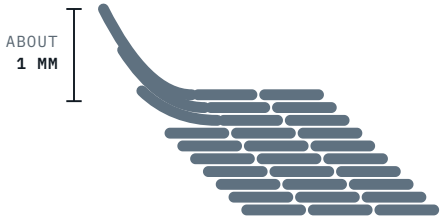
13 OCTOBER 2026 · MARK 1 · BAMBU TPU 85A
The first elbow on the Engineering Plate, about three millimetres tall: the heel of the V, widening as it rises. The nozzle is over its left lip.

What Derek watched

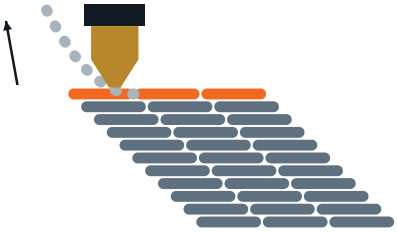
1. **The lip.** About layer 28, a millimetre of the outside edge curled upward on the left.
2. **The nozzle.** The next layers pressed the edge down. It sprang back.
3. **Loose plastic.** Around layer 31 a glob was pulled off the lip and carried elsewhere. A lump lies on the bed here.

> *"Haha, just to test, I dropped part fan from 100 to 50 just now and like instantly we got more curl on that left edge and then the right edge curled too."*
>
> DEREK · DURING THAT PRINT

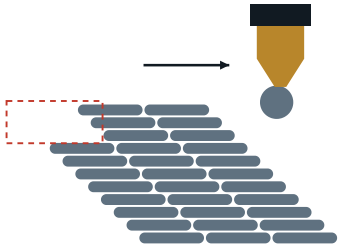
The left lip had curled with the fan at full. Halving it made that worse and brought the right lip up too.



The outer edge of the newest courses lifts.



The next pass presses it flat. It springs back behind the nozzle.



The nozzle catches the raised edge and carries a piece away.

100%
part fan in the TPU 85A recipes, on every layer after the first

0%
part fan on PET-GF's ordinary walls. It comes on, at 70%, only for a road laid over air.

Ten degrees cooler changed nothing

Two prints later the layers ran at 215 °C instead of 225 °C. Derek saw faint curl at layer 21, and at layer 27: "it is about the same as 2 prints ago. I was wrong about it being sudden at 28, I just happened to notice it there."

What we do

The part fan stays at full on soft TPU that leans. That did not stop the curl. What ended it is on the last page of this part, and the fan was at full there too.

STILL OPEN Why less fan curled it more is not known. It was one change to one running print, and the layer was not noted. It has not been tried on the Siraya spool.

IN THE RECORD

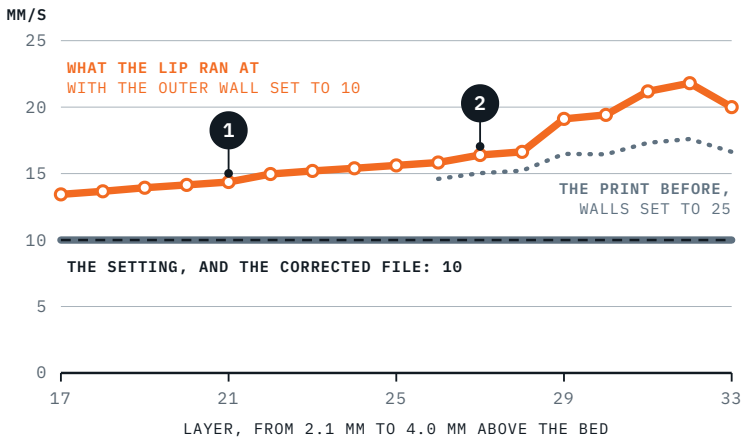
What was seen, and the photograph · Ten degrees cooler · PET-GF's cooling

SPEED

The setting said 10. The lip ran at 16.

A slicer keeps more than one speed for the same wall. Where the wall leans out, Bambu Studio takes its speed from a separate overhang table. Two trials that were meant to print a curling lip slowly printed it faster than the trial before them.

SEEN IN Bambu TPU 85A · Mark 1 · right 0.6 mm nozzle · sliced in Bambu Studio Not tried in: other plastics, whose files have the same table



THE LIP OF THE ELBOW, LAYER BY LAYER

1 Layer 21, 14.4 mm/s: faint curl already. 2 Layer 27, 16.4 mm/s: "about the same as 2 prints ago".

> *"Maybe we can try slowing down that outer edge more?"*
>
> DEREK · 13 OCTOBER 2026, AFTER THE FIRST CURL

So the outer wall went from 25 mm/s to 10 and the elbow was printed again. The edges looked different for a moment, then the same.

> *"Oh, maybe it is curling. I guess I'm not interested in a midpoint when the difference in those numbers is 10 13 17 so small and the difference I'm observing is maybe non existent upon reflection."*
>
> DEREK · LOOKING AT THE SECOND PRINT

He was right. The file had not slowed the lip at all.

13 OCTOBER 2026	OUTER WALL SET TO	THE LIP RAN AT
21:09 · first elbow	25 mm/s	**14.6 to 17.6**
21:45 · "slower"	10 mm/s	**15.8 to 21.8**
23:05 · slower and cooler	10 mm/s	**15.8 to 21.8**
The corrected file	10 mm/s	**10.0**

Millimetres a second at the lip on layers 26 to 33, read from each file.

OVERHANG TABLE	STOCK	ELBOW RECIPE
Second band	50 mm/s	**10 mm/s**
Third band	30 mm/s	**10 mm/s**
Outer wall	200 mm/s	**10 mm/s**

Why

The outer wall speed covers beads that sit squarely on the layer below. A bead that hangs partly over air takes its speed from the overhang table, and two bands of that table still held stock values. The slicer blended them with the wall speed, then stretched each layer to its minimum time of 20 seconds. With the plain walls slower, that slowdown was shared out differently, and the lip came out faster.

The fix

Set those bands to the wall speed, then read the file. In the corrected file all 13,728 outer-wall moves from layer 2 to the top run at 10.0 mm/s.

Before you print

Do not trust the speed box. Pick the feature that matters and read the speed the sliced file commands there, layer by layer. In this repository that is `hsm-gcode speeds FILE --layers 17-33 --at leftmost`.

STILL OPEN Whether a true 10 mm/s cures the curl in Bambu 85A is not known. The corrected Bambu file was never sent as a single elbow. The first lips printed at a real 10 were a different brand, an hour later.

IN THE RECORD

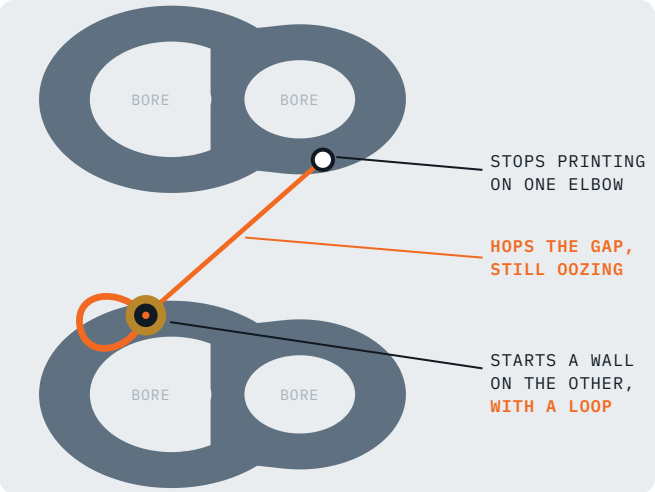
Set against emitted, 21:09 and 21:45 · The 23:05 file · The corrected file · The reader

STOPS AND STARTS

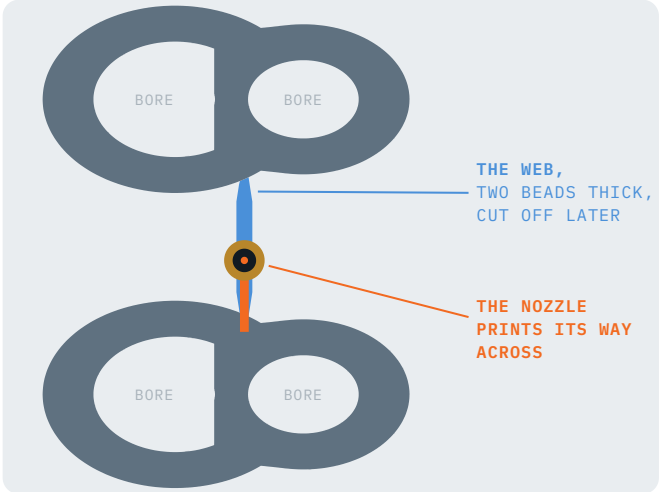
One island at a time.

Each time the nozzle stops laying TPU and starts again elsewhere, a little leaks on the way and a little too much arrives. Two objects on a bed mean that on every layer.

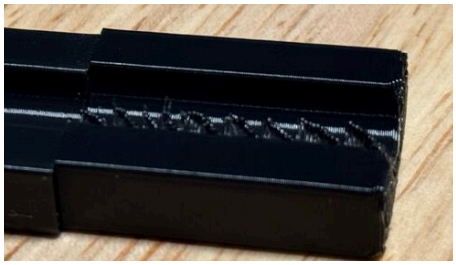
SEEN IN Bambu and Siraya TPU 85A · Mark 1 and Mark2 · right 0.6 mm nozzle Not tried in: 90A, 95A



14 OCTOBER 2026 · TWO ELBOWS, TWO ISLANDS
The bed from above, ten millimetres up. The file made 367 hops between the two elbows in 184 layers.



16 OCTOBER ON · THE SAME TWO, ONE ISLAND
A web drawn between them makes every layer one piece, so the nozzle can cross on plastic it is laying.

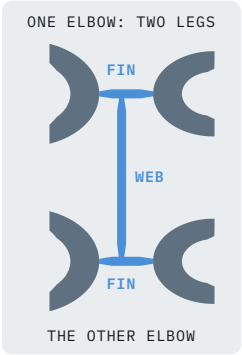


12 OCTOBER 2026 · BAMBU TPU 85A
Two liners printed two millimetres apart. Derek watched a loop form where a wall started, just after the nozzle arrived from the other liner.

> *"With both ASA Aero and TPU I am finding that the distance between objects increases risk of problems on these "unclean start/stop" filaments."*
>
> DEREK · 12 OCTOBER 2026

That day two liners printed 32 mm apart, and one fell over. The next evening two elbows on one plate failed: "you're gonna make life much easier on both of us if you only try to print one at a time."

A timelapse is a hop too. With it on, the head left for the camera after every layer, 184 times in the pair's file. It is off for TPU.



EIGHTEEN MILLIMETRES UP
Each elbow has forked. Two fins and the web keep it one island.

What we do

One TPU object on a plate, or neighbours two millimetres apart. A pair is drawn as one piece, joined by fins. The fins are cut off with a blade: soft TPU stretches instead of snapping.

> *"Why didn't you make the fin between the two objects a vertical fin at the center point of them?"*
>
> DEREK · 16 OCTOBER 2026

The first web had followed one leg. Joining alone was not enough: that pair still came out full of threads. The strings page says why.

STILL OPEN How much the distance matters has not been measured: the liner retry also took off the Z trim. The two-island elbow print of 14 October has no result on record.

IN THE RECORD

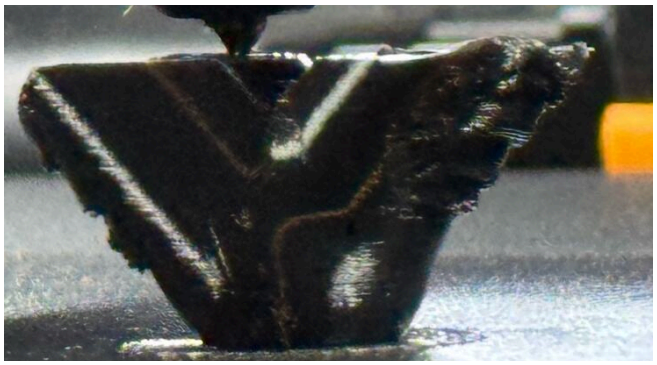
The fallen liner · The loops · Two islands, 367 hops · The joined pair · Hops counted

ORIENTATION

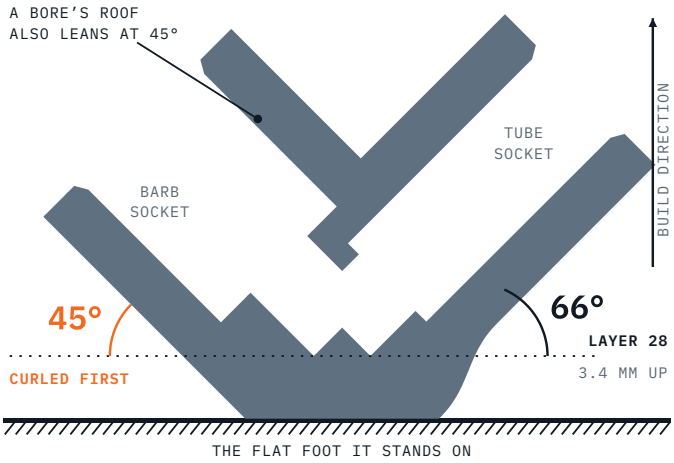
Every underside at 45 degrees or steeper.

The elbow prints standing on its corner like a V, both mouths up. Stood that way, no face looks down at the bed more shallowly than 45 degrees and no bore needs support. Its two sides are still not alike.

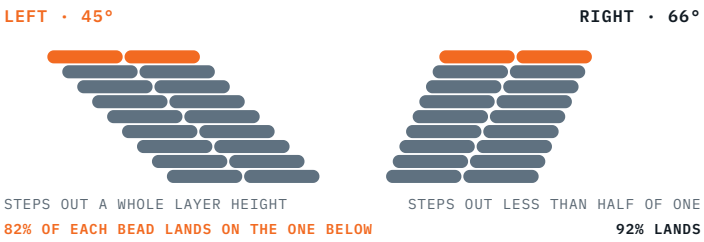
SEEN IN Bambu TPU 85A, black · Mark 1 · glued Engineering Plate · right 0.6 mm nozzle Not tried in: other TPUs



13 OCTOBER 2026 · THE ELBOW ON THE BED
The large socket is on the left. Its outer face leans at 45° from the bed up, and is ragged from the bed up. The right side is smooth low down, where it is steeper, and ragged above.



THE SAME ELBOW CUT DOWN ITS MIDDLE
Both legs lean at 45°. The outsides do not: the tube socket begins with a steeper neck.

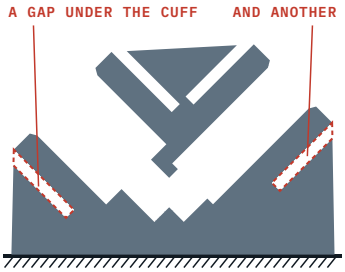


THE TWO OUTSIDE EDGES AT LAYER 28, AT TRUE PROPORTIONS
Read from the slice of the first elbow, 13 October 2026.

> *"Oh, the left edge is the one that gets the bigger tube. Sure you got the same gradual angle on the underside on left and right?"*
>
> DEREK · 13 OCTOBER 2026

The curl was first noticed at layer 28, on the left. Both legs lean the same. Their outsides, at that height, do not. It was Derek who thought to compare them.

Three days earlier, asking for another TPU sleeve: "For printability, I'd like steeper than 45 degrees please."



13 OCTOBER 2026 · STOPPED

Added stock has to be solid to the bed

> *"You added material both above and below, and you left a gap that makes the below useless"*
>
> DEREK · ON THAT ELBOW

Plastic had been added under the legs to hold them up, with a channel left under each cuff. A block that does not touch the thing above it holds nothing up. The elbow that prints today has no added stock. It stands on a flat cut across its heel.

STILL OPEN The shallower side curled first, which fits and proves little: the two sides also differ in size and wall. No TPU part has been printed with every underside steeper than 45° to see if its edges come out clean.

IN THE RECORD

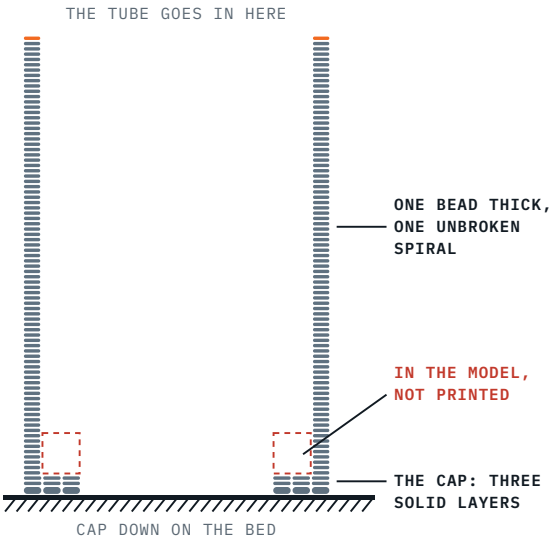
Left against right · The ragged elbow · The stopped one · The elbow today · Steeper than 45

FIRST LAYER

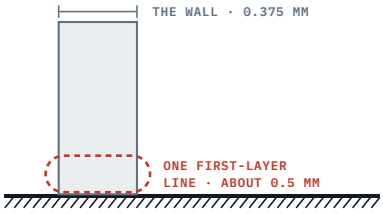
A thin soft wall starts on a cap.

The sleeve that seals the faucet’s tube is a cylinder of soft plastic one bead thick. A wall that thin has almost nothing to stand on, and the first version could not be sliced at all. It now begins with a flat cap, laid first, and the wall rises from that.

SEEN IN Bambu TPU 95A in May 2026, 85A since · the sleeve of 23 October 2026: Mark2, glued Engineering Plate, right 0.6 mm nozzle Not tried in: Siraya 85A



23 OCTOBER 2026 · MARK2 · BAMBU TPU 85A, BLACK
The sleeve as that file printed it, at its true proportions. Derek: “Okay, that came out pretty well.”



22 May 2026: refused

The first sleeve was an open ring with a wall 0.375 mm thick. The slicer would not start it: “empty initial layer”. One first-layer line was wider than the wall.

24 May: cap down

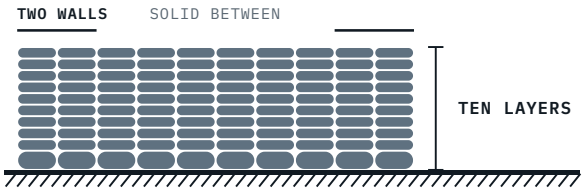
A floor with a hole in it was added to one end and the part was turned to print on it. The first layer is now a flat ring several beads across. It sliced without a raft and printed without trouble, in 95A.

30 May: one bead, one spiral

The wall was thickened to one bead of the 0.62 mm profile and the softer 85A went in. After the cap the nozzle climbs in a single unbroken spiral. A spiral never stops, so the wall cannot string. The cap can: its flat layers had five places where a thread could cross the hole, until the 23 October file laid each of them as one tour.

Drawn thick, printed thin

In spiral mode only the bottom layers are solid. This recipe has 3, which makes a cap 0.66 mm thick. The model draws it 2.1 mm thick. Read the cap from the slice, not from the drawing.



A FLAT GASKET IN SECTION
First layer 0.3 mm, then 0.18 mm layers. Ten of them make 1.92 mm of a gasket drawn 2 mm thick.

Flat gaskets print solid

A gasket is squeezed between two faces, so it is laid flat and filled: 2 walls round every edge and 100% fill between, in Bambu 85A on the stock profile. In June 2026: “gaskets are going well.” The plate of eight printed on Mark2 on 8 October “turned out fine”.

> *“...consider carefully if there’s any lessons we’ve learned from our avoiding threads other places that might apply here...”*
>
> DEREK · 23 OCTOBER 2026, ASKING FOR THE SLEEVE

STILL OPEN “Pretty well” is all that is recorded of the 23 October sleeve. Whether it strings less, how it seats, and whether a cap a third as thick as drawn seals as intended are not written down. Nor is the result of the gasket printed the same evening with every layer as one tour.

IN THE RECORD

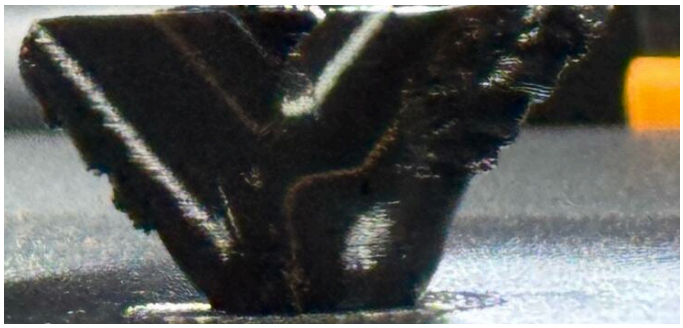
The sleeve’s print log, May to October · The 23 October sleeve · Its cap, read from the file · Gaskets print log · The plate of eight · The gasket of 23 October

TWO BRANDS

Two brands, one hardness.

Two spools both say 85A. One curled its lips on three elbows in an evening. The other printed them clean at its first try, an hour after the last. Six things changed between those two prints, and nobody has gone back to find out which one mattered.

SEEN IN Bambu TPU 85A and Siraya TPU 85A · Mark 1 · glued Engineering Plate, no Z trim · right 0.6 mm nozzle



13 OCTOBER 2026, 21:45 · BAMBU TPU 85A

Rough at every 45° edge. It still fitted: “nice firm friction stretchy fit”.



14 OCTOBER 2026, 00:11 · SIRAYA TPU 85A

“Turned out perfect.” Then, of the inside: “I do see some stringing inside as well.”

	13 OCTOBER 21:09	21:45	23:05	14 OCTOBER 00:11
Spool	Bambu TPU 85A, black	Bambu TPU 85A, black	Bambu TPU 85A, black	**Siraya Tech Flex TPU 85A, clear**
Opened	earlier	earlier	earlier	**that night**
Nozzle after the first layer	225 °C	225 °C	**215 °C**	215 °C
Lip speed that ran, mm/s	14.6 to 17.6	15.8 to 21.8	15.8 to 21.8	**10.0**
Flow ratio	1.00	1.00	1.00	**0.96**
Auxiliary fan	100%	100%	100%	**off**
Retraction	1.0 mm	1.0 mm	1.0 mm	**0.8 mm**
The lips	curled	rough	curled the same	**clean**

Bold marks a change from the print before. The same model, printer, nozzle, plate, layers, six walls and full part fan throughout.

> *“You sure you don’t want to try 215? That is within their range and we’re already lowering the speed from their profile.”*
>
> DEREK · LOADING THE NEW SPOOL

What is known

Heat alone is not it: the third black elbow ran at the same 215 °C and curled. The clear recipe has since printed the pair that went on the pump, on Mark2 on 22 October. Black Bambu 85A printed the faucet sleeve on 23 October, which has no leaning lip, and it “came out pretty well”.

What we do

Elbows print in the clear Siraya on its whole recipe. Sleeves and gaskets print in black Bambu. Neither choice is a verdict on a brand.

STILL OPEN **Which change ended the curl.** The print that would say is one black Bambu elbow, alone, with its lip checked at 10 mm/s. That file was prepared on 13 October and never sent. Black pairs printed at a checked 10 on 15 and 16 October, for the string trials. Nobody judged their lips.

STILL OPEN **Water in the spool.** The black spool went into the dryer after its three curled prints, and whether the cycle finished is not recorded. The clear one came out of a sealed bag.

STILL OPEN **The seal.** “Seal seems as good as before” names no pressure and no time. The elbows have not been held at pressure and watched.

IN THE RECORD

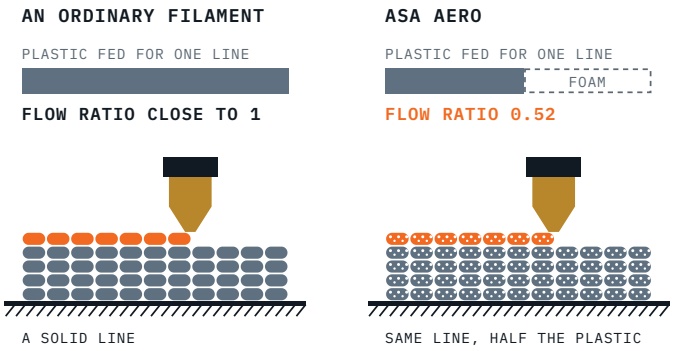
The clear elbow, and what changed · Its recipe and where it came from · The rough black elbow · The black spool in the dryer · The pair on Mark2 · The pressure test still to run

THE PLASTIC

It prints lighter than it feeds.

ASA Aero foams when it is heated. It swells in the nozzle, so the printer feeds about half the plastic of an ordinary line and the foam makes up the rest.

SEEN IN Bambu ASA Aero · Mark2 · glued Engineering plate · right 0.4 mm hardened nozzle Not tried in: Mark 1



HALF THE PLASTIC, THE SAME LINE

The slicer works out the plastic for a line and multiplies it by the flow ratio. The spool weighs 0.99 grams per cubic centimetre, about the same as water. What comes out of the nozzle weighs less. How much less depends on heat, speed and flow, and we have not measured it.

The recipe

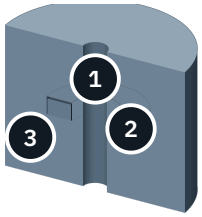
Read from the float's project file as this book is bound. The filament settings are the maker's stock profile for this printer. The walls are ours (see the third page).

Nozzle, right side	**hardened, 0.4 mm, 270 °C**
Plate	**Engineering Plate, with glue, 90 °C**
Chamber	**60 °C, door shut**
Flow ratio	**0.52**
Layer, line	**0.2 mm high, 0.48 mm wide**
Wall loops	**300**
Fill, top skin, bottom skin	**0%, 0, 0**
Part fan	**30 to 50 %**
Most plastic a second	**12 mm³**
First-layer trim on Mark2	**+0.04 mm**

Why the floats are foam

A level float rides in the carbonator and in each flavor reservoir, and it has to float while carrying a magnet. Solid PETG and PET-GF both sink. The float was first drawn as a foam core sealed inside a clear PETG shell, and the first night's prints were that core. The next day it became foam and nothing else.

> *"...starting simpler seems better to me, generally speaking."*
>
> DEREK · 3 OCTOBER 2026



1. **The bore.** It slides on a rod.
2. **A ring magnet,** read through the tank wall.
3. **Foam** above and below, in one piece.

What it needed, found in one night

Mark2, from 22:20 on 2 October 2026. Seven starts in about three hours; the seventh, which Derek started at the printer, ran to the end.



1. **A smooth plate with glue.** Five starts on textured PEI, the plate PET-GF prints on. Of the first: "It did, eventually, adhere to the bed, after more layers were added, but then it peeled off *very* easily." Before the sixth: "Engineering plate is glued and loaded now. Textured plate is a no go."
2. **A head start.** The printer's own short start line "does not adhere to the bed at all. It just raises right up," and was dragged into the part. From the fourth start on, a strip of 12 passes in the front corner prints first.
3. **A dry box.** Dried at 80 °C for 8 hours and kept under 20 % humidity, the maker's figures. The first six starts fed from the AMS HT. Of the seventh: "...we are now using the polymaker drybox and that is working better."

STILL OPEN The strip, the plate and the dry box arrived on three starts in a row, and only the last of them ran to the end, so the record cannot say what each one did. Nothing about the finished foam has been measured. The float is sized on an assumed 0.65 g/cm³, for 5.0 g of spare lift with its 5.09 g magnet aboard. None has been weighed, floated or soaked on the record.

IN THE RECORD

The float's sums · The maker's figures · The first start · The start line · The plate · The print that finished · Drying

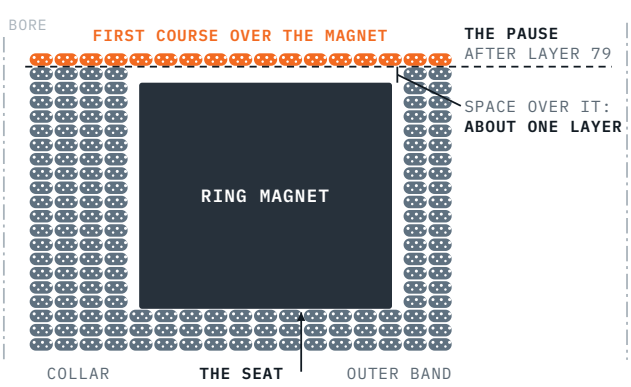
AERO

PRINT AROUND A MAGNET

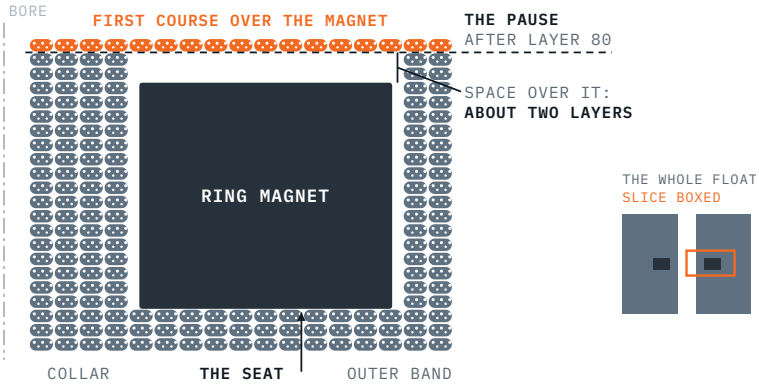
Pause. Seat the magnet. Print on.

The level float has a ring magnet closed inside it, with no lid and no seam. The printer builds the float up to the top of an open pocket and stops. The magnet is pushed in by hand. The printer starts again and lays the rest of the float over it.

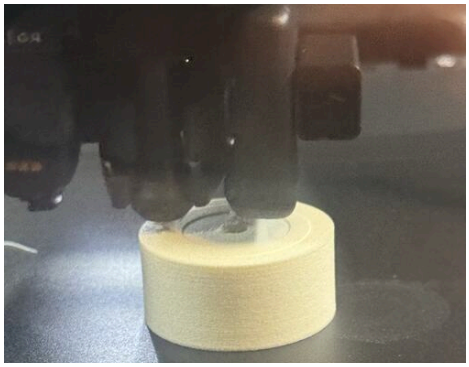
SEEN IN Bambu ASA Aero · Mark2 · glued Engineering plate · right 0.4 mm hardened nozzle Not tried in: Mark 1



3 OCTOBER 2026 · AS IT PRINTED
The pocket walls stopped about one layer above the top of the magnet, so the first course across had almost no room under it.



THE FLOAT AS IT IS DRAWN NOW
The pocket is one layer deeper, 3.4 mm then and 3.6 mm now, and the pause comes one layer later.



3 OCTOBER 2026 · MARK2 · ASA AERO
The float under the toolhead while foam goes on over the magnet. The grey ring is the magnet, part covered.

WHAT THE PRINTER SAYS AT THE PAUSE

Insert one RC62 ring into the open annular pocket, fully down onto its floor. The magnet must sit below both pocket rims. Keep the guide bore open and clear loose strings without moving the float. Resume only with the magnet fully seated.

> *"It worked. Maybe one more layer of space needed above the magnet for next time, as the first layer on the magnet was a bit too tight, but the second layer went down beautifully."*
>
> DEREK · 3 OCTOBER 2026

What happened

The float went to Mark2 at about three in the afternoon. After layer 79 of 140 the printer parked and waited. Derek pushed one ring into the pocket and pressed resume. The pocket was drawn one layer deeper the same day.

Push in, not drop in

The nozzle is steel and passes close over the magnet. Before the print: "...I think it might just work, especially because the magnet actually sits and grips quite well in the ASA Aero." It did. The next day the same pause in PET-GF, in a pump cartridge, left a magnet that "rattles": it had been "too easy to "drop the magnet in" instead of "push the magnet in" as had happened with the aero that worked even better."

STILL OPEN The deeper pocket went onto a plate of three floats on 8 October 2026, and nobody wrote down how it came out. No finished float has been cut open or held under water. The magnet's maker rates it for 80 °C in steady use, and foam at 270 °C is laid straight onto it. What that costs the magnet has not been measured. What is on record: that evening the float worked a reed switch from 20 mm away.

IN THE RECORD

Derek's report and the photograph · The slice that printed · The slice as it is now · The three-float plate · The PET-GF magnet that rattled · Reed reach

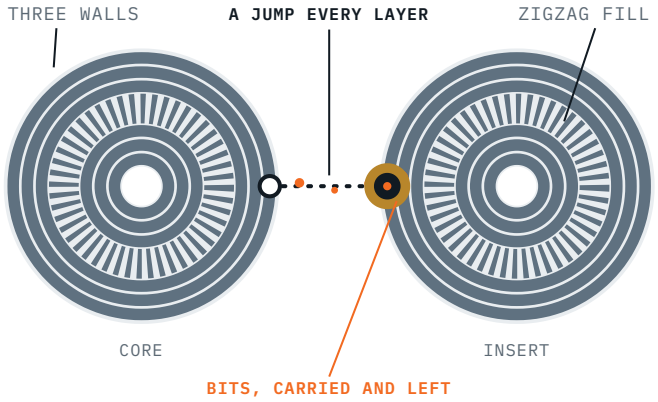
AERO

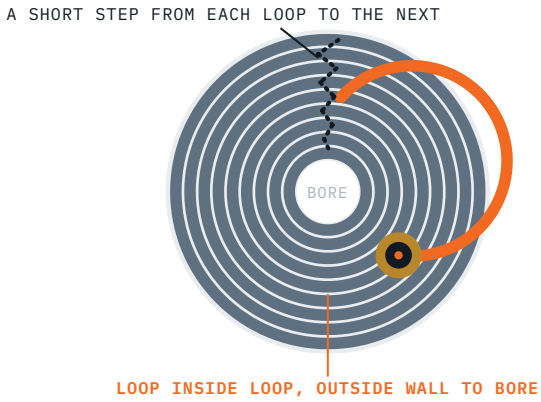
TRAVEL

Every jump carries a bit along.

When the nozzle moves without printing, ASA Aero keeps coming out of it. A bit goes along on every move and is left wherever the nozzle lands. So the float is printed with as few of those moves as its shape allows.

SEEN IN Bambu ASA Aero · Mark2 · glued Engineering plate · right 0.4 mm hardened nozzle Not tried in: Mark 1





THE FIRST NIGHT · 2 AND 3 OCTOBER 2026
The plates that were sent: two parts side by side, each three walls around solid zigzag fill, with a jump between them on every layer. The one print that ran to the end came off with flaking.

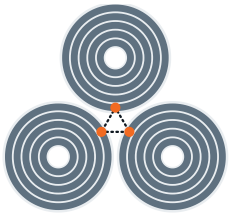
WHAT DEREK ASKED FOR
One part on the plate, and nothing in it but walls. The walls are set to a number too big to reach, so the slicer keeps nesting loops until the part is full.

> *"The one more issue is that every time the nozzle moves and is *not* printing the filament continues to extrude just because of its expansion when heating, and so we end up with bits carried around on each "jump"."*
>
> DEREK · 3 OCTOBER 2026, WATCHING THE FIRST PRINT THAT RAN

> *"So I'm thinking we want to try one piece at a time, and with that piece, just do like 300 walls so that it is all walls and we get concentric the whole way, and no jumps."*
>
> DEREK · THE SAME NIGHT

1605
stops and restarts were still in the all-walls float that printed on 3 October, across 140 layers.

10 mm
was the longest of those moves inside any one layer. All walls is fewer and shorter jumps. It is not none.



Close together

On 8 October 2026 three floats went on one plate, 2 mm apart. The orange dot is where each layer of a float starts and stops. All three face the middle.

Derek asked for it that way: "I'd like the floats to be close together, so that the distance between objects is small." Four days later, printing soft TPU:

> *"With both ASA Aero and TPU I am finding that the distance between objects increases risk of problems on these "unclean start/stop" filaments."*
>
> DEREK · 12 OCTOBER 2026

STILL OPEN Whether the all-walls float came off clean of bits was not written down; what Derek reported of it was the magnet. The plate of three has no written result. His reading that the foam keeps swelling in the heat has not been tested. Nor has his other idea from that night, "maybe we just need to reduce the volume a bit": the flow ratio is still the maker's 0.52.

IN THE RECORD

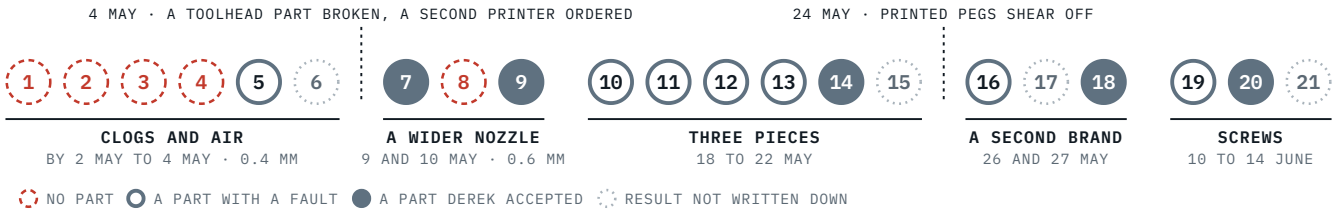
What Derek saw on the first print that ran · The all-walls trial and what travel was left · The float's slice, move by move · Three floats close together · Keeping parts close

SPRING 2026 · THE FAUCET

Twenty-one tries at one faucet.

PET-CF is PET with chopped carbon fibre in it: stiff, matte and hard on nozzles. This is the print log of its first job here, try by try, in Derek's words.

SEEN IN Bambu PET-CF (tries 1 to 15), Polymaker Fiberon PET-CF17 (16 to 21) · H2C · left nozzle · textured PEI



Clogs and air

TRIES 1 TO 6
BY 2 MAY TO 4 MAY
0.4 MM NOZZLES

Two clogs above the hot end. A calibration that failed "over and over". A print that "started printing air" near layer 15 with nothing blocked. The fifth finished, full of gaps. For the sixth the spool was dried twelve more hours and everything was slowed down, and how it came out was never written. On 4 May, clearing a clog with a wire pushed up from below snapped a pincer in the right-hand heating assembly of the toolhead and cut its cable. The spare was sold out. A second printer was ordered that night, and that is why there are two.

A wider nozzle

TRIES 7 TO 9
9 AND 10 MAY
0.6 MM TUNGSTEN CARBIDE

An off-brand 0.6 mm nozzle arrived on 9 May. With it the stock profile went back on and the supports were printed in PET-CF too. "The most recent PET-CF print worked beautifully." Try eight, the shell in one piece, "failed in an interesting way, the support tower fell over" and fused into the top of the faucet. Try nine gave the tower a wide first layer and rooted every support on the bed. "The last print is beautiful! Actual success, and I was able to remove all the supports even those from inside of the tube."

Three pieces

TRIES 10 TO 15
18 TO 22 MAY
TWO PUSH-FIT JOINTS

The gooseneck was cut into three pieces. Drawn with no clearance, the straight joint stopped "just 2 mm short" and the curved one jammed a third of the way in. With half a millimetre and more: "it definitely made a difference, but far too much of one, lol." Three more prints closed in on five hundredths of a millimetre a side. "Pretty good. Or good enough at least for now."

A second brand

TRIES 16 TO 18
26 AND 27 MAY
POLYMAKER FOR BAMBU

The Bambu spool ran out and Polymaker's PET-CF went on with nothing else changed. A tall support came apart and the outer walls had small bare patches. Ten hours in the dryer changed nothing. A hotter nozzle with the fan off, then the first diamond-tipped nozzle and a flow calibration: "They do look better. I still see some, but far less noticeable…" The new brand also printed glossy where the old one was matte.

Screws, a slower top

TRIES 19 TO 21
10 TO 14 JUNE
THE SHELL REDRAWN

The plate under the faucet got three screws (next page) and the tip got a cradle for the display. Everything came out too tight. Try twenty opened the fits: "things fit as well as I need them to for now." One fault was left, strands drooping at the top of the piece that prints standing on end. Try twenty-one slows its last few millimetres to 40 % speed and 20 % acceleration, with the fan on.



10 MAY 2026 · BAMBU PET-CF · 0.6 MM NOZZLE
Try nine on the valve it covers: the first whole faucet shell. The rough patch at the far end is where supports stood. The frame is from the video posted that day, with its title cropped off the top.

> *"the direction that things are falling are actually the direction they are being pulled, and it is the force of the movement that is pulling things apart. So my thought is 'less velocity' => 'less force'."*
>
> DEREK · 14 JUNE 2026, ON TRY TWENTY-ONE

STILL OPEN Try twenty-one was sliced on 14 June and its result was never written down, so that thought is untested on the record. Try seven changed the nozzle, the profile and the support plastic together, and which of them ended the clogs was never separated.

IN THE RECORD

The print log · Why a second printer · Clearing a clog without breaking the toolhead · The last PET-CF project · The video post

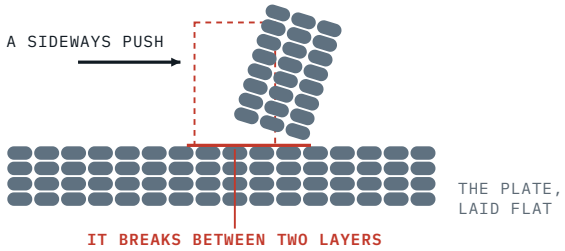
WHAT IT LEFT BEHIND

A plastic's lessons stay with it.

From May to August 2026 the faucet was PET-CF. Then the machine moved to PET-GF, glass fibre in place of carbon. What PET-CF had taught about layers and about the printer went along. What it had taught about itself did not.

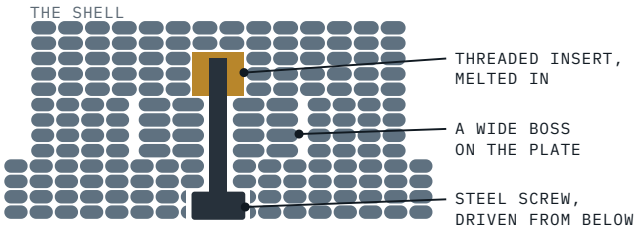
SEEN IN PET-CF, May and June 2026 · PET-GF from 15 August 2026 · left nozzle of an H2C · textured PEI

A PEG PRINTED WITH THE PLATE



23 TO 25 MAY 2026 · PET-CF
To save two screws the plate was given two printed pegs. One snapped off as the plate went in. Thinner ones snapped too. "they still don't fit, and they snap so easy because of the orientation of the layer lines. This is just not workable."

A SCREW THROUGH A WIDE BOSS



10 JUNE 2026 · PET-CF, AND PET-GF TODAY
Three wide bosses, each with a steel screw through it. The faucet is still held to its plate this way. This lesson was about layers, and it moved to PET-GF with nothing to change.

Stayed with PET-CF

Its settings. PET-GF's profile began as a copy of PET-CF's, so the first PET-GF print ran on PET-CF's plate, chamber and flow limit. Ten days later the faucet's copy was checked against the maker's sheet. The plate and the chamber were both hotter than PET-GF is meant to have, and they came down. One thing never changed: the name.

	PET-CF 14 JUNE 2026	PET-GF 15 AUGUST 2026	PET-GF NOW
Nozzle heat	**280 °C**	**290 °C**	**280 °C**
Plate	**100 °C**	**100 °C**	**80 °C**
Chamber heater	**50 °C**	**50 °C**	**0 °C**
Part fan	**0 %**	**0 %**	**0 %, 70 % on overhangs**
Most plastic a second	**5 mm³**	**5 mm³**	**18 mm³**
The slicer calls it	**PET-CF**	**PET-CF**	**PET-CF**

Its fits. The push-fit joints settled at five hundredths of a millimetre a side on 21 May. On 10 June, in the second brand, the same joints were too tight and went to a quarter of a millimetre on the diameter. PET-GF's fits were found again from nothing.

Carried on paper, and wrong

"No snaps." On 21 May the snap fits came off the pump case, on the reasoning that PET-CF is too stiff and brittle to flex. No snap that broke is on the record. On 25 August the same reasoning was written down as a rule for PET-GF: a catch goes home "never by deflecting". On 16 September the rule was deleted and the faucet's display cover got two sprung wings. The next day: "They are tough though, and do spring back."

> *"...those broad long thin (but not too thin!) walls provide the necessary give and spring, more so than anything else anyone else has tried, in PET-GF so far."*
>
> DEREK · 20 SEPTEMBER 2026

Went along to PET-GF

The left nozzle, and a hard one. Carbon and glass both wear a nozzle away. Both print from the left hotend, the one that takes tungsten carbide and diamond tips.

Hands off a clog from below. The toolhead comes apart from the top, in order. Learning that broke one and bought the second printer.

A second brand is a second plastic. On 17 May the choice was written up as PET-CF from any maker. Nine days later the second maker's spool failed on the first one's settings.

The dryer, and its limit. Both dry in the same dryer. Drying did not fix what was not damp.

15 August 2026. The first PET-GF print was the top front quarter of the enclosure, and on 24 August the faucet's material was changed to match. About six hours into that first print Derek wrote that it was "working a treat so far" and called what had printed "very beautiful wall surface."

STILL OPEN No PET-CF snap was ever printed and broken on the record, so "too brittle to snap" is still a reasoning, not a result. PET-GF's snaps have been judged in the hand. None has a measured force or a count of cycles.

IN THE RECORD

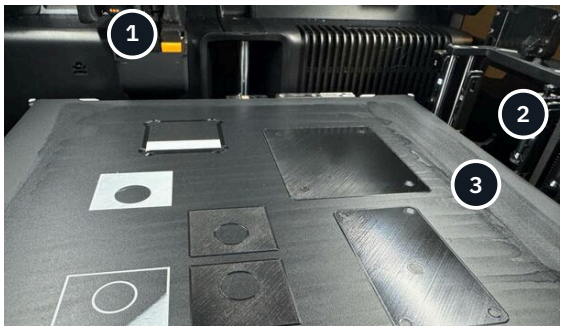
The pegs, in the print log · Snaps off the pump case · The rule written for PET-GF · The rule deleted · The cover that springs · The first PET-GF print · PET-CF's plate and chamber come down · Which nozzle goes where

THE MACHINES

Each plastic has its side, plate and feed.

Both printers are Bambu Lab H2Cs, with two hotends each. A print settled which side, plate and feed every plastic gets.

SEEN IN Mark 1 and Mark2 · every plastic the machine is made of · each row of the table is one dated print



6 NOVEMBER 2026 · MARK 1 · PETG
Inside Mark 1, just after a five-colour print was stopped in its first layers.

1. **The left hotend is fixed.** It is the hotend of Bambu's H2D, so nozzles sold for that printer fit it: hardened steel, tungsten carbide, diamond.
2. **The right hotend swaps.** It is heated by induction and gripped by pincers, and the printer trades it for another from a rack during a print. Only hotends made for the H2C fit. Here spares marked 0.2 wait in the rack.
3. **The plate is changed for the plastic.** This one is the smooth Engineering Plate, spread with glue. The other kind is textured PEI.

Mark 1 was ordered on 22 March 2026. It was called H2C until 6 October and still answers to that name on the network. **Mark2** was ordered on 4 May. Neither stays loaded one way: plates, nozzles and spools are changed for the job, so every job's record names its own.

Which plastic goes where

PLASTIC · ITS PRINT	SIDE AND NOZZLE	PLATE	FED FROM	DRIED IN
PET-GF Front-bottom · Mark2 · 11 November 2026	**Left, 0.4 mm**, hardened. A second colour: right 0.4 mm.	Textured PEI Plate	The left outside spool: 1 kg straight out of the dryer, 3 kg out of a sealed box with an axle.	SUNLU E2, only if the spool has taken up moisture.
PETG CLEAR Reservoirs · Mark 1 · 26 October 2026	**Left, 0.8 mm.**	Engineering Plate, glued	An AMS 2 Pro, which carries on to its next spool when one runs out.	The same AMS 2 Pro, spool in place.
PETG COLOURS Labels · Mark 1 · 8 November 2026	**0.2 mm, both sides.** Black left; four colours on four right hotends.	Engineering Plate, glued	Black from the AMS HT, the other four from an AMS 2 Pro.	Those two units.
TPU Pump elbows · Mark2 · 22 October 2026	**Right, 0.6 mm.**	Engineering Plate, glued	The right outside spool, in a sealed box it turns freely in. No AMS feeds TPU.	AMS HT.
ASA AERO Level float · Mark2 · 3 October 2026	**Right, 0.4 mm**, hardened.	Engineering Plate, glued	The right outside spool, from a sealed drybox.	SUNLU E2.
PET-CF Faucet, since retired · May and June 2026	**Left, 0.6 mm** tungsten carbide, later diamond.	Textured plate	The SUNLU E2.	SUNLU E2.

Fibre goes left, mostly. Glass and carbon fibre wear a nozzle away. Tungsten carbide and diamond nozzles stand up to them and are made only for the left side, so the ledger's rule is that abrasive plastic runs there. The record also holds PET-GF through the right side's hardened steel: a whole white faucet on Mark2 on 15 October 2026, and the second colour of every label.

STILL OPEN Nothing in the record measures the wear on a right-side nozzle after PET-GF.

> *"So far, the diamond PCD has been the toughest and most reliable, in other sizes."*
>
> DEREK · 11 SEPTEMBER 2026

IN THE RECORD

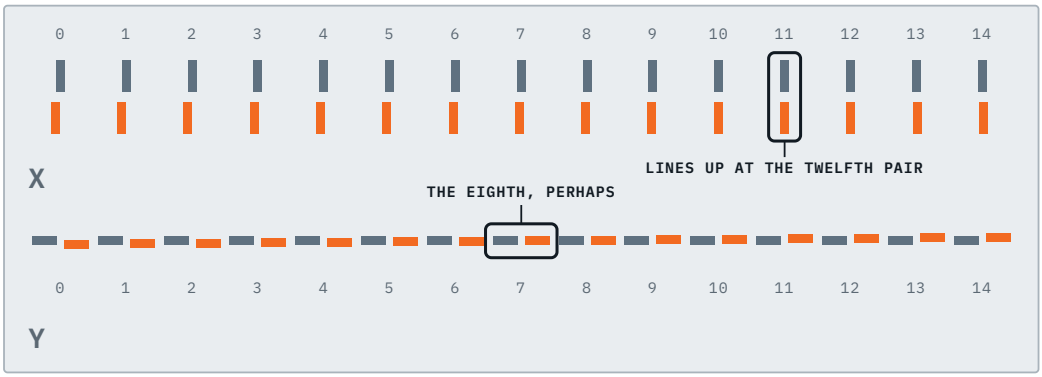
What fits which side, what dries where · Every piece, its printer and hotend · The white faucet on the right

TWO NOZZLES ON ONE PART

Measure where the second nozzle lands.

Two nozzles sharing a part must lay their lines side by side. The printer measures the gap between its own nozzles before every job. On Mark2 that still left white lettering out of its black pockets.

SEEN IN PET-GF · Mark2 · textured PEI · both 0.4 mm · 29 September 2026 Not tried in: Mark 1, or any other nozzle pair

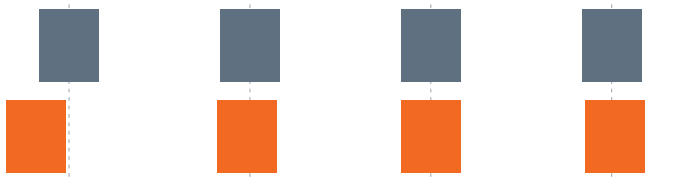


-0.5 mm
to move the right nozzle's paths along the bed, left to right.

+0.7 mm
to move them front to back.

Thirteen minutes and under six grams of plastic to find out.

THE COUPON, DRAWN AS IT PRINTED
Slate is the left nozzle, laid where it was drawn. **Orange** is the right nozzle. No photograph of the printed coupon was kept: this is its geometry, with the right nozzle moved by the error Derek read off it.



How it is read

Each pair is drawn end to end, with the right nozzle's line 0.05 mm further over at every pair along the row. The pair that prints as one straight line names the shift. The lines are 0.9 mm wide, so neighbours differ by an eighteenth of a line.

The first coupon ran out of range: its best pairs were the last in each row. A second carried the range further. It numbers its pairs from 0 and Derek counted from 1; his "fourth from the right" settled which he meant.



4 OCTOBER 2026 · MARK2 · PET-GF
Two-colour pieces printed with the correction. Of the CO2 and SODA rings, printed that day, the record says the white letters sit in their recesses with no visible offset.

Where the correction goes

Into the slice, never the drawing. The slicer is told the right nozzle sits that far from where the printer believes, and it moves every right-nozzle path to suit. The model stays as drawn. Each corrected file is compared, path by path, with an uncorrected slice of itself.

What it belongs to

One printer and one pair of hotends. Change a hotend, or how the printer calibrates, and the coupon is printed again.

> *"The X-Y alignment between the two nozzles seems really good."*
>
> DEREK · 8 NOVEMBER 2026 · MARK 1, PETG, 0.2 MM, NO COUPON

STILL OPEN The front-to-back reading is tentative: slight oozing made neighbouring pairs hard to tell apart. No print has measured the error left after the correction.

IN THE RECORD

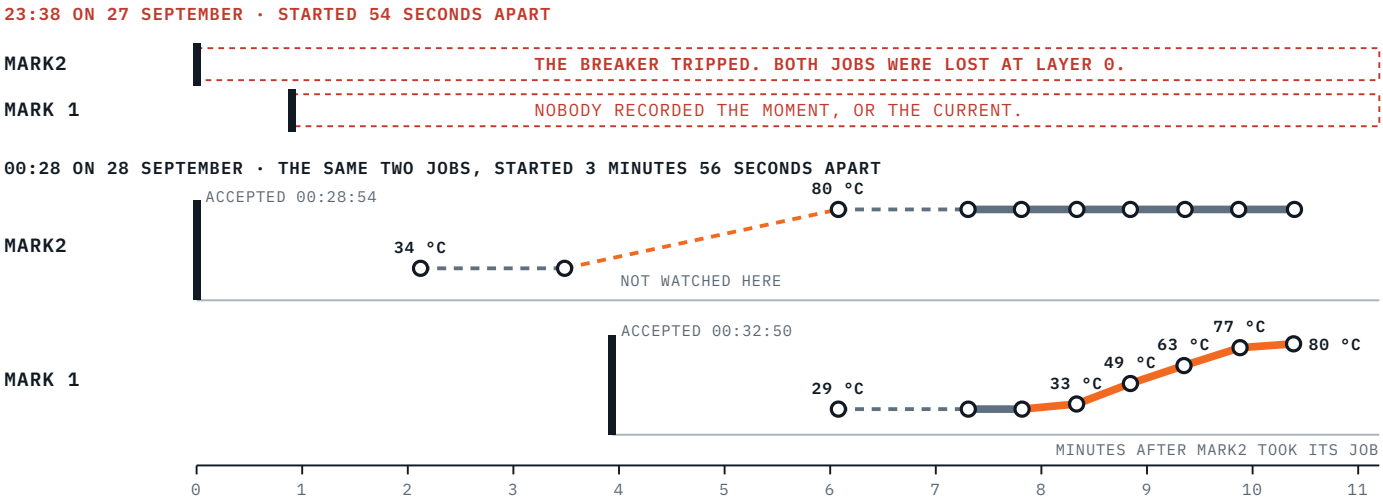
The coupon and how to read it · The reading · The first coupon · The nameplate that showed it

POWER

Start the second printer three minutes after the first.

Both printers plug into one household circuit. Started within a minute of each other, they tripped its breaker and both jobs were lost. Started four minutes apart, the same two jobs ran to the end.

SEEN IN Mark 1 and Mark2 on one circuit · both printing black PET-GF from the left 0.4 mm · the night of 27 September 2026 Not tried in: any gap between one minute and four



WHAT THE PRINTERS REPORTED

Each dot is a bed temperature one of the printers reported. **Orange** is a bed heating, **slate** a bed holding its temperature. With the gap, Mark2's bed was already at 80 °C when Mark 1's began to climb.

What happened

Two display covers, one on each printer, were sent 54 seconds apart late on 27 September 2026. The breaker tripped. When the power came back both printers reported their job failed at layer 0, and neither picked it up again. Derek had seen the same thing when he started them by hand.

A cut mid-print is different

Two days earlier both printers had been unplugged by accident. Mark 1 was part of the way through a quarter of the enclosure. It carried on when the power returned, and Derek could see no mark where it had stopped.

What we do

After one printer accepts a job, wait at least three minutes before starting or resuming the other. A resume counts as a start. After a power cut, read both printers first: a job that is still running needs nothing sent.

Who keeps the time

The sending tool stops two sends colliding in the app. It does not keep this interval. That is done by hand, and both start times go in the record. The enclosure quarter of 11 November 2026 records that Mark 1 had started 5227 seconds before it.

54 seconds

between the two starts that tripped the breaker.

3:56

minutes and seconds between the two that ran.

STILL OPEN Three minutes is Derek's figure, from experience. One run at just under four is all that is on record. Nothing has measured the current, and the readings only fit the two beds heating together. They do not prove it.

IN THE RECORD

The trial · Its bed readings · The rule as written · The power cut two days before · Both start times in a job record

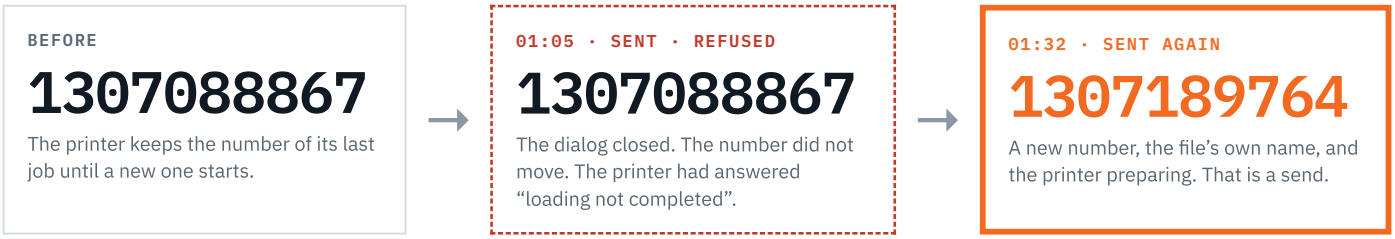
H2C

SENDING A JOB

A send is proven only by a new job number.

The printer starts a job only when Bambu's own program asks. So an agent presses that program's buttons, then listens to the printer itself. Several times the press did nothing, and nothing on the screen said so.

SEEN IN Mark2 · PET-GF jobs · sent through Bambu Connect · autumn 2026 Not tried in: the same refusals on Mark 1



4 OCTOBER 2026 · MARK2 · ONE LABEL RING, SENT TWICE

FOUR WAYS A SEND HAS GONE WRONG

21 SEPTEMBER 2026 · MARK2

It vanished

Two sends went out while the printer was loading filament, its nozzle held hot with no job. Each time the dialog closed and nothing started. The third, after the heat was off, ran. Three more vanished over the next two days, for no reason anyone found.

4 OCTOBER 2026 · MARK2

Refused: still hot

Just after a spool was loaded, the printer's heater targets already read zero. Its nozzle was still at 232 °C, and it refused the job. Twenty-seven minutes later it took the same file. That is the send drawn above.

12 OCTOBER 2026 · MARK2

Refused: seen before

A file that had been started and cancelled was refused as invalid when it was sent again. The identical file under a new name was accepted.

20 SEPTEMBER 2026

The other printer

The send dialog offered Mark 1 three times after Mark2's page had been opened. The tool now reads the name in the dialog and changes it before it presses anything else.

How an agent sends a job

1. **Read both printers.** Finished and failed both mean idle. A heater still set above zero means wait.
2. **Open the checked file in Bambu Connect** by pressing its controls the way a screen reader would. The printer ignores a start from any program Bambu has not signed.
3. **Check the dialog:** the printer named, the spool on each nozzle by type and colour, timelapse and bed levelling on, both calibrations on auto.
4. **Press Send once.**
5. **Listen to the printer.** A new job number, the file's name, preparing or running, no error. An unclear send is never simply repeated.

> *"If I tell you to print something, you print something. I will not tell you to print something if the plate is not clear."*
>
> DEREK · 22 SEPTEMBER 2026

> *"There is no queue. That is some agent's invention..."*
>
> DEREK · 23 SEPTEMBER 2026

So nobody asks whether the bed is clear, or how much is left on the spool. A request to print is the answer to both.

STILL OPEN Why three sends vanished with nothing loading, and why a cancelled file's name is refused, are not known.

IN THE RECORD

How sending works, and each failure · The refused send · The sends that vanished

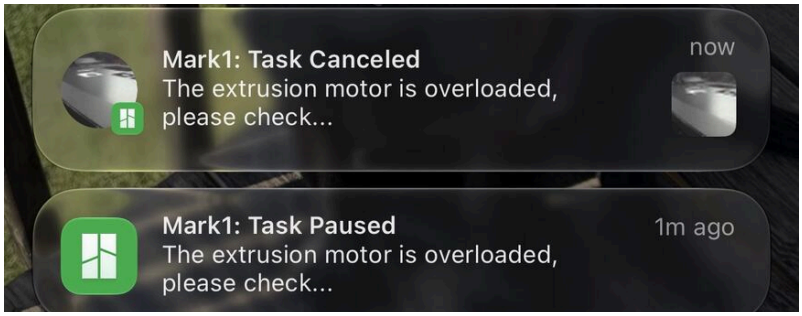
H2C

"THE EXTRUSION MOTOR IS OVERLOADED"

One message. Three causes.

The printer feels how hard its motor has to push the filament, and pauses when that is too hard. The message is the same whatever is in the way. We have had three different things in the way.

SEEN IN recorded on Mark 1, three times · PET-GF on a left 0.4 mm, twice · white PETG on a right 0.2 mm, once



6 NOVEMBER 2026 · MARK 1 · ON DEREK'S PHONE
All the printer says. The same words arrived for all three causes below.



7 NOVEMBER 2026 · MARK 1 · WHITE PETG
The hotend Derek took off after the second stop. It looked like the cause, and a spare went on.

1 Filament left loaded and idle

28 SEPTEMBER 2026 · MARK 1 · PET-GF · LEFT 0.4 MM

A print paused before its first layer. The printer had sat for a while with filament in it.

What fixed it. Cut the filament at the spool. Unload at the touchscreen. Pull the whole cut piece out. Load the fresh end and resume. The print ran its 252 layers.

Derek had done this a few times by then. He suspects the length left in the feed path takes up moisture. Nothing has tested that.

2 A hotend that would not pass plastic

11 SEPTEMBER 2026 · "THE H2C", AS MARK 1 WAS CALLED · PET-GF · LEFT 0.4 MM

The print paused. Derek cleaned the outside of the hotend and resumed, and it paused again. His words for the message that day were "extruder motor overheating".

What fixed it. Another hotend. He swapped in the spare, resumed, and the same job finished.

The hotend that came off has not been cold-pulled, so "clogged" is still a suspicion.

3 A nozzle too close to the plate

6 TO 8 NOVEMBER 2026 · MARK 1 · WHITE PETG · RIGHT 0.2 MM · GLUED ENGINEERING PLATE

Four tries stopped on this message, at about the same spot in the first layer. Drying the spools twice, a spare hotend and rethreading the filament changed nothing.

What fixed it. Lifting the right-side nozzles 0.05 mm off the plate. The fifth try finished.

The PETG part tells this one in full.

Never push a clog from below.

In May 2026, during the first PET-CF faucet attempts, filament jammed in the feed gears above the right hotend of the only printer. A probe pushed up from below to free it snapped the pincers that grip that hotend and cut the heater's flat cable.

On the evening of 4 May a replacement extruder was ordered, and an hour later a second printer. That printer is Mark2.

THE WAY IN, FROM THE FRONT

1. Take off the back fan.
2. Unplug the hotend's flat cable.
3. Unscrew its power leads.
4. Only then undo its mounting screws.
5. Lift off the plate in front of the motors. The jam is there.

> *"Well, it broke the same way again."*
>
> DEREK · 7 NOVEMBER 2026 · AFTER THE SPARE HOTEND

A fouled hotend can be the result of the trouble. On the fourth stop he asked for one change only, and it was the one that worked.

IN THE RECORD

The idle-filament recovery · The swapped hotend · The lifted nozzle · The clog and the way in

THE RULES, RANKED BY THE RECORD

Seven things that had to be said again.

Since late September 2026, every change to the machine is saved with the words that asked for it. Read back in order, they are a ledger of what the written record kept failing to pass on. Each mark is one commit that carries Derek saying it, and each date opens that commit.

9 TIMES

Fewer edges, more material.

ANY PLASTIC

"...an attempt to use less material when less material is basically the opposite of our goal." 19 OCT

5 OCT · 8 OCT · 10 OCT · 19 OCT · 19 OCT · 25 OCT · 27 OCT · 1 NOV · 12 NOV

7 TIMES

Blocks of support do not come out of a cavity. Short trees rooted inside it do.

PET-GF

*"lol, lmfao, it is *every* support here ... there's no way I'm going to be able to do anything with this except throw it away"* 9 OCT

4 OCT · 9 OCT · 9 OCT · 11 OCT · 25 OCT · 6 NOV · 8 NOV

6 TIMES

What one print proved has to reach the next print.

ANY PLASTIC

*"Please start me a complete faucet with ***EVERYTHING*** we've learned."* 15 OCT

5 OCT · 14 OCT · 15 OCT · 26 OCT · 7 NOV · 11 NOV

5 TIMES

Change one thing. Do not print the control again.

ANY PLASTIC

"...get back to our normal settings, and pick one specific thing to change to address the problem we are seeing." 9 OCT

9 OCT · 15 OCT · 8 NOV · 9 NOV · 12 NOV

4 TIMES

Fan on every bridge and overhang, and nowhere else.

PET-GF

"We can use 100% cooling if we want, but only on the places that actually need it..." 17 OCT

15 OCT · 17 OCT · 6 NOV · 12 NOV

4 TIMES

A first-layer trim belongs to one printer, one plate and one plastic.

EACH PLASTIC

*"The +0.04 trim was a *special* case for a problem with a *specific* print - not a change to the rule for Mark1."* 8 OCT

8 OCT · 13 OCT · 25 OCT · 6 NOV

3 TIMES

Stay off the edges of a textured plate that has not been printed on.

TEXTURED PEI

"You can't print that close to the edge of the bed on a plate that hasn't been seasoned properly first." 6 OCT

4 OCT · 6 OCT · 10 OCT

AND FROM THE NIGHT THIS BOOK WAS ASKED FOR

12 November 2026.

> *"The only reason a bridge works is because it *is* a bridge that actually has a column supporting both ends."*
>
> DEREK

> *"Do not use finer layers anywhere - that is not an option - that is a completely different variable"*
>
> DEREK

2026

Nine months.

Dates are those of the commit or purchase that records each event.

● The project ● PET-GF ● PETG ● TPU ● ASA Aero ● PET-CF ● The H2C

March

22 ● The first printer is ordered.

April

2 ● The first test prints: three snap-fit struts.
5 ● A section of general printing advice is deleted from the repository.
23 ● A printed pressure vessel is abandoned.

May

2 ● The faucet's print log opens, in PET-CF.
4 ● A hotend is wrecked clearing a clog. The second printer is ordered.
10 ● Faucet attempt nine is logged as a success.
22 ● The first reservoir attempt in PETG.
30 ● The first reservoir that holds water.
30 ● An 85A TPU seal seats.

June

14 ● A float is printed around a magnet for the first time.
18 ● The first test coupon cut from the enclosure.

August

15 ● The first enclosure piece in PET-GF.
20 ● The enclosure's exterior moves to a 0.4 mm nozzle.
28 ● A ladder of first-layer trims.

September

13 ● Agents are given the printer controls.
14 ● Sixteen specimens start the support trials.
23 ● Enclosure quadrants come off both printers.

October

3 ● The float printed wholly in ASA Aero is accepted.
4 ● A front-top is rejected for its supports.
9 ● A back-top is thrown away for the same reason.
13 ● The first 85A pump elbow fits.
17 ● White covers crack. The fan is limited to bridges and overhangs.
22 ● The elbows print without strings.
26 ● A reservoir holds water at the last overlap setting.

November

9 ● The handhold roofs become corbels.
10 ● Support D is adopted across PET-GF.
12 ● A pocket roof is printed over air. This book is asked for.

HOW THIS BOOK IS KEPT

The page you read is the file an agent reads.

The book lives in the machine's repository, beside the parts it describes. Each page is one plain file.

Where it is

The pages are `hardware/printing-the-machine/NN-title.html`, numbered in reading order. `README.md` in the same folder is the index: every page's rule, by plastic, and where to start for the job in hand.

Where its pictures come from

Every photograph was taken at the bench and saved with the print it shows. `art/manifest.json` lists each one with the file and revision it was copied from. Pictures of the model are drawn from the same geometry the printers get.

Where its numbers come from

Settings in the current recipes are read from their project files each time the book is bound; `guide-inputs.json` names those files. Numbers from trials are as they were measured, with a date.

To add a page

Start from a print: its date, printer, plastic, plate and nozzle, and a photograph. Say what happened, then the rule it taught, then how far that has been tested. Put it in the part for its plastic. Link the record.

To bind it

`tools/run hsm-manual-guides bind printing-the-machine` draws every page, stops on any whose content overflows, and writes the PDF this was read from.

What is not in it yet

How far PET-GF leans, and its bridges and fins. Screws and heat-set inserts. Why clear PETG moved to a glued plate, and PETG's molds. Reading back the file the printer receives. Each has prints on record and no page.

Who made it

The prints, the photographs and the judgements are Derek's, made with AI agents over nine months of 2026. The pages were written and drawn from that record by Claude (Fable 5.2) in November 2026.

> *"I'd like the repo to be a clear memory to all agents, a written memory that builds FDM print expertise progressively as and when needed. And a PDF that gives a visual high level of it all, so I can see too."*
>
> DEREK · 12 NOVEMBER 2026